

4)

Given the points are

$$A(1, 5, 0) \quad B(6, 6, 4) \quad C(0, 9, 5)$$

Then

$$\begin{aligned} |AB| &= \sqrt{(6-1)^2 + (6-5)^2 + (4-0)^2} \\ &= \sqrt{25+1+16} \\ &= \sqrt{42} \end{aligned}$$

$$\begin{aligned} |BC| &= \sqrt{(0-6)^2 + (9-6)^2 + (5-4)^2} \\ &= \sqrt{36+9+1} \\ &= \sqrt{46} \end{aligned}$$

$$\begin{aligned} |CA| &= \sqrt{(1-0)^2 + (5-9)^2 + (0-5)^2} \\ &= \sqrt{1+16+25} \\ &= \sqrt{42} \end{aligned}$$

Since $|AB| = |CA|$

So it is an isosceles triangle.

5) Given the points are
 $A(4, 9, 4) \quad B(0, 11, 2) \quad C(1, 0, 1)$

Then

$$\begin{aligned} |AB| &= \sqrt{(0-4)^2 + (11-9)^2 + (2-4)^2} \\ &= \sqrt{16+4+4} \\ &= \sqrt{24} \end{aligned}$$

$$\begin{aligned} |BC| &= \sqrt{(1-0)^2 + (0-11)^2 + (1-2)^2} \\ &= \sqrt{1+121+1} \\ &= \sqrt{123} \end{aligned}$$

$$\begin{aligned} |CA| &= \sqrt{(1-4)^2 + (0-9)^2 + (1-4)^2} \\ &= \sqrt{9+81+9} \\ &= \sqrt{99} \end{aligned}$$

Clearly

$$|BC|^2 = (|AB|)^2 + (|CA|)^2$$

$$123 = 24 + 99$$

$$123 = 123$$

Therefore the given triangle is right angle.

6) Given the points are

$$A(1, 0, 2) \quad B(4, 3, 2) \quad C(0, 7, 6)$$

Then

$$\begin{aligned} |AB| &= \sqrt{(4-1)^2 + (3-0)^2 + (2-2)^2} \\ &= \sqrt{9+9} = \sqrt{18} \end{aligned}$$

$$\begin{aligned} |BC| &= \sqrt{(0-4)^2 + (7-3)^2 + (6-2)^2} \\ &= \sqrt{16+16+16} \\ &= \sqrt{48} \end{aligned}$$

$$\begin{aligned} |CA| &= \sqrt{(1-0)^2 + (0-7)^2 + (2-6)^2} \\ &= \sqrt{1+49+16} \\ &= \sqrt{66} \end{aligned}$$

Clearly

$$|AB|^2 + |BC|^2 = |CA|^2$$

$$18 + 48 = 66$$

$$66 = 66$$

Therefore the given triangle is right angled.

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⑦ Given the points are

$$A(2, 3, 4) \quad B(8, -1, 2) \quad C(-4, 1, 0)$$

Then

$$\begin{aligned} |AB| &= \sqrt{(8-2)^2 + (-1-3)^2 + (2-4)^2} \\ &= \sqrt{36 + 16 + 4} \\ &= \sqrt{56} \end{aligned}$$

$$\begin{aligned} |BC| &= \sqrt{(-4-8)^2 + (1+1)^2 + (0-2)^2} \\ &= \sqrt{144 + 4 + 4} \\ &= \sqrt{152} \end{aligned}$$

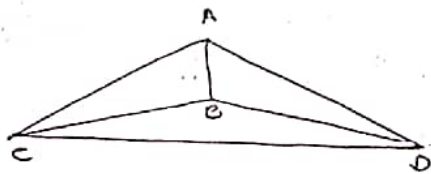
$$\begin{aligned} |AC| &= \sqrt{(2+4)^2 + (3-1)^2 + (4-0)^2} \\ &= \sqrt{36 + 4 + 16} \\ &= \sqrt{56} \end{aligned}$$

Since $|AB| = |AC|$
it is an isosceles triangle.

⑧ we are given to prove

$$\text{that the points } A(1, 6, 1) \\ B(1, 3, 4) \quad C(4, 3, 1) \quad D(0, 2, 0)$$

are the vertices of a regular tetrahedron.



Now

$$\begin{aligned} |AB| &= \sqrt{(1-1)^2 + (3-6)^2 + (4-1)^2} \\ &= \sqrt{0 + 9 + 9} = \sqrt{18} \\ &= 3\sqrt{2} \end{aligned}$$

$$\begin{aligned} |AC| &= \sqrt{(4-1)^2 + (3-6)^2 + (1-1)^2} \\ &= \sqrt{9 + 9 + 0} \\ &= \sqrt{18} \end{aligned}$$

$$|AC| = 3\sqrt{2}$$

$$\begin{aligned} |AD| &= \sqrt{(0-1)^2 + (2-6)^2 + (0-1)^2} \\ &= \sqrt{1 + 16 + 1} \\ &= \sqrt{18} = 3\sqrt{2} \end{aligned}$$

Similarly

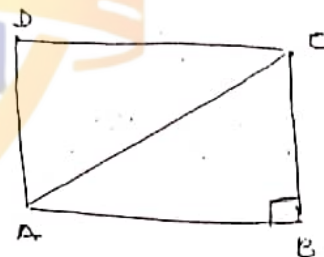
$$|BC| = |CD| = |DB| = 3\sqrt{2}$$

Hence the given points are the vertices of the regular tetrahedron.

⑨ we are given to prove

that the points $A(3, -1, 3)$
 $B(1, -1, 2) \quad C(2, 1, 0) \quad D(4, 1, 1)$
are the vertices of a rectangle.

Firstly



Now

$$\begin{aligned} |AB| &= \sqrt{(1-3)^2 + (-1+1)^2 + (2-3)^2} \\ &= \sqrt{4 + 0 + 1} = \sqrt{5} \end{aligned}$$

$$\begin{aligned} |CD| &= \sqrt{(4-2)^2 + (1-1)^2 + (1-0)^2} \\ &= \sqrt{4 + 0 + 1} = \sqrt{5} \end{aligned}$$

$$\therefore \text{Hencefore } |AB| = |CD|$$

Again

$$\begin{aligned} |AC| &= \sqrt{(2-3)^2 + (1+1)^2 + (0-3)^2} \\ &= \sqrt{1 + 4 + 9} \\ &= \sqrt{14} \end{aligned}$$

$$|BD| = \sqrt{(4-1)^2 + (1+1)^2 + (1-2)^2}$$

$$= \sqrt{9+4+1}$$

$$= \sqrt{14}$$

Therefore $|AC| = |BD|$

Andly

To prove that the angle B is a right angle.

For this

$$|BC| = \sqrt{(2-1)^2 + (1+1)^2 + (0-2)^2}$$

$$= \sqrt{1+4+4}$$

$$= \sqrt{9}$$

So

$$(|AC|)^2 = (|AB|)^2 + (|BC|)^2$$

$$(\sqrt{14})^2 = (\sqrt{5})^2 + (3)^2$$

$$14 = 5+9$$

$$14 = 14$$

So, B is a right angle

Hence ABCD is a rectangle.

⑫ Let the given points

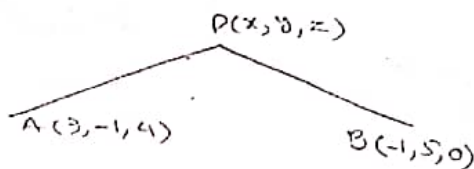
A(3, -1, 4) & B(-1, 5, 0)

Let the required point be

P(x, y, z)

Then by the given condition,

$$|AP| = |BP|$$



$$\sqrt{(x-3)^2 + (y+1)^2 + (z-4)^2} = \sqrt{(x+1)^2 + (y-5)^2 + (z-0)^2}$$

Squaring both sides

$$(x-3)^2 + (y+1)^2 + (z-4)^2 = (x+1)^2 + (y-5)^2 + (z-0)^2$$

$$x^2 + 9 - 6x + y^2 + 1 + 2y + z^2 + 16 - 8z =$$

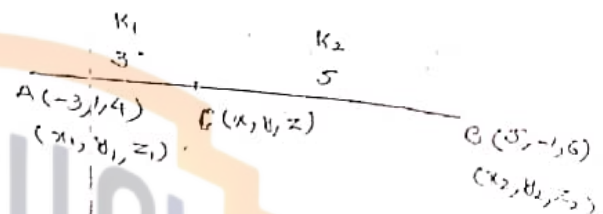
$$x^2 + 1 + 2x + y^2 + 25 - 10y + z^2$$

$$-8x + 12y - 8z = 0$$

$$\text{or } 2x - 3y + 2z = 0$$

⑪

we are need to find the point dividing AB in the ratio 3:5



$$x = \frac{K_1 x_2 + K_2 x_1}{K_1 + K_2} = \frac{3(5) + 5(-3)}{3+5}$$

$$= \frac{15-15}{8} = \frac{0}{8} = 0$$

$$y = \frac{K_1 y_2 + K_2 y_1}{K_1 + K_2} = \frac{3(-1) + 5(1)}{3+5} = \frac{1}{4}$$

$$z = \frac{K_1 z_2 + K_2 z_1}{K_1 + K_2} = \frac{3(6) + 5(4)}{3+5} = \frac{38}{8}$$

$$= \frac{19}{4}$$

Required pt $(0, \frac{1}{4}, \frac{19}{4})$



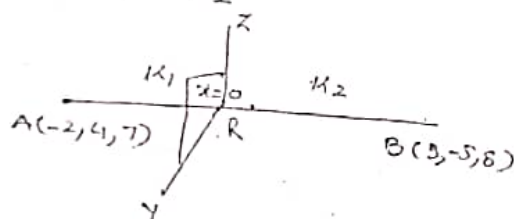
(12)

Suppose the YZ -plane divides the AB in the ratio $k_1:k_2$

So x must be zero at the YZ -plane

x -Coordinate of the point R

$$x = \frac{k_1 x_2 + k_2 x_1}{k_1 + k_2}$$



$$0 = \frac{3k_1 - 2k_2}{k_1 + k_2}$$

$$3k_1 - 2k_2 = 0$$

$$3k_1 = 2k_2$$

$$\frac{k_1}{k_2} = \frac{2}{3}$$

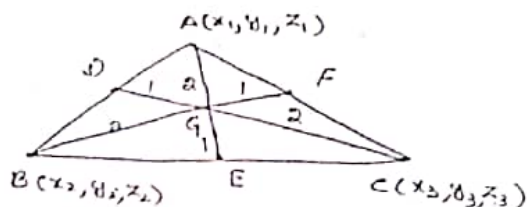
i.e. $k_1:k_2 = 2:3$

(13)

For $i=1, 2, 3$

Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$

$C(x_3, y_3, z_3)$



$$D = \text{mid pt of } AB = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

$$E = \text{mid pt of } BC = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}, \frac{z_2 + z_3}{2} \right)$$

$$F = \text{mid pt of } AC = \left(\frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2}, \frac{z_1 + z_3}{2} \right)$$

Now

Coordinates of G dividing AB

$$= \left(\frac{1(x_2) + 2\left(\frac{x_1 + x_2}{2}\right)}{1+2}, \frac{1(y_2) + 2\left(\frac{y_1 + y_2}{2}\right)}{1+2}, \frac{1(z_2) + 2\left(\frac{z_1 + z_2}{2}\right)}{1+2} \right)$$

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

Coordinates of G dividing BC

$$= \left(\frac{1(x_3) + 2\left(\frac{x_1 + x_2}{2}\right)}{1+2}, \frac{1(y_3) + 2\left(\frac{y_1 + y_2}{2}\right)}{1+2}, \frac{1(z_3) + 2\left(\frac{z_1 + z_2}{2}\right)}{1+2} \right)$$

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

Coordinates of G dividing AC

$$= \left(\frac{1(x_1) + 2\left(\frac{x_2 + x_3}{2}\right)}{1+2}, \frac{1(y_1) + 2\left(\frac{y_2 + y_3}{2}\right)}{1+2}, \frac{1(z_1) + 2\left(\frac{z_2 + z_3}{2}\right)}{1+2} \right)$$

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

We find that Centroid of the triangle

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

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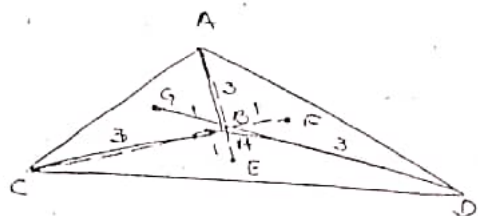
(11)

For $i = 1, 2, 3, 4$

Let $A(x_1, y_1, z_1)$ $B(x_2, y_2, z_2)$

$C(x_3, y_3, z_3)$ and $D(x_4, y_4, z_4)$

be the vertices of tetrahedron



$$E = \text{Centroid of } \triangle BCD \text{ Face} = \left(\frac{x_2 + x_3 + x_4}{3}, \frac{y_2 + y_3 + y_4}{3}, \frac{z_2 + z_3 + z_4}{3} \right)$$

$$F = \text{Centroid of } \triangle ABD \text{ Face} = \left(\frac{x_1 + x_2 + x_4}{3}, \frac{y_1 + y_2 + y_4}{3}, \frac{z_1 + z_2 + z_4}{3} \right)$$

$$G = \text{Centroid of } \triangle ABC \text{ Face} = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

The coordinates of H dividing CF in the ratio 3:1

$$\left(\frac{1(x_3) + 3\left(\frac{x_1 + x_2 + x_4}{3}\right)}{1+3}, \frac{1(y_3) + 3\left(\frac{y_1 + y_2 + y_4}{3}\right)}{1+3}, \frac{1(z_3) + 3\left(\frac{z_1 + z_2 + z_4}{3}\right)}{1+3} \right)$$

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

The coordinates of H dividing AD in the ratio 3:1

$$\left(\frac{1(x_1) + 3\left(\frac{x_2 + x_3 + x_4}{3}\right)}{1+3}, \frac{1(y_1) + 3\left(\frac{y_2 + y_3 + y_4}{3}\right)}{1+3}, \frac{1(z_1) + 3\left(\frac{z_2 + z_3 + z_4}{3}\right)}{1+3} \right)$$

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

The coordinates of H dividing BE in the ratio 3:1

$$\left(\frac{1(x_4) + 3\left(\frac{x_1 + x_2 + x_3}{3}\right)}{1+3}, \frac{1(y_4) + 3\left(\frac{y_1 + y_2 + y_3}{3}\right)}{1+3}, \frac{1(z_4) + 3\left(\frac{z_1 + z_2 + z_3}{3}\right)}{1+3} \right)$$

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

So the centroid of tetrahedron is

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

The Parametric Equation of the

line passing through the points
 $(1, -2, 0)$ and $(5, -10, 1)$
 (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$x = x_1 + t(x_2 - x_1)$$

$$y = y_1 + t(y_2 - y_1)$$

$$z = z_1 + t(z_2 - z_1)$$

So $x = 1 + t(5 - 1) \Rightarrow x = 1 + 4t$

$$y = -2 + t(-10 + 2) \Rightarrow y = -2 - 8t$$

$$z = 0 + t(1 - 0) \Rightarrow z = t$$

The Symmetric Form of the

line passing through the points

$(1, -2, 0)$ and $(5, -10, 1)$ is given
 (x_1, y_1, z_1) and (x_2, y_2, z_2)

by
$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

So
$$\frac{x - 1}{5 - 1} = \frac{y + 2}{-10 + 2} = \frac{z - 0}{1 - 0}$$

$$\frac{x - 1}{4} = \frac{y + 2}{-8} = \frac{z - 0}{1}$$

Now the Direction Ratios of

the given line are

$$x_2 - x_1 = 5 - 1 = 4$$

$$y_2 - y_1 = -10 + 2 = -8$$

$$z_2 - z_1 = 1 - 0 = 1$$

Therefore the Direction Cosines are

$$\left(\frac{4}{\sqrt{(4)^2 + (-8)^2 + (1)^2}}, \frac{-8}{\sqrt{(4)^2 + (-8)^2 + (1)^2}}, \frac{1}{\sqrt{(4)^2 + (-8)^2 + (1)^2}} \right)$$

$$\frac{4}{\sqrt{16 + 64 + 1}}, \frac{-8}{\sqrt{16 + 64 + 1}}, \frac{1}{\sqrt{16 + 64 + 1}}$$

$$\frac{4}{9}, \frac{-8}{9}, \frac{1}{9}$$

Direction Angles

$$\cos \alpha = \frac{4}{9} \Rightarrow \alpha = \cos^{-1}\left(\frac{4}{9}\right)$$

$$\cos \beta = \frac{-8}{9} \Rightarrow \beta = \cos^{-1}\left(\frac{-8}{9}\right)$$

$$\cos \gamma = \frac{1}{9} \Rightarrow \gamma = \cos^{-1}\left(\frac{1}{9}\right)$$

(2) The Parametric Equation of the

line passing through the points

$P(6, 5, -3)$ and $Q(4, 1, 1)$ is
 (x_1, y_1, z_1) and (x_2, y_2, z_2)

$$x = x_1 + t(x_2 - x_1)$$

$$y = y_1 + t(y_2 - y_1)$$

$$z = z_1 + t(z_2 - z_1)$$

So $x = 6 + t(4 - 6) \Rightarrow x = 6 - 2t$

$$y = 5 + t(1 - 5) \Rightarrow y = 5 - 4t$$

$$z = -3 + t(1 + 3) \Rightarrow z = -3 + 4t$$

The Symmetric Form of the

line passing through the pts.

(x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

So
$$\frac{x - 6}{4 - 6} = \frac{y - 5}{1 - 5} = \frac{z + 3}{1 + 3}$$

$$\frac{x - 6}{-2} = \frac{y - 5}{-4} = \frac{z + 3}{4}$$

$$\frac{x - 6}{1} = \frac{y - 5}{2} = \frac{z + 3}{-2}$$

Now the Direction Ratios of

the given line are

$$x_2 - x_1 = 4 - 6 = -2$$

$$y_2 - y_1 = 1 - 5 = -4$$

$$z_2 - z_1 = 1 + 3 = 4$$

Therefore the Direction Cosines are

$$\left(\frac{-2}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}}, \frac{-4}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}}, \frac{4}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}} \right)$$

$$\left(\frac{-2}{\sqrt{4+16+16}}, \frac{-4}{\sqrt{4+16+16}}, \frac{4}{\sqrt{4+16+16}} \right)$$

$$-\frac{2}{6}, -\frac{4}{6}, \frac{4}{6}$$

$$-\frac{1}{3}, -\frac{2}{3}, +\frac{2}{3}$$

Direction Angles

$$\cos \alpha = -\frac{1}{3}$$

$$\Rightarrow \alpha = \cos^{-1}\left(-\frac{1}{3}\right)$$

$$\cos \beta = -\frac{2}{3}$$

$$\Rightarrow \beta = \cos^{-1}\left(-\frac{2}{3}\right)$$

$$\cos \gamma = \frac{2}{3}$$

$$\Rightarrow \gamma = \cos^{-1}\left(\frac{2}{3}\right)$$

2) The Parametric Equation of the line passing through the points $P(1, -5, 1)$ and $Q(4, -5, 4)$ is

$$x = x_1 + t(x_2 - x_1)$$

$$y = y_1 + t(y_2 - y_1)$$

$$z = z_1 + t(z_2 - z_1)$$

$$\text{So } x = 1 + t(4-1)$$

$$\Rightarrow x = 1 + 3t$$

$$y = -5 + t(-5+5)$$

$$\Rightarrow y = -5 + 0t$$

$$z = 1 + t(4-1)$$

$$\Rightarrow z = 1 + 3t$$

The Symmetric Form of the line passing through the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-1}{4-1} = \frac{y+5}{-5+5} = \frac{z-1}{4-1}$$

$$\frac{x-1}{3} = \frac{y+5}{0} = \frac{z-1}{3}$$

$$\frac{x-1}{1} = \frac{y+5}{0} = \frac{z-1}{1}$$

Now the Direction Ratios of the given line are

$$x_2 - x_1 = 4 - 1 = 3$$

$$y_2 - y_1 = -5 + 5 = 0$$

$$z_2 - z_1 = 4 - 1 = 3$$

Therefore the Direction Cosines are

$$\left(\frac{3}{\sqrt{(3)^2 + (0)^2 + (3)^2}}, \frac{0}{\sqrt{(3)^2 + (0)^2 + (3)^2}}, \frac{3}{\sqrt{(3)^2 + (0)^2 + (3)^2}} \right)$$

$$\left(\frac{3}{\sqrt{18}}, 0, \frac{3}{\sqrt{18}} \right)$$

$$\left(\frac{3}{3\sqrt{2}}, 0, \frac{3}{3\sqrt{2}} \right)$$

$$\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$$

Direction Angles

$$\cos \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

$$\cos \beta = 0 \Rightarrow \beta = \cos^{-1}(0) = 90^\circ$$

$$\cos \gamma = \frac{1}{\sqrt{2}} \Rightarrow \gamma = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

(4)

The Parametric Equation of the line passing through the points $P(3, 5, 7)$ and $Q(6, -8, 10)$
 (x_1, y_1, z_1) (x_2, y_2, z_2)

is

$$x = x_1 + t(x_2 - x_1)$$

$$y = y_1 + t(y_2 - y_1)$$

$$z = z_1 + t(z_2 - z_1)$$

So

$$x = 3 + t(6 - 3) \Rightarrow x = 3 + 3t$$

$$y = 5 + t(-8 - 5) \Rightarrow y = 5 - 13t$$

$$z = 7 + t(10 - 7) \Rightarrow z = 7 + 3t$$

The Symmetric Form of the line passing through the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

$$\frac{x - 3}{6 - 3} = \frac{y - 5}{-8 - 5} = \frac{z - 7}{10 - 7}$$

$$\frac{x - 3}{3} = \frac{y - 5}{-13} = \frac{z - 7}{3}$$

Now the Direction Ratios of

the given line are

$$x_2 - x_1 = 6 - 3 = 3$$

$$y_2 - y_1 = -8 - 5 = -13$$

$$z_2 - z_1 = 10 - 7 = 3$$

Therefore the Direction Cosines

are

$$\left(\frac{3}{\sqrt{9+169+9}}, \frac{-13}{\sqrt{9+169+9}}, \frac{3}{\sqrt{9+169+9}} \right)$$

$$\left(\frac{3}{\sqrt{187}}, \frac{-13}{\sqrt{187}}, \frac{3}{\sqrt{187}} \right)$$

Direction Angles

$$\cos \alpha = \frac{3}{\sqrt{187}}$$

$$\Rightarrow \alpha = \cos^{-1}\left(\frac{3}{\sqrt{187}}\right)$$

$$\cos \beta = \frac{-13}{\sqrt{187}}$$

$$\Rightarrow \beta = \cos^{-1}\left(\frac{-13}{\sqrt{187}}\right)$$

$$\cos \gamma = \frac{3}{\sqrt{187}}$$

$$\Rightarrow \gamma = \cos^{-1}\left(\frac{3}{\sqrt{187}}\right)$$

(5)

i. Direction Angles $0^\circ, 90^\circ, 90^\circ$
 of x-axis

Direction Cosines $\cos 0^\circ, \cos 90^\circ, \cos 90^\circ$
 of x-axis $\Rightarrow 1, 0, 0$

ii. Direction Angles $90^\circ, 0^\circ, 90^\circ$
 of y-axis

Direction Cosines $\cos 90^\circ, \cos 0^\circ, \cos 90^\circ$
 of y-axis $\Rightarrow 0, 1, 0$

iii. Direction Angles $90^\circ, 90^\circ, 0^\circ$
 of z-axis

Direction Cosines $\cos 90^\circ, \cos 90^\circ, \cos 0^\circ$
 of z-axis $\Rightarrow 0, 0, 1$

⑥

Given that α, β, γ are
and $\cos \alpha, \cos \beta, \cos \gamma$ are direction cosines
direction angles, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

$$- \sin^2 \alpha - \sin^2 \beta - \sin^2 \gamma = 1 - 3$$

$$- \sin^2 \alpha - \sin^2 \beta - \sin^2 \gamma = -2$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

⑦

γ $\alpha = 45^\circ, \beta = 60^\circ$
 $\gamma = ?$

then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\cos^2 45^\circ + \cos^2 60^\circ + \cos^2 \gamma = 1$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \cos^2 \gamma = 1$$

$$\frac{1}{2} + \frac{1}{4} + \cos^2 \gamma = 1$$

$$\cos^2 \gamma = 1 - \frac{1}{2} - \frac{1}{4}$$

$$\cos^2 \gamma = \frac{1}{4}$$

$$\cos \gamma = \pm \frac{1}{2}$$

$$\cos \gamma = \frac{1}{2}$$

$$\cos \gamma = -\frac{1}{2}$$

$$\gamma = \cos^{-1}\left(\frac{1}{2}\right)$$

$$\gamma = \cos^{-1}\left(-\frac{1}{2}\right)$$

$$\gamma = 60^\circ$$

$$\gamma = 120^\circ$$

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(11)

Direction cosines of two lines are

$$l + m + n = 0 \quad \text{--- (1)}$$

$$l^2 + m^2 - n^2 = 0 \quad \text{--- (2)}$$

From (1) $n = -l - m$
 $n = -(l + m)$

put value of n in Eq (2)

$$l^2 + m^2 - (l + m)^2 = 0$$

$$l^2 + m^2 - (l^2 + m^2 + 2lm) = 0$$

$$\cancel{l^2} + \cancel{m^2} - \cancel{l^2} - \cancel{m^2} - 2lm = 0$$

$$-2lm = 0$$

Either $l = 0$ --- (3)

or $m = 0$ --- (4)

From (3) & (2)

$$l + m + n = 0$$

$$l + 0 + n = 0$$

$$\frac{l}{0} = \frac{m}{-1} = \frac{n}{1}$$

$$\frac{l}{0} = \frac{m}{1} = \frac{n}{-1}$$

$$\Rightarrow a = 0, \quad b = 1, \quad c = -1$$

l, m, n are proportional to
 $0, 1, -1$

Direction cosines are

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\frac{0}{\sqrt{0+1+1}}, \frac{1}{\sqrt{0+1+1}}, \frac{-1}{\sqrt{0+1+1}}$$

$$0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$$

From (1) & (4)

$$l + m + n = 0$$

$$0 + l + m + 0 = 0$$

$$\frac{l}{1} = \frac{m}{-1} = \frac{n}{1}$$

$$\frac{l}{-1} = \frac{m}{0} = \frac{n}{1}$$

$$\Rightarrow a = -1, \quad b = 0, \quad c = 1$$

l, m, n are proportional to
 $-1, 0, 1$

Direction cosines are

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\frac{-1}{\sqrt{1+0+1}}, \frac{0}{\sqrt{1+0+1}}, \frac{1}{\sqrt{1+0+1}}$$

$$-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$$

Therefore, direction cosines of two lines are

$$0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \quad \text{or} \quad -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$$

if θ is the angle between these lines

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= 0(-\frac{1}{\sqrt{2}}) + \frac{1}{\sqrt{2}}(0) + (-\frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}})$$

$$= 0 + 0 - \frac{1}{2}$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \cos^{-1}(-\frac{1}{2})$$

$$\theta = 120^\circ$$

(7)

Direction Cosines of two lines are

$$l+m+n=0$$

①

$$2lm+2ln-mn=0$$

②

From ①

$$n = -l-m$$

$$n = -(l+m)$$

Put value of n in Eq ②

$$2lm+2l[-(l+m)]-m[-(l+m)]=0$$

$$2lm-2l^2-2lm+lm+m^2=0$$

$$-2l^2+lm+m^2=0$$

$$2l^2-lm-m^2=0$$

$$2l^2-2lm+lm-m^2=0$$

$$2l(l-m)+m(l-m)=0$$

$$(l-m)(2l+m)=0$$

$$l-m=0$$

③

$$2l+m=0$$

④

From ① & ③

$$l+m+n=0$$

$$l-l+m+0n=0$$

$$\frac{l}{1} = \frac{m}{-1} = \frac{n}{1}$$

$$\frac{l}{1} = \frac{m}{1} = \frac{n}{-2}$$

are proportional

$$\Rightarrow a=1, b=1, c=-2$$

l, m, n are proportional

to $1, 1, -2$

Direction Cosines are

$$\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

$$\frac{1}{\sqrt{1+1+4}}, \frac{1}{\sqrt{1+1+4}}, \frac{-2}{\sqrt{1+1+4}}$$

$$\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$$

From ① & ④

$$l+m+n=0$$

$$2l+m+0n=0$$

$$\frac{l}{1} = \frac{m}{-1} = \frac{n}{2}$$

$$\frac{l}{-1} = \frac{m}{2} = \frac{n}{-1}$$

$$\Rightarrow a=-1, b=2, c=-1$$

l, m, n are proportional

to $-1, 2, -1$

Direction Cosines are

$$\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

$$\frac{-1}{\sqrt{1+4+1}}, \frac{2}{\sqrt{1+4+1}}, \frac{-1}{\sqrt{1+4+1}}$$

$$-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}$$

Therefore, Direction Cosines of two lines are

$$\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \text{ and } -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}$$

θ is the angle between these lines

$$\cos \theta = \frac{1}{6} \left(-\frac{1}{\sqrt{6}} \right) + \frac{1}{\sqrt{6}} \left(\frac{2}{\sqrt{6}} \right) + \left(-\frac{2}{\sqrt{6}} \right) \left(-\frac{1}{\sqrt{6}} \right)$$

$$= -\frac{1}{6} + \frac{2}{6} + \frac{2}{6}$$

$$= \frac{3}{6}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \cos^{-1} \left(\frac{1}{2} \right)$$

$$\therefore \theta = 60^\circ$$

(11)

(10)

Firstly for line L

The Symmetric Form of the line

L through the points A(2, 1, 3)

 (x_1, y_1, z_1)

and B(-1, 2, -4)

 (x_2, y_2, z_2)

A

B

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-2}{-1-2} = \frac{y-1}{2-1} = \frac{z-3}{-4-3}$$

$$\frac{x-2}{-3} = \frac{y-1}{1} = \frac{z-3}{-7}$$

By putting the symmetric form of the straight line

equal to some parameter 't'

we get the parametric eq

$$\frac{x-2}{-3} = \frac{y-1}{1} = \frac{z-3}{-7} = t$$

$$\frac{x-2}{-3} = t, \quad \frac{y-1}{1} = t, \quad \frac{z-3}{-7} = t$$

$$x-2 = -3t, \quad y-1 = t, \quad z-3 = -7t$$

$$x = 2-3t, \quad y = 1+t, \quad z = 3-7t$$

pt on AB $(2-3t, 1+t, 3-7t)$

Now for Line M

The symmetric form of the line M passing through the

points P(5, 1, -2) Q(0, 4, 3)

 (x_1, y_1, z_1) (x_2, y_2, z_2)

P

Q

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-5}{0-5} = \frac{y-1}{4-1} = \frac{z+2}{3+2}$$

$$\frac{x-5}{-5} = \frac{y-1}{+3} = \frac{z+2}{+5}$$

By putting the symmetric form of the straight line equal to some parameter 's' we get the parametric eq

$$\frac{x-5}{-5} = \frac{y-1}{3} = \frac{z+2}{5} = s$$

$$\left. \begin{array}{l} \frac{x-5}{-5} = s \\ x-5 = -5s \\ x = 5-5s \end{array} \right\} \left. \begin{array}{l} \frac{y-1}{3} = s \\ y-1 = 3s \\ y = 1+3s \end{array} \right\} \left. \begin{array}{l} \frac{z+2}{5} = s \\ z+2 = 5s \\ z = -2+5s \end{array} \right\}$$

pt on PQ

$$(5-5s, 1+3s, -2+5s)$$

we need to find whether lines L, M are intersect or not. The lines L & M will intersect if

pt on AB = pt on PQ

$$2-3t = 5-5s \quad \text{--- i}$$

$$1+t = 1+3s \quad \text{--- ii}$$

$$3-7t = -2+5s \quad \text{--- iii}$$

From i & ii

$$t = -\frac{7}{4}$$

$$s = -\frac{3}{4}$$

Put in iii

$$3 + \frac{63}{4} = -2 - \frac{15}{4}$$

$$\frac{78}{4} \neq -5$$

These lines do not intersect.

(11)

Firstly for line L

$$\vec{r} = (3\hat{i} + 2\hat{j} - \hat{k}) + t(6\hat{i} - 4\hat{j} - 3\hat{k})$$

This is vector Eq of line
in form

$$\vec{r} = \vec{a} + t\vec{b}$$

where $\vec{a} = 3\hat{i} + 2\hat{j} - \hat{k}$ is any pt
and $\vec{b} = 6\hat{i} - 4\hat{j} - 3\hat{k}$ is collinear or parallel
vector.

$$\text{pt } (3, 2, -1)$$

$$\text{DR } [6, -4, -3]$$

The symmetric form of line L

$$\frac{x-3}{6} = \frac{y-2}{-4} = \frac{z+1}{-3}$$

By putting, the symmetric
form of the straight line
equal to some parameter 't'
we get the parametric Eq

$$\frac{x-3}{6} = \frac{y-2}{-4} = \frac{z+1}{-3} = t$$

$$\begin{cases} \frac{x-3}{6} = t \\ \frac{y-2}{-4} = t \\ \frac{z+1}{-3} = t \end{cases} \Rightarrow \begin{cases} x-3 = 6t \\ y-2 = -4t \\ z+1 = -3t \end{cases} \Rightarrow \begin{cases} x = 3+6t \\ y = 2-4t \\ z = -1-3t \end{cases}$$

$$\text{pt on line L } (3+6t, 2-4t, -1-3t)$$

Now for line M

$$\vec{r} = (5\hat{i} + 4\hat{j} + 7\hat{k}) + s(14\hat{i} - 6\hat{j} + 2\hat{k})$$

This is vector Eq of line in
form $\vec{r} = \vec{a} + s\vec{b}$

where $\vec{a} = 5\hat{i} + 4\hat{j} + 7\hat{k}$ is any pt
and $\vec{b} = 14\hat{i} - 6\hat{j} + 2\hat{k}$ is
collinear or parallel vector

$$\text{pt } (5, 4, 7)$$

$$\text{DR } [14, -6, 2]$$

The symmetric form of the
line M

$$\frac{x-5}{14} = \frac{y-4}{-6} = \frac{z-7}{2}$$

By putting, the symmetric
form of the straight line
equal to some parameter 's'
we get the parametric Eq

$$\frac{x-5}{14} = \frac{y-4}{-6} = \frac{z-7}{2} = s$$

$$\begin{cases} \frac{x-5}{14} = s \\ \frac{y-4}{-6} = s \\ \frac{z-7}{2} = s \end{cases} \Rightarrow \begin{cases} x-5 = 14s \\ y-4 = -6s \\ z-7 = 2s \end{cases} \Rightarrow \begin{cases} x = 5+14s \\ y = 4-6s \\ z = 7+2s \end{cases}$$

$$\text{pt on line M } (5+14s, 4-6s, 7+2s)$$

The lines L & M intersecting

$$\begin{cases} 3+6t = 5+14s & \text{--- i} \\ 2-4t = 4-6s & \text{--- ii} \\ -1-3t = 7+2s & \text{--- iii} \end{cases}$$

Solve as ii

$$t = -2$$

$$s = -1$$

Put in iii

$$-1+6 = 7+2s$$

$$s = 5$$

$$\text{LHS} = \text{RHS}$$

Hence lines intersect.

pt of intersection

$$\text{Put } t = -2 \text{ in (A)}$$

$$(-9, 10, 5)$$

(12)

Firstly for line L.

The symmetric form of the straight line L passing through the pt A (2, -1, 0) and parallel to a vector $\vec{b} = [4, 3, -2]$ is

$$\frac{x-2}{4} = \frac{y+1}{3} = \frac{z-0}{-2}$$

By putting, the symmetric form of straight line equal to some parameter "t"

We get the parametric eq.

$$\frac{x-2}{4} = \frac{y+1}{3} = \frac{z-0}{-2} = t$$

$$\begin{aligned} \frac{x-2}{4} = t & \Rightarrow \frac{y+1}{3} = t, \quad \frac{z-0}{-2} = t \\ 2+4t & \quad y+1=3t \quad z=0-2t \\ 2+4t & \quad y=-1+3t \quad z=0-2t \end{aligned}$$

pt on line L $(2+4t, -1+3t, 0-2t)$

Now for line M

The symmetric form of the straight line M passing through pt P (-1, 3, 5) and parallel to a vector $\vec{c} = [1, 7, 3]$ is

$$\frac{x+1}{1} = \frac{y-3}{7} = \frac{z-5}{3}$$

Putting, the symmetric form of the straight line M equal to some parameter "s"

We get the parametric eq.

$$\frac{x+1}{1} = \frac{y-3}{7} = \frac{z-5}{3} = s$$

$$\frac{x+1}{1} = s, \quad \frac{y-3}{7} = s, \quad \frac{z-5}{3} = s$$

$$\begin{cases} x+1 = s \\ y-3 = 7s \\ z-5 = 3s \end{cases} \Rightarrow \begin{cases} x = -1+s \\ y = 3+7s \\ z = 5+3s \end{cases}$$

pt on line M $(-1+s, 3+7s, 5+3s)$

These lines intersect if

pt on L = pt on M

$$2+4t = -1+3s \quad \text{--- i}$$

$$-1+3t = 3+7s \quad \text{--- ii}$$

$$-2t = 5+3s \quad \text{--- iii}$$

Solve i & ii

$$s = -1$$

$$t = -1$$

put in (iii)

$$-1-3 = 3-7$$

$$-4 = -4$$

Hence lines intersect

Point of intersection

put $t = -1$ in (A)

We get $(-2, -4, 2)$

(13)

$$P = (3, -2, 1)$$

$$L = \begin{cases} x = 1+t \\ y = 3-2t \\ z = -2+2t \end{cases}$$

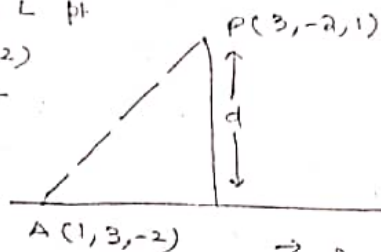
pt

From the Eq

of line L pt

$$A(1, 3, -2)$$

lies on L



$$\vec{b} = [1, -2, 2]$$

(direction vector of line L)

$$\vec{AP} = (3, -2, 1) - (1, 3, -2)$$

$$= [2, -5, 3]$$

$$\vec{AP} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 3 \\ 1 & -2 & 2 \end{vmatrix}$$

$$= -4\hat{i} - \hat{j} + \hat{k}$$

$$|\vec{AP} \times \vec{b}| = \sqrt{16+1+1} = \sqrt{18} = 3\sqrt{2}$$

$$|\vec{b}| = \sqrt{1+4+4} = \sqrt{9} = 3$$

Hence

$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$$

$$= \frac{3\sqrt{2}}{3}$$

$$= \sqrt{2}$$

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(14)

$$P = (0, -2, 1)$$

$$L: \frac{x-1}{4} = \frac{y+3}{-2} = \frac{z+1}{5}$$

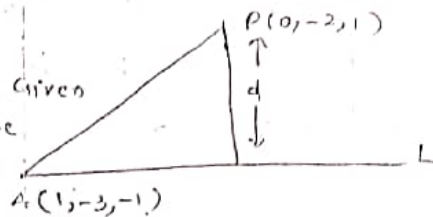
From the Given

Eq of line

$$pt(1, -3, -1)$$

lies on

line L



Direction

$$\text{vector of line L} = \vec{b} = [4, -2, 5]$$

$$\vec{AP} = (0, -2, 1) - (1, -3, -1) = [-1, 1, 2]$$

$$\vec{AP} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 4 & -2 & 5 \end{vmatrix}$$

$$\vec{AP} \times \vec{b} = 9\hat{i} + 13\hat{j} - 2\hat{k}$$

$$|\vec{AP} \times \vec{b}| = \sqrt{(9)^2 + (13)^2 + (-2)^2} = \sqrt{81 + 169 + 4} = \sqrt{254}$$

$$|\vec{b}| = \sqrt{16 + 4 + 25} = \sqrt{45}$$

Hence

$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$$

$$d = \frac{\sqrt{254}}{\sqrt{45}}$$

(15)

The edges of the parallelepiped are OA, OB, OC having lengths a, b, c respectively.

Show the coordinates of

O, A, B, C, D, E, F, O' are

$O(0,0,0), A(a,0,0), B(0,b,0)$

$C(0,0,c), D(a,b,0), E(0,b,c)$

$F(a,0,c), O'(a,b,c)$

Thus the four diagonals will be

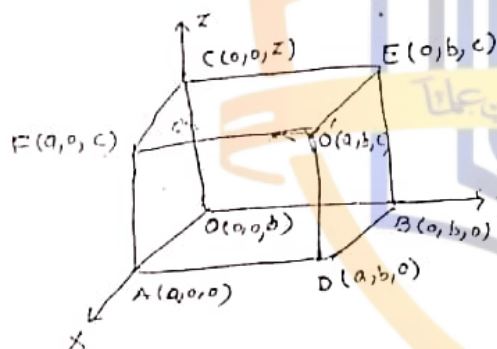
OO', AE, CD, BF

D, Ratios of OO' are $[a, b, c]$

" " AE are $[-a, b, c]$

" " CD are $[a, b, -c]$

" " BF are $[a, -b, c]$



∴ cosines of OO' are $\left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$

" " AE are $\left(\frac{-a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$

" " CD are $\left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{-c}{\sqrt{a^2+b^2+c^2}}\right)$

" " BF are $\left(\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{-b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$

Thus the angle between

OO' and AE is $\cos^{-1}\left(\frac{-a^2+b^2+c^2}{a^2+b^2+c^2}\right)$

and the angle between

OO' and CD is $\cos^{-1}\left(\frac{a^2+b^2-c^2}{a^2+b^2+c^2}\right)$

The angle between OO' and BF is $\cos^{-1}\left(\frac{a^2-b^2+c^2}{a^2+b^2+c^2}\right)$

Similarly, the angles between the remaining diagonals are found, which will be one of

$$\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$$

(16) Let us consider a unit cube i.e. a cube whose edge is 1.

Then the coordinates of

O, A, B, C, D, E, F, O'

$O(0,0,0), A(1,0,0), B(0,1,0)$

$C(0,0,1), D(1,1,0), E(0,1,1)$

$F(1,0,1), O'(1,1,1)$

Thus the four diagonals will be

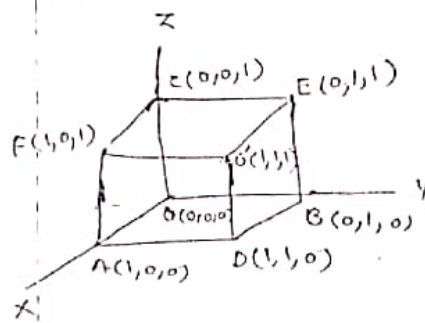
OO', AE, CD, BF

D, Ratios of OO' are $[1, 1, 1]$

" " of AE are $[-1, 1, 1]$

" " of CD are $[1, 1, -1]$

" " of BF are $[1, -1, 1]$



Direction cosines of OO' are $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

" " of AE are $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

" " of CD are $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$

" " of BF are $\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Let l, m, n be the direction cosines of the line which makes angles of measure $\alpha, \beta, \gamma, \delta$ with the diagonals AO', AE, CD, BF are

then

$$\cos \alpha = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \quad \text{--- i}$$

$$\cos \beta = -\frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \quad \text{--- ii}$$

$$\cos \gamma = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} - \frac{n}{\sqrt{3}} \quad \text{--- iii}$$

$$\cos \delta = \frac{l}{\sqrt{3}} - \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \quad \text{--- iv}$$

Squaring and adding i, ii, iii, iv,

we get

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \left(\frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \right)^2 + \left(-\frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \right)^2 + \left(\frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} - \frac{n}{\sqrt{3}} \right)^2 + \left(\frac{l}{\sqrt{3}} - \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} \right)^2$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{1}{3} (l+m+n)^2 + \frac{1}{3} (-l+m+n)^2 + \frac{1}{3} (l+m-n)^2 + \frac{1}{3} (l-m+n)^2$$

$$= \frac{1}{3} \left[l^2 + m^2 + n^2 + 2lm + 2ln + 2mn + l^2 + m^2 + n^2 - 2lm - 2ln + 2mn + l^2 + m^2 + n^2 + 2lm - 2ln - 2mn + l^2 + m^2 + n^2 - 2lm + 2ln - 2mn \right]$$

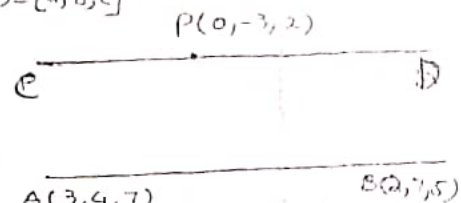
$$\therefore l^2 + m^2 + n^2 = 1$$

$$= \frac{1}{3} [1 + 1 + 1 + 1] = \frac{4}{3}$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

(17)

Let DR of $CD = [a, b, c]$



DR of AB

$$[2-3, 7-4, 5-7] = [-1, 3, -2]$$

We are need to find the Eq of line CD

For this

$$\text{pt on CD} = P(0, -3, 2) \quad (\text{Given})$$

As CD is parallel to AB

$$\text{So DR of CD} = \text{DR of AB}$$

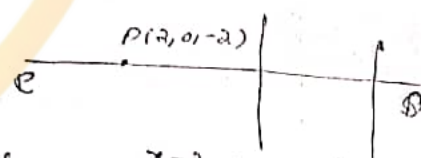
$$[a, b, c] = [-1, 3, -2]$$

Eq of CD

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\frac{x-0}{-1} = \frac{y+3}{3} = \frac{z-2}{-2}$$

(18)



We are need to find the Eq of line CD

For this

$$\text{pt on CD} = P(2, 0, -2) \quad (\text{Given})$$

D.R. of CD

Let a, b, c be D.R. of CD

and req line CD is $\perp$ to two lines whose D.R. are $2, 2, 2$ and $3, -1, 2$.

Then by condition of perpendicularity we have,

$$2a + 2b + 2c = 0$$

$$3a - 1b + 2c = 0$$

By the method of Cross-multiplication, we get

$$\frac{a}{\begin{vmatrix} 2 & 2 \\ -1 & 2 \end{vmatrix}} = \frac{b}{-\begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix}}$$

$$\frac{a}{4+2} = \frac{b}{-(4-6)} = \frac{c}{-2-6}$$

$$\frac{a}{6} = \frac{b}{2} = \frac{c}{-8}$$

Multiplying by 2

$$\frac{a}{3} = \frac{b}{1} = \frac{c}{-4}$$

$$\Rightarrow a = 3, b = 1, c = -4$$

Hence Eq of CD

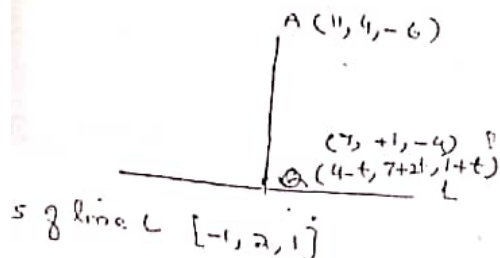
$$\frac{x-2}{3} = \frac{y-0}{1} = \frac{z+2}{-4}$$

are need to find the

of line passing through $A(11, 4, -6)$

and perpendicular to the

$$\left. \begin{aligned} x &= 4-t \\ y &= 7+2t \\ z &= -1+t \end{aligned} \right\} : L$$



Let Q be the foot of perpendicular

Any pt on the given line has parametric coordinates

$$(4-t, 7+2t, 1+t)$$

Let this pt be Q

$$\text{DR of AQ} [4-t-11, 7+2t-4, -1+t+6] \\ [-7-t, 3+2t, 5+t]$$

AQ is to L,

then by condition of perpendicularity

$$-1(-7-t) + 2(2t+3) + 1(t+5) = 0$$

$$7+t+4t+6+t+5=0$$

$$6t+18=0$$

$$6t = -18$$

$$t = -3$$

Hence Q becomes (7, 1, -4)

Hence Eq of AQ passing through

$$A(11, 4, -6) \text{ and } Q(7, 1, -4) \\ \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-11}{11-7} = \frac{y-4}{4-1} = \frac{z+6}{-6+4}$$

$$\frac{x-11}{4} = \frac{y-4}{3} = \frac{z+6}{-2}$$

(20)

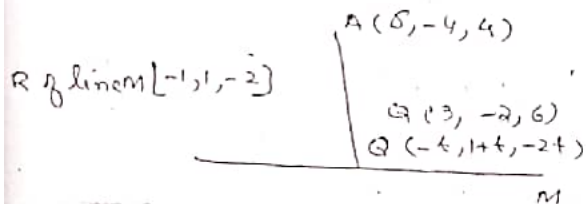
we have to find the Eq

of line passing through $A(5, -4, 4)$

and perpendicular to the

line $M: \begin{cases} x = -t \\ y = 1+t \\ z = -2t \end{cases}$ or $\frac{x}{-1} = \frac{y-1}{1} = \frac{z+2}{-2}$

on



Let Q be the foot of perpendicular,

any pt on the given line has

parametric coordinates

$$(-t, 1+t, -2t)$$

Let this pt be Q

$$R \text{ of } AQ: [-t-5, t+5, -2t-4]$$

$AQ \perp M$

Then by condition of perpendicularity

$$-(-t-5) + 1(t+5) - 2(-2t-4) = 0$$

$$t+5+t+5+4t+8=0$$

$$6t+18=0$$

$$6t=-18$$

$$t=-3$$

Q becomes $(3, -2, 6)$

Eq of AQ passing through

$(5, -4, 4)$ & $Q(3, -2, 6)$

(x_1, y_1, z_1) & (x_2, y_2, z_2)

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-5}{5-3} = \frac{y+4}{-4+2} = \frac{z+4}{4-6}$$

$$\frac{x-5}{2} = \frac{y+4}{-2} = \frac{z+4}{-2}$$

(21)

$$P(x_1, y_1, z_1)$$

$$L: \frac{x-d}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$$

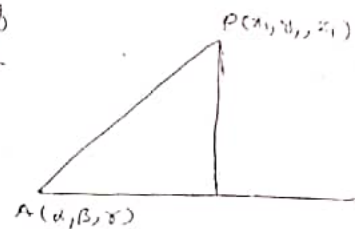
Let

From the Eq of

line L pt

$A(\alpha, \beta, \gamma)$

lies on L



Direction

vector of

line L

$$= \vec{b} = [l, m, n]$$

$$\vec{AP} = (x_1, y_1, z_1) - (\alpha, \beta, \gamma)$$

$$= [x_1-\alpha, y_1-\beta, z_1-\gamma]$$

$$\vec{AP} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1-\alpha & y_1-\beta & z_1-\gamma \\ l & m & n \end{vmatrix}$$

$$= \hat{i} [n(y_1-\beta) - m(z_1-\gamma)] + \hat{j} [l(z_1-\gamma) - m(x_1-\alpha)] + \hat{k} [m(x_1-\alpha) - l(y_1-\beta)]$$

Hence

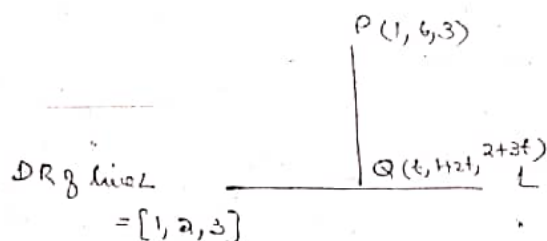
$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|} = \frac{\sqrt{\{n(y_1-\beta) - m(z_1-\gamma)\}^2 + \{l(z_1-\gamma) - m(x_1-\alpha)\}^2 + \{m(x_1-\alpha) - l(y_1-\beta)\}^2}}{\sqrt{l^2 + m^2 + n^2}}$$

(22)

we have to find the
Equation of line passing
through $P(1, 6, 3)$ and
perpendicular to the straight

$$\text{line: } \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

$$\text{or } \begin{cases} x = t \\ y = 1 + 2t \\ z = 2 + 3t \end{cases}$$



Let Q be the foot of perpendicular

any pt on the given line
has parametric coordinates

$$(t, 1+2t, 2+3t)$$

Let this point be Q

$$\text{DR of } PQ = [t-1, 2t-5, 3t-1]$$

$$PQ \perp \text{ to } L$$

then by condition of perpendicularity

$$1(t-1) + 2(2t-5) + 3(3t-1) = 0$$

$$t - 1 - 10 + 4t - 3 + 9t = 0$$

$$14t - 14 = 0$$

$$14t = 14$$

$$t = 1$$

$$Q \Rightarrow \text{becomes } (1, 3, 5)$$

E.g. PQ passing through $P(1, 6, 3)$
and $Q(1, 3, 5)$

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-1}{1-1} = \frac{y-6}{3-6} = \frac{z-3}{5-3}$$

$$\frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$$

(23)

Necessary Condition

$$\text{Let } P_1(x_1, y_1, z_1), P_2(x_2, y_2, z_2)$$

$$P_3(x_3, y_3, z_3) \text{ be collinear}$$

$$\vec{P_1P_2} \text{ \& } \vec{P_2P_3} \text{ are collinear}$$

So there exist a real number

λ such that

$$\vec{P_1P_2} = \lambda \vec{P_2P_3}$$

$$[x_2-x_1, y_2-y_1, z_2-z_1] = \lambda [x_3-x_2, y_3-y_2, z_3-z_2]$$

$$[x_2-x_1, y_2-y_1, z_2-z_1] = [\lambda(x_3-x_2), \lambda(y_3-y_2), \lambda(z_3-z_2)]$$

$$\Rightarrow x_2-x_1 = \lambda(x_3-x_2)$$

$$y_2-y_1 = \lambda(y_3-y_2)$$

$$z_2-z_1 = \lambda(z_3-z_2)$$

$$\Rightarrow \frac{x_2-x_1}{x_3-x_2} = \lambda, \frac{y_2-y_1}{y_3-y_2} = \lambda, \frac{z_2-z_1}{z_3-z_2} = \lambda$$

$$\Rightarrow \frac{x_2-x_1}{x_3-x_2} = \frac{y_2-y_1}{y_3-y_2} = \frac{z_2-z_1}{z_3-z_2}$$

$$\Rightarrow \frac{x_1-x_2}{x_2-x_3} = \frac{y_1-y_2}{y_2-y_3} = \frac{z_1-z_2}{z_2-z_3}$$

is condition for three points
another

to be collinear.

Sufficient Condition

$$\text{Let } \frac{x_1-x_2}{x_2-x_3} = \frac{y_1-y_2}{y_2-y_3} = \frac{z_1-z_2}{z_2-z_3} = k$$

$$\frac{x_1-x_2}{x_2-x_3} = k, \frac{y_1-y_2}{y_2-y_3} = k, \frac{z_1-z_2}{z_2-z_3} = k$$

$$\frac{x_2-x_1}{x_3-x_2} = k, \frac{y_2-y_1}{y_3-y_2} = k, \frac{z_2-z_1}{z_3-z_2} = k$$

$$\Rightarrow x_2-x_1 = k(x_3-x_2)$$

$$y_2-y_1 = k(y_3-y_2)$$

$$z_2-z_1 = k(z_3-z_2)$$

$$\Rightarrow [x_2 - x_1, y_2 - y_1, z_2 - z_1] = [k(x_3 - x_2), k(y_3 - y_2), k(z_3 - z_2)]$$

$$\Rightarrow [x_2 - x_1, y_2 - y_1, z_2 - z_1] = k [x_3 - x_2, y_3 - y_2, z_3 - z_2]$$

$$\vec{P_1 P_2} = k \vec{P_2 P_3}$$

the points
this proves that P_1, P_2, P_3 are
collinear...

(14) we know that

if two lines having direction
cosines l_1, m_1, n_1 and l_2, m_2, n_2
are perpendicular, then

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

thus we have

$$l_1 l_3 + m_1 m_3 + n_1 n_3 = 0$$

$$l_2 l_3 + m_2 m_3 + n_2 n_3 = 0$$

$$\text{Also } l_1^2 + m_1^2 + n_1^2 = 1$$

$$l_2^2 + m_2^2 + n_2^2 = 1$$

$$l_3^2 + m_3^2 + n_3^2 = 1$$

let θ_1 be the angle between
the lines having direction
cosines l_1, m_1, n_1 and
 $l_1 + l_2 + l_3, m_1 + m_2 + m_3, n_1 + n_2 + n_3$
then

$$\cos \theta_1 = \frac{l_1(l_1 + l_2 + l_3) + m_1(m_1 + m_2 + m_3) + n_1(n_1 + n_2 + n_3)}{\sqrt{l_1^2 + m_1^2 + n_1^2} \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{l_1 l_1 + l_1 l_2 + l_1 l_3 + m_1 m_1 + m_1 m_2 + m_1 m_3 + n_1 n_1 + n_1 n_2 + n_1 n_3}{1 \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_1 = \frac{(l_1 l_2 + m_1 m_2 + n_1 n_3)}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{1 + 0 + 0}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_1 = \frac{1}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

let θ_2 be the angle between
the lines having direction
cosines l_2, m_2, n_2 and $l_1 + l_2 + l_3, m_1 + m_2 + m_3, n_1 + n_2 + n_3$, then

$$\cos \theta_2 = \frac{l_2(l_1 + l_2 + l_3) + m_2(m_1 + m_2 + m_3) + n_2(n_1 + n_2 + n_3)}{\sqrt{l_2^2 + m_2^2 + n_2^2} \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{(l_2^2 + m_2^2 + n_2^2) + (l_2 l_1 + m_2 m_1 + n_2 n_1) + (l_2 l_3 + m_2 m_3 + n_2 n_3)}{\sqrt{1} \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_2 = \frac{1 + 0 + 0}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_2 = \frac{1}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

let θ_3 be the angle between
the lines having direction
cosines l_3, m_3, n_3 and $l_1 + l_2 + l_3, m_1 + m_2 + m_3, n_1 + n_2 + n_3$, then

$$\cos \theta_3 = \frac{l_3(l_1 + l_2 + l_3) + m_3(m_1 + m_2 + m_3) + n_3(n_1 + n_2 + n_3)}{\sqrt{l_3^2 + m_3^2 + n_3^2} \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$= \frac{(l_3^2 + m_3^2 + n_3^2) + (l_3 l_1 + m_3 m_1 + n_3 n_1) + (l_3 l_2 + m_3 m_2 + n_3 n_2)}{\sqrt{1} \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_3 = \frac{1 + 0 + 0}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

$$\cos \theta_3 = \frac{1}{\sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}}$$

Hence all angles are congruent.

(25)

The variable line is shown in two adjacent positions, in the figure.

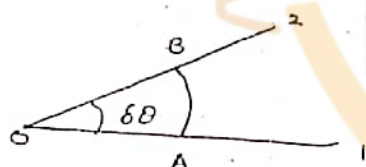
In position (1), it has direction cosines

$$l, m, n$$

In position (2), it has direction cosines

$$l + \delta l, m + \delta m, n + \delta n$$

$\delta \theta$ is the angle between two points of the line



Let $\hat{OA}$, $\hat{OB}$ be unit vectors in the direction of two positions of line

$$\hat{OA} = [l, m, n]$$

$$\hat{OB} = [l + \delta l, m + \delta m, n + \delta n]$$

Therefore the pt A has coordinates (l, m, n)

The pt B has coordinates $(l + \delta l, m + \delta m, n + \delta n)$

Draw arc AB of circle of radius 1

$$\text{arc } AB = 1 \cdot \delta \theta = \delta \theta$$

Because $\delta \theta$ is very small

$$\text{then } \text{arc } AB = AB$$

$$AB = \sqrt{(l + \delta l - l)^2 + (m + \delta m - m)^2 + (n + \delta n - n)^2}$$

$$\delta \theta = \sqrt{(\delta l)^2 + (\delta m)^2 + (\delta n)^2}$$

$$\text{or } (\delta \theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$$

proved.

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PUACP

① Equation of a plane through $P(2, 1, 1)$
 (x_1, y_1, z_1) with normal $n = [a, b, c]$ is

$$a(x-2) + b(y-1) + c(z-1) = 0 \quad \text{--- (1)}$$

As the other two points
 $(6, 3, 1)$ & $(-2, 1, 2)$ pass
 through the plane (lie on
 the plane)

Eq (1) will give two equations

$$a(6-2) + b(3-1) + c(1-1) = 0$$

$$a(-2-2) + b(1-1) + c(2-1) = 0$$

$$4a + 2b + 0c = 0$$

$$-4a + 0b + 1c = 0$$

By the method of cross
 multiplication

$$\frac{a}{2} = \frac{b}{0} = \frac{c}{4} \quad \text{--- (2)}$$

$$\frac{a}{2} = \frac{b}{-4} = \frac{c}{8}$$

$$\frac{a}{1} = \frac{b}{-2} = \frac{c}{4}$$

$$\Rightarrow a=1, \quad b=-2, \quad c=4$$

Put in (1)

$$1(x-2) - 2(y-1) + 4(z-1) = 0$$

$$x - 2 - 2y + 2 + 4z - 4 = 0$$

$$x - 2y + 4z - 4 = 0$$

② Equation of a plane through
 $P(1, -1, 2)$ with normal $n = [a, b, c]$
 (x_1, y_1, z_1)

$$a(x-1) + b(y+1) + c(z-2) = 0 \quad \text{--- (1)}$$

As the other two points
 $(5, -8, -2)$ & $(4, 1, 0)$ lie
 on the plane

Eq (1) will give two equations

$$a(-3-1) + b(-2+1) + c(-2-2) = 0$$

$$a(6-1) + b(0+1) + c(1-2) = 0$$

$$-4a - b + 4c = 0$$

$$5a + b - c = 0$$

by the method of cross
 multiplication

$$\frac{a}{-1} = \frac{b}{4} = \frac{c}{5}$$

$$\frac{a}{1-4} = \frac{b}{-(4-20)} = \frac{c}{(-4+5)}$$

$$\frac{a}{-3} = \frac{b}{16} = \frac{c}{1}$$

$$\Rightarrow a=-3, \quad b=16, \quad c=1$$

Put in (1)

$$-3(x-1) + 16(y+1) + 1(z-2) = 0$$

$$-3x + 3 + 16y + 16 + z - 2 = 0$$

$$-3x + 16y + z + 17 = 0$$

$$3x - 16y - z - 17 = 0$$

③ Equation of a plane through
 $P(-1, 1, 1)$ with normal $n = [a, b, c]$
 (x_1, y_1, z_1) is

$$a(x+1) + b(y-1) + c(z-1) = 0$$

As the other two points

$(5, -8, -2)$, $(4, 1, 0)$ lie on the
 plane

Eq (1) will give two equations

$$a(5+1) + b(-8-1) + c(-2-1) = 0$$

$$a(4+1) + b(1-1) + c(0-1) = 0$$

$$6a - 9b - 3c = 0$$

$$5a + 0b - c = 0$$

by the method of cross
 multiplication

$$\frac{a}{\begin{vmatrix} -9 & -3 \\ 0 & -1 \end{vmatrix}} = \frac{b}{\begin{vmatrix} 6 & -3 \\ 5 & -1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 6 & -9 \\ 5 & 0 \end{vmatrix}}$$

$$\frac{a}{9} = \frac{b}{-(-6+15)} = \frac{c}{0+45}$$

$$\frac{a}{9} = \frac{b}{-9} = \frac{c}{15}$$

$$\frac{a}{9} = \frac{b}{-1} = \frac{c}{15}$$

$$\Rightarrow a=1 \quad b=-1, \quad c=15$$

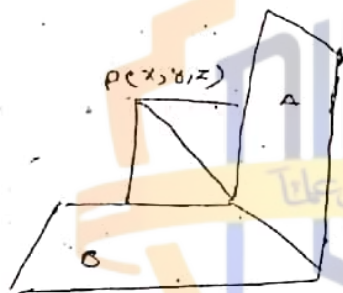
put in (1)

$$1(x+1) - (y-1) + 5(z-1) = 0$$

$$x+1-y+1+5z-5=0$$

$$x-y+5z-3=0$$

(4)



Let $P(x, y, z)$ be a point on the planes is equidistant from plane A and plane B.

Then the distance from this point to the two planes must be equal.

so the eqs of two bisecting planes are

$$\frac{2x+y+2z-5}{\sqrt{(2)^2+(1)^2+(2)^2}} = \pm \frac{3x+2y-6z+1}{\sqrt{(3)^2+(2)^2+(-6)^2}}$$

$$\frac{2x+y+2z-5}{3} = \pm \frac{3x+2y-6z+1}{7}$$

$$7(2x+y+2z-5) \pm 3(3x+2y-6z+1)$$

$$14x+7y+14z-35 = \pm(9x+6y-18z+3)$$

$$14x+7y+14z-35 = 9x+6y-18z+3$$

$$5x+y+32z-38=0$$

or

$$14x+7y+14z-35 = -9x-6y+18z-3$$

$$23x+13y-4z-32=0$$

(5)

$$3x-4y+5z=0$$

$$\text{Dividing by } \sqrt{(3)^2+(-4)^2+(5)^2}$$

$$= 5\sqrt{2}$$

we get the normal form of the Equation

$$\frac{3}{5\sqrt{2}}x - \frac{4}{5\sqrt{2}}y + \frac{5}{5\sqrt{2}}z = 0 \quad \text{--- (1)}$$

of the form $lx+my+nz=p$

$$d \quad 2x-y-2z=5$$

$$\text{Dividing by } \sqrt{(2)^2+(-1)^2+(2)^2} = 3$$

we get the normal form of the Equation.

$$\frac{2}{3}x - \frac{1}{3}y - \frac{2}{3}z = \frac{5}{3}$$

Direction cosines of normals of planes are $(\frac{2}{3\sqrt{2}}, -\frac{1}{3\sqrt{2}}, -\frac{2}{3\sqrt{2}})$ and $(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3})$. Measure of angle between the

planes

$$\cos \theta = \frac{\frac{2}{3\sqrt{2}} \cdot \frac{2}{3} + (-\frac{1}{3\sqrt{2}}) \cdot (-\frac{1}{3}) + (-\frac{2}{3\sqrt{2}}) \cdot (-\frac{2}{3})}{1}$$

$$= \frac{2}{5\sqrt{2}} + \frac{1}{15\sqrt{2}} - \frac{2}{3\sqrt{2}}$$

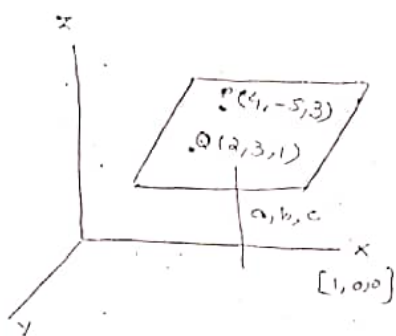
$$\cos \theta = \frac{6+4-10}{15\sqrt{2}} = 0$$

$$\theta = \cos^{-1}(0)$$

$$\theta = \pi/2$$



(C) 4,



Equation of the plane parallel to x -axis with DR $[1, 0, 0]$

Equation of the plane through the pt $(4, -5, 3)$ with normal $n = [a, b, c]$ is

$$a(x-4) + b(y+5) + c(z-3) = 0 \quad (1)$$

It passes through another pt $(2, 3, 1)$ lie on the plane. Eq (1) will give.

$$a(2-4) + b(3+5) + c(1-3) = 0$$

$$-2a + 8b - 2c = 0 \quad (2)$$

The normal $[a, b, c]$ of the plane $\perp$ to x -axis to the vector $[1, 0, 0]$

$$so \quad 1a + 0b + 0c = 0 \quad (3)$$

Solve (2) & (3)

$$\frac{a}{\begin{vmatrix} 8 & -2 \\ 0 & 0 \end{vmatrix}} = \frac{b}{\begin{vmatrix} -2 & -2 \\ 1 & 0 \end{vmatrix}} = \frac{c}{\begin{vmatrix} -2 & 8 \\ 1 & 0 \end{vmatrix}}$$

$$\frac{a}{0} = \frac{b}{-2} = \frac{c}{-8}$$

$$\frac{a}{0} = \frac{b}{1} = \frac{c}{+4}$$

$$\Rightarrow a=0, b=1, c=+4$$

Put in (1)

$$0(x-4) + 1(y+5) + 4(z-3) = 0$$

$$y+5+4z-12=0$$

$$y+4z-7=0$$

(C) 5

Equation of plane parallel to y -axis with DR $[0, 1, 0]$

Equation of the plane through the pt $(4, -5, 3)$ with normal $[a, b, c]$ is

$$a(x-4) + b(y+5) + c(z-3) = 0 \quad (1)$$

Also another pt $(2, 3, 1)$ lie on the plane.

Eq (1) will give

$$a(2-4) + b(3+5) + c(1-3) = 0$$

$$-2a + 8b - 2c = 0 \quad (2)$$

The normal $[a, b, c]$ of the plane $\perp$ to y -axis to the vector $[0, 1, 0]$.

$$so \quad 0a + 1b + 0c = 0 \quad (3)$$

Solve (2) & (3)

$$\frac{a}{\begin{vmatrix} 8 & -2 \\ 0 & 0 \end{vmatrix}} = \frac{b}{\begin{vmatrix} -2 & -2 \\ 0 & 0 \end{vmatrix}} = \frac{c}{\begin{vmatrix} -2 & 8 \\ 0 & 0 \end{vmatrix}}$$

$$\frac{a}{0} = \frac{b}{0} = \frac{c}{-2}$$

$$\frac{a}{1} = \frac{b}{0} = \frac{c}{-1}$$

$$\Rightarrow a=1, b=0, c=-1$$

Put in (1)

$$1(x-4) + 0(y+5) - 1(z-3) = 0$$

$$x-4-z+3=0$$

$$x-z-1=0$$

iii)

Equation of the plane through
parallel to x -axis with DR $[0, 0, 1]$

Eq of the plane through the
pt $(4, -5, 3)$ with normal
 $[a, b, c]$ is

$$a(x-4) + b(y+5) + c(z-3) = 0 \quad \text{--- (1)}$$

Also another pt $(2, 3, 1)$ lie
on the plane

$$a(2-4) + b(3+5) + c(1-3) = 0$$

$$-2a + 8b - 2c = 0 \quad \text{--- (2)}$$

The normal $[a, b, c]$ of
the plane $\perp$ to x -axis
to the vector $[0, 0, 1]$

$$\text{So } 0a + 0b + 1c = 0 \quad \text{--- (3)}$$

Solve (2) & (3)

$$\frac{a}{18-2} = \frac{b}{-2-2} = \frac{c}{-2-8}$$

$$\frac{a}{8} = \frac{b}{-4} = \frac{c}{-10}$$

$$\frac{a}{4} = \frac{b}{1} = \frac{c}{5}$$

$$\Rightarrow a=4, \quad b=1, \quad c=5$$

put in (1)

$$4(x-4) + 1(y+5) + 5(z-3) = 0$$

$$4x - 16 + y + 5 + 5z - 15 = 0$$

$$4x + y + 5z - 26 = 0$$

(i)

Eq of plane:

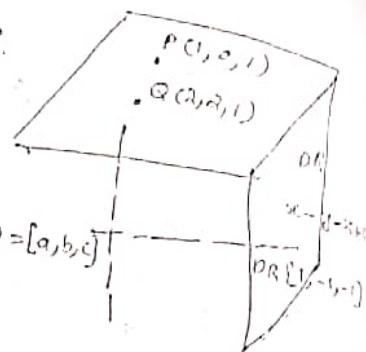
through the

pt $P(1, 0, 1)$

with

normal

$[a, b, c]$ is



$$a(x-1) + b(y-0) + c(z-1) = 0$$

Also another pt $(2, 2, 1)$ lie

on the plane

$$a(2-1) + b(2-0) + c(1-1) = 0$$

$$1a + 2b + 0c = 0 \quad \text{--- (2)}$$

The normal $[a, b, c]$ of the
plane $\perp$ to normal of the
given plane $[1, -1, -1]$

$$\text{So } 1a - 1b - 1c = 0 \quad \text{--- (3)}$$

Solve (2) & (3)

$$\frac{a}{2-1} = \frac{b}{-1-1} = \frac{c}{-1-1}$$

$$\frac{a}{1} = \frac{b}{-2} = \frac{c}{-2}$$

$$\frac{a}{-2} = \frac{b}{1} = \frac{c}{-3}$$

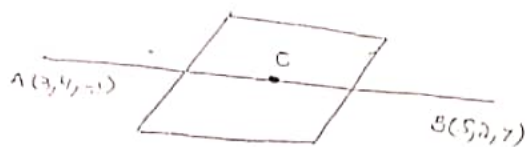
$$\Rightarrow a=-2, \quad b=1, \quad c=-3$$

put in (1)

$$-2(x-1) + 1(y-0) - 3(z-1) = 0$$

$$-2x + 2 + y - 3z + 3 = 0$$

$$2x - y + 3z - 5 = 0$$



$$C = \text{Mid pt of } AB = \left(\frac{3+5}{2}, \frac{4+2}{2}, \frac{-1+7}{2} \right) = (4, 3, 3)$$

$$\text{DR of line joining } AB = [5-3, 2-4, 7-(-1)] = [2, -2, 8]$$

These are Direction ratios of normal to the required plane

Equation of the plane through the pt $(4, 3, 3)$ with normal (a, b, c) is

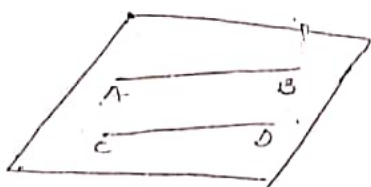
$$2(x-4) - 2(y-3) + 8(z-3) = 0$$

$$2x - 8 - 2y + 6 + 8z - 24 = 0$$

$$2x - 2y + 8z - 26 = 0$$

$$x - y + 4z - 13 = 0$$

(1)



Two lines will intersect if they are coplanar and non-parallel

First we show that lines are coplanar

The Eq of the plane through A, B, C is

$$\begin{vmatrix} x & y & z & 1 \\ 3 & 4 & -1 & 1 \\ 5 & 2 & 7 & 1 \\ 4 & 3 & 3 & 1 \end{vmatrix} = 0$$

$$\begin{matrix} R_1 - R_2 \\ R_2 - R_3 \\ R_3 - R_4 \end{matrix} \begin{vmatrix} x & y+1 & z & 0 \\ -2 & -5 & 1 & 0 \\ 1 & 3 & -2 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0$$

Expanding

$$\begin{vmatrix} x & y+1 & z \\ -2 & -5 & 1 \\ 1 & 3 & -2 \end{vmatrix} = 0$$

$$x(10-3) - (y+1)(4-2) + z(-6+5) = 0$$

$$7x - 3y - 3 - z = 0$$

$$7x - 3y - z - 3 = 0 \quad \text{--- (2)}$$

If lines are coplanar, the given point $(3, 3, 6)$ must satisfy Eq (2)

put this pt in (2)

$$21 - 9 - 6 - 3 = 0$$

$$0 = 0$$

Thus lines are coplanar

Now we show that lines are non-parallel

$$\text{DR of } AB = [2, 5, -1]$$

$$\text{DR of } CD = [2, 2, 8]$$

DR of AB & DR of CD are not proportional

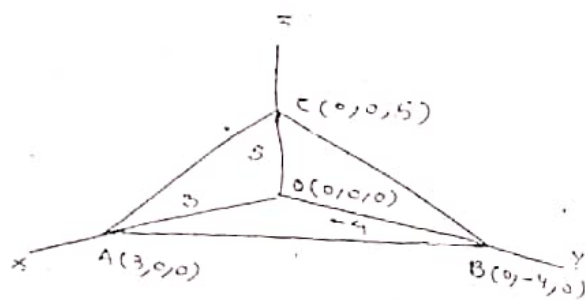
$$\frac{2}{2} \neq \frac{5}{2} \neq \frac{-1}{8}$$

Thus lines are not parallel

∴ These lines intersect

(10)

The vertices of a tetrahedron are $O(0,0,0)$, $A(3,0,0)$, $B(0,-4,0)$ & $C(0,0,5)$



We are to find Equations of planes of its four faces

i. Equation of OAB Face

It lies on xy-plane

so Eq is $z=0$

ii. Equation of OBC Face

It lies on yz-plane

so Eq is $x=0$

iii. Equation of OAC Face

It lies on xz-plane

so Eq is $y=0$

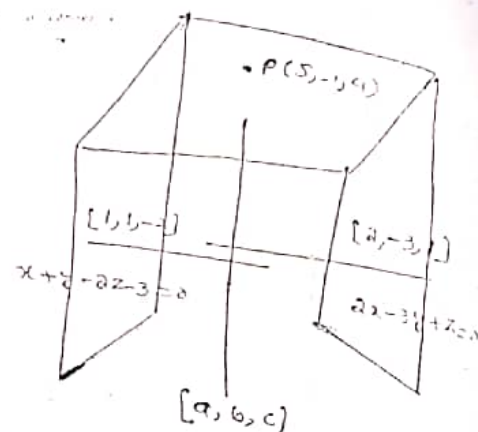
iv. Equation of Face ABC

in intercept form

$$\frac{x}{3} + \frac{y}{-4} + \frac{z}{5} = 1$$

(11)

(26)



Equation of the plane through the pt $(5, -1, 4)$ with normal $[a, b, c]$ is

$$a(x-5) + b(y+1) + c(z-4) = 0 \quad \text{--- (1)}$$

Since it is perpendicular to each of the planes, we have

$$a + b - 4c = 0$$

$$2a - 3b + 1c = 0$$

Solving

$$\frac{a}{1-3} = \frac{b}{-2} = \frac{c}{1}$$

$$\frac{a}{1-6} = \frac{b}{-(1+4)} = \frac{c}{-3-2}$$

$$\frac{a}{-5} = \frac{b}{-5} = \frac{c}{-5}$$

$$\frac{a}{1} = \frac{b}{1} = \frac{c}{1}$$

$$\Rightarrow a=1, b=1, c=1$$

put in (1)

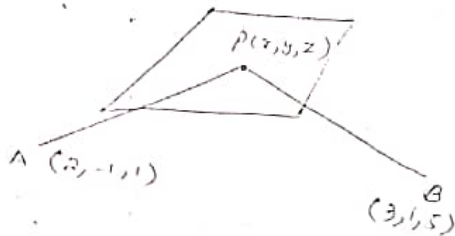
$$1(x-5) + 1(y+1) + 1(z-4) = 0$$

$$x-5 + y+1 + z-4 = 0$$

$$x + y + z - 8 = 0$$

62

Let $P(x, y, z)$ be any point on the plane. Since this point is equidistant from $A(2, -1, 1)$ and $B(3, 1, 5)$



Therefore $PA = PB$

$$\sqrt{(x-2)^2 + (y+1)^2 + (z-1)^2} = \sqrt{(x-3)^2 + (y-1)^2 + (z-5)^2}$$

Squaring both sides, we get

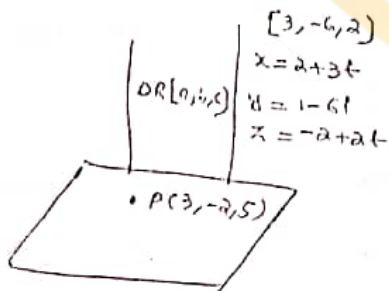
$$(x-2)^2 + (y+1)^2 + (z-1)^2 = (x-3)^2 + (y-1)^2 + (z-5)^2$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 + z^2 - 2z + 1 = x^2 - 6x + 9 + y^2 - 2y + 1 + z^2 - 10z + 25$$

$$x^2 + 9 - 6x + y^2 + 1 - 2y + z^2 + 25 - 10z$$

$$2x + 4y - 8z - 29 = 0$$

63



Equation of the plane through $P(3, -2, 5)$ with normal $[a, b, c]$

$$a(x-3) + b(y+2) + c(z-5) = 0$$

DR of normal = DR of given line

$$[a, b, c] = [3, -6, 2]$$

617

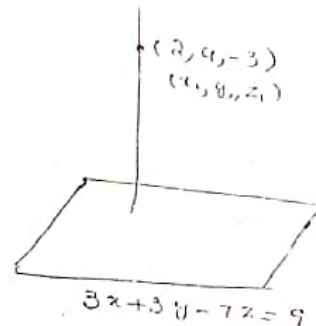
11

$$3(x-3) - 6(y+2) + 2(z-5) = 0$$

$$3x - 9 - 6y - 12 + 2z - 10 = 0$$

$$3x - 6y + 2z - 31 = 0$$

11



Since the line is $\perp$ to the plane, therefore it can be regarded as Normal to the plane

Therefore its DR $[3, 3, -7]$

Eq in symmetric form is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = t$$

$$\frac{x-2}{3} = \frac{y-4}{3} = \frac{z+3}{-7} = t$$

$$\text{as } \frac{x-2}{3} = t, \frac{y-4}{3} = t, \frac{z+3}{-7} = t$$

$$x-2=3t, y-4=3t, z+3=-7t$$

$$x=2+3t, y=4+3t, z=-3-7t$$

are parametric equations of the line.

(15)

The Eq. of plane in Normal Form is

$$lx + my + nz = p$$

In this case, the distance of the plane from the origin is 7
i.e. $p = 7$
therefore, Eq. is

$$lx + my + nz = 7$$

We are to find; the members of the family $lx + my + nz = 7$ of the family is parallel to the plane $x + y + z + 5 = 0$

then their normals are also parallel

By condition of parallelism

$$\frac{l}{1} = \frac{m}{1} = \frac{n}{1} = k \text{ (say)}$$

$$l = k, \quad m = k, \quad n = k$$

Squaring and adding, we get

$$l^2 + m^2 + n^2 = k^2$$

$$k^2 + k^2 + k^2 = 1$$

$$3k^2 = 1$$

$$k^2 = \frac{1}{3}$$

$$k = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow l = \pm \frac{1}{\sqrt{3}}, \quad m = \pm \frac{1}{\sqrt{3}}, \quad n = \pm \frac{1}{\sqrt{3}}$$

Hence the members of family of planes which are parallel to the plane $x + y + z + 5 = 0$ is

$$\pm \frac{1}{\sqrt{3}}x \pm \frac{1}{\sqrt{3}}y \pm \frac{1}{\sqrt{3}}z = 7$$

(16)

(20)

Eq. of two plane

through the

pt $(3, 4, 5)$

with normal

$[a, b, c]$

$$a(x-3) + b(y-4) + c(z-5) = 0$$

As this plane makes

x -intercept equal to -5

so $y = 0$

$z = 0$

Therefore it passes through the

pt $(-5, 0, 0)$

$$a(-5-3) + b(0-4) + c(0-5) = 0$$

$$-8a - 4b - 5c = 0$$

$$8a + 4b + 5c = 0 \quad \text{--- (2)}$$

Also this plane is $\perp$ to the plane $2x + 3y - z = 8$

therefore, their normals will be perpendicular,

By the condition of perpendicularity

$$2a + 3b - 1c = 0$$

solve (2) & (3)

By cross multiplication method

$$\frac{a}{-19} = \frac{b}{+18} = \frac{c}{16}$$

put in (1)

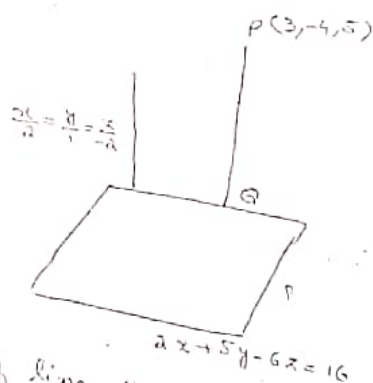
$$-19(x-3) + 18(y-4) + 16(z-5) = 0$$

$$-19x + 18y + 16z + 57 + 72 + 80 = 0$$

$$-19x + 18y + 16z - 9.5 = 0$$

$$19x - 18y - 16z + 9.5 = 0$$

(11)



Equation of line through $P(3, -4, 5)$ and parallel to the line having DR $[2, 1, -2]$

$$\frac{x-3}{2} = \frac{y+4}{1} = \frac{z-5}{-2} = t$$

$$\frac{x-3}{2} = t, \quad \frac{y+4}{1} = t, \quad \frac{z-5}{-2} = t$$

$$x-3=2t, \quad y+4=t, \quad z-5=-2t$$

$$x=3+2t, \quad y=-4+t, \quad z=5-2t$$

parametric Eq is also coordinates of any pt

Let this pt Q $(3+2t, -4+t, 5-2t)$

Q lies on the plane, then

$$2(3+2t) + 5(-4+t) - 6(5-2t) = 16$$

$$6+4t-20+5t-30+12t=16$$

$$21t=60$$

$$t = \frac{60}{21} = \frac{20}{7}$$

$$\text{Thus } Q\left(\frac{61}{7}, -\frac{8}{7}, -\frac{5}{7}\right)$$

$$|Q| = \sqrt{\left(\frac{61}{7}-3\right)^2 + \left(-\frac{8}{7}+4\right)^2 + \left(-\frac{5}{7}-5\right)^2}$$

$$= \sqrt{\left(\frac{40}{7}\right)^2 + \left(\frac{20}{7}\right)^2 + \left(\frac{40}{7}\right)^2}$$

$$= \sqrt{\frac{1600}{49} + \frac{400}{49} + \frac{1600}{49}}$$

$$= \sqrt{\frac{3600}{49}} = \frac{60}{7}$$

(12)

These lines intersect, if the equations

$$3+2t = -3+4s \quad \text{--- (1)}$$

$$2+t = 5-4s \quad \text{--- (2)}$$

$$-2-3t = 6-5s \quad \text{--- (3)}$$

have simultaneous solutions.

Solving (1) & (2)

$$\text{we get } -t = -1$$

$$s = 1$$

Put in (3)

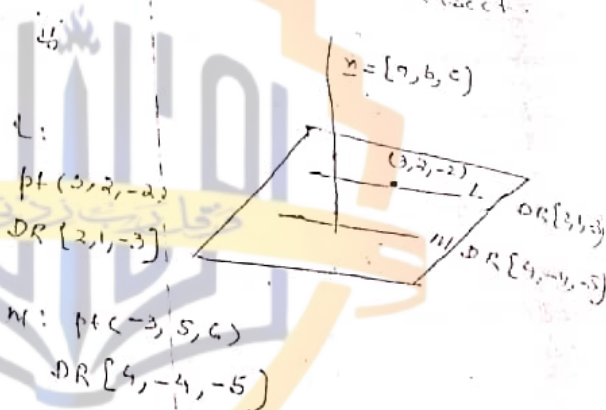
$$-2-3(-1) = 6-5(1)$$

$$-2+3 = 1$$

$$1 = 1$$

Last eq (3) is satisfied by these values

Hence, two lines intersect.



L: $P(3, -4, 5)$
DR $[2, 1, -2]$

M: $P(-3, 5, 6)$
DR $[4, -4, -5]$

Eq of the plane through P(3, -4, 5)

with normal $[a, b, c]$

$$a(x-3) + b(y+4) + c(z-5) = 0$$

$$\text{normal vector of the plane is perpendicular to } L \text{ \& } M$$

$$\text{So } 2a+1b-3c=0$$

$$4a-4b-5c=0$$

$$\frac{a}{717} = \frac{b}{72} = \frac{c}{712}$$

$$(1) \Rightarrow 17(x-3) + 2(y+4) + 12(z-5) = 0$$

$$17x - 51 + 2y + 8 + 12z - 60 = 0$$

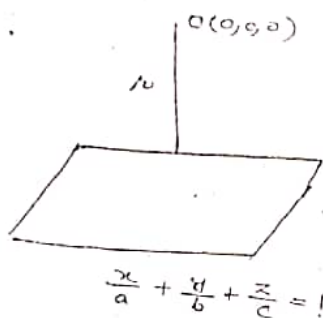
$$17x + 2y + 12z - 31 = 0$$



(19)

Eq of the plane with a, b, c as intercepts on x -axis, y -axis, z -axis, so

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$



Distance " r " of the plane from $O(0,0,0)$ is

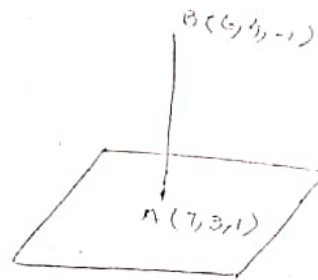
$$r = \frac{|\frac{a}{a} + \frac{0}{b} + \frac{0}{c} - 1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$r = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\frac{1}{r} = \sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\frac{1}{r^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

(20)



The line AB is normal to both the required planes

$$DR \text{ of } AB = [1, -1, 2]$$

$$DCS \text{ of } AB = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)$$

$l, \quad m, \quad n$

Equations of planes at distance 3 units from the origin are

$$lx + my + nz = \pm 3$$

where l, m, n are direction cosines of the normal to the planes

Since AB is normal to two planes, we have

$$\frac{1}{\sqrt{6}}x - \frac{1}{\sqrt{6}}y + \frac{2}{\sqrt{6}}z = \pm 3$$

$$x - y + 2z = \pm 3\sqrt{6}$$

①

$$P_1: 4x + 4y - 5z = 12$$

$$P_2: 8x + 12y - 13z = 32$$

∴ direction ratios of the normal to the planes are

$$[4, 4, -5] \text{ and } [8, 12, -13]$$

Planes intersect, if planes are not parallel

planes are non-parallel if their normals are not proportional

Clearly, normals are not proportional

$$\text{Thus } \frac{4}{8} \neq \frac{4}{12} \neq \frac{-5}{-13}$$

∴ planes intersect.

∴ To transform into Eq

$$P_1: 4x + 4y - 5z - 12 = 0$$

$$P_2: 8x + 12y - 13z - 32 = 0$$

of a line to the symmetrical form.

we require

the coordinates of any pt on it

put $z=0$ in two eqs

$$4x + 4y - 12 = 0 \quad \text{--- (i)}$$

$$8x + 12y - 32 = 0 \quad \text{--- (ii)}$$

Solving two equations

$$\begin{array}{r} 8x + 4y - 24 = 0 \\ 8x + 12y - 32 = 0 \\ \hline -4y + 8 = 0 \end{array}$$

$$y = 2$$

∴

$$4x + 4(2) - 12 = 0$$

$$4x + 8 - 12 = 0$$

$$4x - 4 = 0$$

$$x = 1$$

therefore $(1, 2, 0)$ is a pt on the line

∴ let a, b, c be the direction ratios of the line.

Since the line lies in both the planes.

It is $\perp$ to the normals to both of them.

$$\text{DR of } P_1 [4, 4, -5]$$

$$\text{DR of } P_2 [8, 12, -13]$$

(the direction ratios of the normals to the planes)

$$\begin{aligned} \text{we have } 4a + 4b - 5c &= 0 \\ 8a + 12b - 13c &= 0 \end{aligned}$$

$$\frac{a}{\begin{vmatrix} 4 & -5 \\ 8 & -13 \end{vmatrix}} = \frac{b}{\begin{vmatrix} 4 & -5 \\ 8 & -13 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 4 & 4 \\ 8 & 12 \end{vmatrix}}$$

$$\frac{a}{2} = \frac{b}{3} = \frac{c}{4}$$

Equation of line of intersection in the symmetric form is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-0}{4}$$

② To transform the Eq

$$x+y+z+1=0 = 4x+y-2z+2$$

of a line to the symmetrical form.

we require

i. the coordinates of any pt on it

put $z=0$ in two Equations

$$x+y+1=0 \quad \text{--- i}$$

$$4x+y+2=0 \quad \text{--- ii}$$

solving i & ii

$$\begin{array}{r} x+y+1=0 \\ 4x+y+2=0 \\ \hline -3x-1=0 \end{array}$$

$$-3x=1$$

$$x=-\frac{1}{3}$$

$$\text{ii) } -\frac{1}{3}+y+1=0$$

$$y+\frac{2}{3}=0$$

$$y=-\frac{2}{3}$$

Therefore, $(-\frac{1}{3}, -\frac{2}{3}, 0)$ is a point on the line.

ii. Let a, b, c be the direction ratios of the line.

Since the line lies in both the planes.

It is $\perp$ to the normals to both of them

$$\text{DR of } P_1 [1, 1, 1]$$

$$\text{DR of } P_2 [4, 1, -2]$$

$$\begin{aligned} \text{we have } 1a+1b+1c &= 0 \\ 4a+1b-2c &= 0 \end{aligned}$$

$$\frac{a}{\begin{vmatrix} 1 & -1 \\ 4 & -2 \end{vmatrix}} = \frac{b}{\begin{vmatrix} 1 & 1 \\ 4 & -2 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}}$$

$$\frac{a}{-3} = \frac{b}{-5} = \frac{c}{-3}$$

$$\frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$$

Equation of line of intersection in the symmetrical form is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\frac{x+1/3}{1} = \frac{y+2/3}{-2} = \frac{z-0}{1}$$

③

$$L: x+2y-z-7=0$$

$$y+z-2x-6=0$$

line is the intersection of planes

Let a_1, b_1, c_1 be the direction ratios of the line.

Since the normals to the two planes are perpendicular to the line of intersection.

normals of two planes are

$$[1, 2, -1]$$

$$\text{or } [-2, 1, 1]$$

we have

$$1a_1+2b_1-1c_1=0$$

$$-2a_1+1b_1+1c_1=0$$

$$\frac{a_1}{\begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix}} = \frac{b_1}{\begin{vmatrix} 1 & -1 \\ -2 & 1 \end{vmatrix}} = \frac{c_1}{\begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix}}$$

$$\frac{a}{3} = \frac{b}{1} = \frac{c}{5}$$

$$M: 3x+6y-3z-8=0$$

$$2x-y-z=0$$

line is the line of intersection of planes.

Let a_2, b_2, c_2 be the direction ratios of the line.

Since the normals to the two planes are perpendicular to the line of intersection.

normals of two planes are

$$\begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix}$$

$$\text{and } \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$$

$$\begin{aligned} \text{we have } 3a_2 + 6b_2 - 3c_2 &= 0 \\ 2a_2 - 1b_2 - 1c_2 &= 0 \end{aligned}$$

$$\frac{a_2}{\begin{vmatrix} 6 & -3 \\ -1 & -1 \end{vmatrix}} = \frac{b_2}{\begin{vmatrix} 3 & -3 \\ 2 & -1 \end{vmatrix}} = \frac{c_2}{\begin{vmatrix} 3 & 6 \\ 2 & -1 \end{vmatrix}}$$

$$\frac{a_2}{-6+3} = \frac{b_2}{-(-3+6)} = \frac{c_2}{(-3-12)}$$

$$\frac{a_2}{-3} = \frac{b_2}{-3} = \frac{c_2}{-15}$$

$$\frac{a_2}{3} = \frac{b_2}{1} = \frac{c_2}{5}$$

so

$$\text{DR of } L \quad [3 \ 1 \ 5]$$

$$\text{DR of } M \quad [3 \ 1 \ 5]$$

we find

$$\frac{3}{3} = \frac{1}{1} = \frac{5}{5}$$

$$1 = 1 = 1$$

direction ratios of line

L & M are proportional

Therefore, lines are parallel.

$$\begin{aligned} L: \quad x + 2y - 1 &= 0 \\ 2y - x - 1 &= 0 \end{aligned}$$

line is the line of intersection of planes

Let a_1, b_1, c_1 be the direction ratios of the line.

Since the normals to two planes are perpendicular to the line of intersection

normals of two planes are

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\text{and } \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

we have

$$1a_1 + 2b_1 + 0c_1 = 0$$

$$0a_1 + 2b_1 - 1c_1 = 0$$

$$\frac{a_1}{-2} = \frac{b_1}{1} = \frac{c_1}{2}$$

$$M: \quad x - y - 1 = 0$$

$$x - 2x - 3 = 0$$

line is the line of intersection of planes.

Let a_2, b_2, c_2 be the direction ratios of the line.

Since the normals to two planes are perpendicular to the line of intersection.

normals of two planes are

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\text{and } \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\text{we have } 1a_2 - 1b_2 + 0c_2 = 0$$

$$1a_2 + 0b_2 - 2c_2 = 0$$

$$\frac{a_2}{2} = \frac{b_2}{2} = \frac{c_2}{1}$$

$$\text{DR of } L \quad [-2, 1, 2]$$

$$\text{DR of } M \quad [2, 2, 1]$$

using condition for perpendicularity

$$= a_1a_2 + b_1b_2 + c_1c_2 = (-2)(2) + (1)(2) + (2)(1) = -4 + 2 + 2 = 0$$

Therefore, lines are perpendicular.

$$P(1, 2, 3)$$

L:

$$x - y + 2z - 5 = 0$$

$$3x + y + z + 6 = 0$$

Let a_1, b_1, c_1 be the direction ratios of the line.

Since the normals to two planes are perpendicular to the line.

Normal of two planes

$$\text{are } [1, -1, 2]$$

$$\text{and } [3, 1, 1]$$

We have

$$1a_1 - 1b_1 + 2c_1 = 0$$

$$3a_1 + 1b_1 + 1c_1 = 0$$

$$\frac{a_1}{\begin{vmatrix} -1 & 2 \\ 1 & 1 \end{vmatrix}} = \frac{b_1}{\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}} = \frac{c_1}{\begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix}}$$

$$\frac{a_1}{-3} = \frac{b_1}{5} = \frac{c_1}{4}$$

∴ Required line = DR of L

$$[a, b, c] = [-3, 5, 4]$$

Eq of line is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

(6)

Equation of the plane through the given line which is the line of intersection of the planes

$$x + y - z = 0$$

$$\text{and } 2x - y + 3z - 5 = 0$$

$$(x + y - z) + k(2x - y + 3z - 5) = 0$$

$$x + y - z + 2kx - ky + 3kz - 5k = 0 \quad (1)$$

$$(1+2k)x + (1-k)y + (3k-1)z - 5k = 0$$

Normal to the plane (1) has

Direction ratios

$$[1+2k, 1-k, 3k-1] \quad (2)$$

is if the plane is $\perp$ to yz -plane

Normal to the yz -plane is

x -axis whose direction ratios

$$\text{are } [1, 0, 0] \quad (3)$$

As the plane is $\perp$ to yz -plane

then their normals will also be perpendicular

∴ from (2) & (3)

$$1(1+2k) + 0(1-k) + 0(3k-1) = 0$$

$$1+2k = 0$$

$$2k = -1$$

$$k = -\frac{1}{2}$$

Put in (1)

$$(x + y - z) - \frac{1}{2}(2x - y + 3z - 5) = 0$$

$$2x + 2y - 2z - 2x + y - 3z + 5 = 0$$

$$3y - 5z + 5 = 0$$

Ans

ii) if the plane is perpendicular to zx -plane

Normal to the zx -plane is y -axis, whose direction ratios are $\{0, 1, 0\}$ — (3)

As the plane is $\perp$ to zx -plane then their normals will also be perpendicular

so from (1) & (3)

$$0(1+2k) + 1(1-k) + 0(3k-1) = 0$$

$$1-k = 0$$

$$k = 1$$

Put in (1)

$$(x+y-z) + 1(2x-y+3z-5) = 0$$

$$3x + 2z - 5 = 0$$

Ans

iii) if this plane is perpendicular to xy -plane

Normal to the xy -plane is z -axis, whose direction ratios are $\{0, 0, 1\}$ — (4)

As the plane is $\perp$ to xy -plane

then their normals will also be perpendicular

so from (1) & (4)

$$0(1+2k) + 0(1-k) + 1(3k-1) = 0$$

$$3k-1 = 0$$

$$3k = 1$$

$$k = \frac{1}{3}$$

Put in (1)

$$(x+y-z) + \frac{1}{3}(2x-y+3z-5) = 0$$

$$3x + 3y - 3z + 2x - y + 3z - 5 = 0$$

$$5x + 2y - 5 = 0$$

(7)

$$L: x = 2t$$

$$y = 3t$$

$$z = 4t$$

and intersecting planes

$$x + y + z = 0$$

$$2y - z = 0$$



Equation of the plane through the given line which is the line of intersection of the planes

$$x + y + z = 0$$

$$2y - z = 0$$

is

$$(x+y+z) + k(2y-z) = 0$$

— (1)

$$x + y + z + 2ky - kz = 0$$

$$x + (1+2k)y + (1-k)z = 0$$

Normal to the plane (1) has direction ratios

$$\{1, 1+2k, 1-k\}$$

— (7)

Since plane (1) contains the line $x = 2t, y = 3t, z = 4t$ having DR $\{2, 3, 4\}$ — (2)

Normal to the plane (1) $\perp$ to the normal of line

so

$$2(1) + 3(1+2k) + 4(1-k) = 0$$

$$2 + 3 + 6k + 4 - 4k = 0$$

$$2k + 9 = 0$$

$$2k = -9$$

$$k = -\frac{9}{2}$$

Put in (1)

$$\Rightarrow (x+y+z) - \frac{9}{2}(2y-z) = 0$$

$$2x + 2y + 2z - 18y + 9z = 0$$

$$2x - 16y + 11z = 0$$



$$x\text{-intercept} = a = 5$$

$$y\text{-intercept} = b = 2$$

$$z\text{-intercept} = c = ?$$

Equation of family of planes

$$\frac{x}{5} + \frac{y}{2} + \frac{z}{c} = 1 \quad \text{--- (1)}$$

The plane $\frac{x}{5} + \frac{y}{2} + \frac{z}{c} = 1$ is perpendicular to the plane $3x - 2y + z - 4 = 0$

Then their normals $(\frac{1}{5}, \frac{1}{2}, \frac{1}{c})$ & $\{3, -2, 1\}$ will be perpendicular. Then

$$\frac{1}{5}(3) + \frac{1}{2}(-2) + \frac{1}{c}(1) = 0$$

$$\frac{3}{5} - 1 + \frac{1}{c} = 0$$

$$-\frac{2}{5} + \frac{1}{c} = 0$$

$$\frac{1}{c} = \frac{2}{5}$$

$$c = \frac{5}{2}$$

$$\text{(iv)} \Rightarrow \frac{x}{5} + \frac{y}{2} + \frac{z}{5/2} = 1$$

$$\frac{x}{5} + \frac{y}{2} + \frac{2z}{5} = 1$$

Also another pt $(3, 0, \frac{1}{3})$ lies on the plane

$$a(3-5) + b(0-2) + c(\frac{1}{3}-1) = 0$$

$$1a + 3b + \frac{2}{3}c = 0$$

$$3a + 9b + 2c = 0 \quad \text{--- (2)}$$

Also normal to the plane will lie to the line

$$\text{Then } 6a + 3b + 2c = 0 \quad \text{--- (3)}$$

Solving (2) & (3), by cross multiplication method

$$\frac{a}{\begin{vmatrix} 9 & -2 \\ 3 & 2 \end{vmatrix}} = \frac{b}{-\begin{vmatrix} 3 & -2 \\ 6 & 2 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 3 & 9 \\ 6 & 2 \end{vmatrix}}$$

$$\frac{a}{18+6} = \frac{b}{-(6+12)} = \frac{c}{9-54}$$

$$\frac{a}{24} = \frac{b}{-18} = \frac{c}{-45}$$

$$\frac{a}{8} = \frac{b}{-6} = \frac{c}{-15}$$

Putting (1)

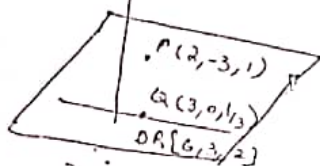
$$8(x-5) - 6(y+3) - 15(z-1) = 0$$

$$8x - 40 - 6y - 18 - 15z + 15 = 0$$

$$8x - 6y - 15z - 43 = 0$$

Ans

$$n = [a, b, c]$$



$$x-3 = 2y = 3z-1$$

$$\frac{x-3}{1} = \frac{y-0}{1/2} = \frac{z-1/3}{1/3}$$

$$\frac{x-3}{6} = \frac{y-0}{3} = \frac{z-1/3}{2}$$

on the

straight

$(2, -3, 1)$

normal $[a, b, c]$

$+b(y+3)+c(z-1)=0$

--- (7)

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(10)

Equation of the plane through
the given line which is
the line of intersection
of the planes

$$2x - y + 3z = 0$$

$$x + 2y - 2z - 3 = 0$$

is

$$(2x - y + 3z) + k(x + 2y - 2z - 3) = 0$$

— (1)

$$2x - y + 3z + kx + 2ky - 2kz - 3k = 0$$

$$(2+k)x + (-1+2k)y + (3-2k)z - 3k = 0$$

If this plane is at unit distance
from the origin, then
(0,0,0)

$$\frac{|0+0+0-3k|}{\sqrt{(2+k)^2 + (-1+2k)^2 + (3-2k)^2}} = 1$$

Cross multiply & simplify

$$\sqrt{(2+k)^2 + (-1+2k)^2 + (3-2k)^2} = |-3k|$$

Squaring

$$(2+k)^2 + (-1+2k)^2 + (3-2k)^2 = 9k^2$$

$$4+k^2+4k+1+4k^2-4k+9+4k^2-12k=9k^2$$

$$14-12k=0$$

$$-12k = -14$$

$$k = \frac{14}{12}$$

$$k = \frac{7}{6}$$

put in (1)

$$(2x - y + 3z) + \frac{7}{6}(x + 2y - 2z - 3) = 0$$

$$12x - 6y + 18z + 7x + 14y - 14z - 21 = 0$$

$$19x + 8y + 4z - 21 = 0$$

normals of two planes are

$$[1, 2, 3]$$

$$[2, 3, 4]$$

we have

$$1a + 2b + 3c = 0$$

$$2a + 3b + 4c = 0$$

$$\frac{a}{\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix}} = \frac{b}{-\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}}$$

$$\frac{a}{8-9} = \frac{b}{-(4-6)} = \frac{c}{3-4}$$

$$\frac{a}{-1} = \frac{b}{2} = \frac{c}{-1}$$

Eq of line L in symmetric form is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\frac{x-2}{-1} = \frac{y+3}{2} = \frac{z-0}{-1} = t$$

$$\frac{x-2}{-1} = t \quad \frac{y+3}{2} = t \quad \frac{z-0}{-1} = t$$

$$\left. \begin{aligned} x-2 &= -t \\ y+3 &= 2t \\ z &= -t \end{aligned} \right\} \begin{aligned} x &= 2-t \\ y &= -3+2t \\ z &= -t \end{aligned}$$

parametric Eq of line

coordinates of any pt

Let this pt is $A(2-t, -3+2t, -t)$

$$D.R. of OA = [2-t, -3+2t, -t]$$

$$D.R. of L = [-1, 2, -1]$$

OA is $\perp$ L.

$$\text{So } -1(2-t) + 2(-3+2t) + 1(-t) = 0$$

$$-2+t-6+4t-t=0$$

$$6t-8=0$$

$$6t=8$$

$$t = 8/6$$

$$t = 4/3$$

$$\text{So } A(2-\frac{4}{3}, -3+2(\frac{4}{3}), -\frac{4}{3})$$

$$A(\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}), O(0,0,0)$$

$$Eg \text{ of OA } x_2, y_2, z_2 \quad (x_1, y_1, z_1)$$

$$\frac{x-0}{2/3} = \frac{y-0}{-1/3} = \frac{z-0}{-4/3}$$

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{-4}$$

$$O(0,0,0)$$

A

$$L: x+2y+3z+4=0$$

$$2x+3y+4z+5=0$$

To transform the Eq

$$x+2y+3z+4=0$$

$$2x+3y+4z+5=0$$

of a line to the symmetrical form.

we require

is the coordinate of any pt on it

put zero in two eqs

$$x+2y+4=0$$

$$2x+3y+5=0$$

Solving it as

$$2x+4y+8=0$$

$$x+3y+5=0$$

$$y+3=0$$

$$y=-3$$

$$\text{is } x+2(-3)+4=0$$

$$x-6+4=0$$

$$x-2=0$$

$$x=2$$

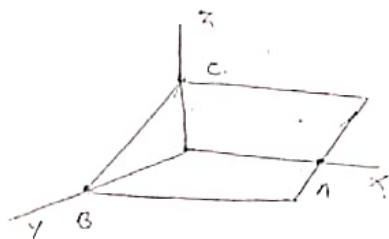
Therefore $(2, -3, 0)$ is a pt on the line.

Let a, b, c be the direction ratios of the line

Since the line lies in both the planes

Since the normal to two planes are perpendicular to the line of intersection





Let the variable plane which is at constant distance "p" from the origin is

$$lx + my + nz = p$$

The plane intersects the axes in the points A, B, C.

The x-coordinate of 'A' is obtained by putting $y=0, z=0$ in the above equation

$$\text{Thus } lx = p$$

$$x = \frac{p}{l} \text{ is the x-coordinate of A}$$

$$\text{Coordinates of A } \left(\frac{p}{l}, 0, 0 \right)$$

$$\text{Coordinates of B } \left(0, \frac{p}{m}, 0 \right)$$

$$\text{Coordinates of C } \left(0, 0, \frac{p}{n} \right)$$

Equation of the plane through "A"

$\left(\frac{p}{l}, 0, 0 \right)$ and // to YZ-plane is

$$x = \frac{p}{l} \text{ or } l = \frac{p}{x}$$

Equation of the plane through

$B \left(0, \frac{p}{m}, 0 \right)$ and // to XZ-plane is

$$y = \frac{p}{m} \text{ or } m = \frac{p}{y}$$

Equation of the plane through

$C \left(0, 0, \frac{p}{n} \right)$ and // to XY-plane

$$\text{is } z = \frac{p}{n} \text{ or } n = \frac{p}{z}$$

Since

$$l^2 + m^2 + n^2 = 1$$

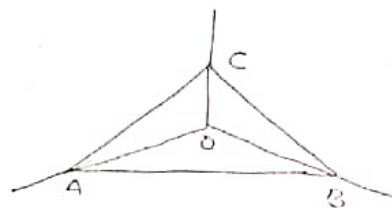
$$\left(\frac{p}{x} \right)^2 + \left(\frac{p}{y} \right)^2 + \left(\frac{p}{z} \right)^2 = 1$$

$$\frac{p^2}{x^2} + \frac{p^2}{y^2} + \frac{p^2}{z^2} = 1$$

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$$

$$\bar{x}^2 + \bar{y}^2 + \bar{z}^2 = \bar{p}^2$$

(13)



Let the variable plane which is at distance "p" from the origin is

$$lx + my + nz = p$$

The plane intersects the axes in the points A, B, C

$$\text{Coordinates of A } \left(\frac{p}{l}, 0, 0 \right)$$

$$\text{Coordinates of B } \left(0, \frac{p}{m}, 0 \right)$$

$$\text{Coordinates of C } \left(0, 0, \frac{p}{n} \right)$$

$$\text{Coordinates of O } \left(0, 0, 0 \right)$$

Centroid of tetrahedron

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

$$\left(\frac{p}{4l}, \frac{p}{4m}, \frac{p}{4n} \right)$$

$$\therefore (x, y, z) = \left(\frac{p}{4l}, \frac{p}{4m}, \frac{p}{4n} \right)$$

$$x = \frac{p}{4l}, \quad y = \frac{p}{4m}, \quad z = \frac{p}{4n}$$

$$\text{or } l = \frac{p}{4x}, \quad m = \frac{p}{4y}, \quad n = \frac{p}{4z}$$

$$\text{Since } l^2 + m^2 + n^2 = 1$$

$$\left(\frac{p}{4x} \right)^2 + \left(\frac{p}{4y} \right)^2 + \left(\frac{p}{4z} \right)^2 = 1$$

$$\frac{p^2}{16x^2} + \frac{p^2}{16y^2} + \frac{p^2}{16z^2} = 1$$

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$$

$$\bar{x}^2 + \bar{y}^2 + \bar{z}^2 = 16\bar{p}^2$$

$$L: \frac{x+3}{2} = \frac{y-4}{-7} = \frac{z}{3}$$

$$P: 4x + 2y + 2z - 9 = 0$$



$$DR \text{ of line } [2, -7, 3]$$

$$DR \text{ of plane } [4, 2, 2]$$

using condition of perpendicularity

$$\Rightarrow 4(2) + 2(-7) + 2(3)$$

$$\Rightarrow 8 - 14 + 6$$

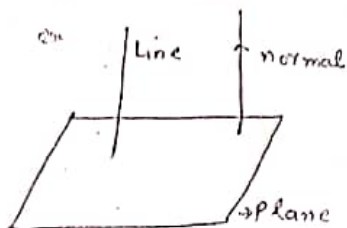
$$\Rightarrow 14 - 14$$

$$\Rightarrow 0$$

Since the normal to the plane is $\perp$ to the line, this line is parallel to the given plane.

$$L: \frac{x}{1} = \frac{y}{2} = \frac{z}{3}$$

$$P: 4x + 8y + 12z + 9 = 0$$



$$DR \text{ of plane } [4, 8, 12]$$

$$DR \text{ of line } [1, 2, 3]$$

using condition of parallelism

$$\frac{1}{4} = \frac{2}{8} = \frac{3}{12}$$

$$\frac{1}{4} = \frac{1}{4} = \frac{1}{4}$$

So the normal to plane is normal to the line. This line is $\perp$ to the plane.

(5)

$$x = mz + a$$

$$y = nz + b$$

$$x - a = mz$$

$$y - b = nz$$

$$\frac{x-a}{m} = z$$

$$\frac{y-b}{n} = z$$

$$\frac{x-a}{m} = \frac{y-b}{n} = \frac{z}{1} = t$$

which is symmetric form of the line

$$\frac{x-a}{m} = t \quad \frac{y-b}{n} = t \quad \frac{z}{1} = t$$

$$x - a = mt \Rightarrow y - b = nt, \quad z = t$$

$$x = a + mt \Rightarrow y = b + nt, \quad z = t$$

Any point on the line is

$$(a + mt, b + nt, t)$$

If this point lies in the plane

$$Ax + By + Cz + D = 0$$

$$A(a + mt) + B(b + nt) + Ct + D = 0$$

$$Aa + Am t + Bb + Bn t + Ct + D = 0$$

$$(Aa + Bb + D) + (Am + Bn + C)t = 0$$

This line will be in the plane if the above equation is satisfied for every value of t .

This requires that

$$Am + Bn + C = 0$$

$$Aa + Bb + D = 0$$

which are required condition

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(1)

Given That

$$\text{Line: } \left. \begin{aligned} x &= 1+t \\ y &= t \\ z &= -1+t \end{aligned} \right\} \text{--- (1)}$$

a plane $x+y+z=3$

line intersect plane at p

so $1+t+t-1+t=3$

$$3t=3$$

$$t=1$$

Put in (1)

$$x=1+t, y=1, z=-1+1$$

$$x=2, y=1, z=0$$

which is (2, 1, 0) common point to the line and plane.

(2)

Equation of the plane through

$P(x_1, y_1, z_1)$

with normal

$$[A, B, C]$$

$$A(x-x_1)+B(y-y_1)+C(z-z_1)=0$$

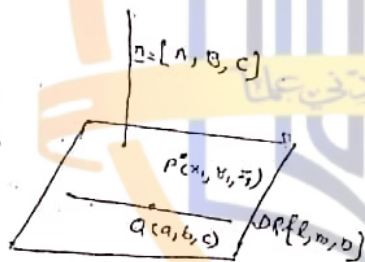
it passes through (a, b, c)

$$A(a-x_1)+B(b-y_1)+C(c-z_1)=0$$

normal of the plane has direction ratios $[A, B, C]$ is perpendicular to the plane having direction ratios $[l, m, n]$

so $Al+Bm+Cn=0$

(3)



Eliminating (1) (2) (3) from (1) (2) (3)

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a-x_1 & b-y_1 & c-z_1 \\ l & m & n \end{vmatrix} = 0$$

(3)

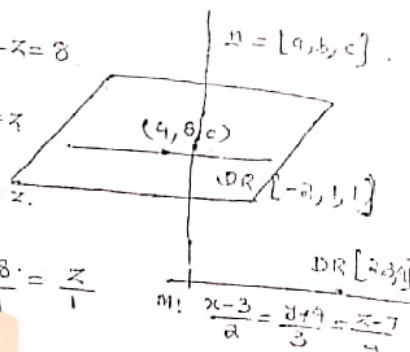
L:

$$x+2z=4 \Rightarrow y-z=8$$

$$x-4=-2z, y-8=z$$

$$\frac{x-4}{-2}=z, y-8=z$$

$$\frac{x-4}{-2} = \frac{y-8}{1} = \frac{z}{1}$$



Eq of plane through $P(4, 8, 0)$ with normal $[a, b, c]$

$$a(x-4)+b(y-8)+c(z-0)=0$$

(1)

Normal of the plane is $\perp$ to

to L + M

$$-2a+1b+1c=0$$

$$2a+3b+4c=0$$

$$\frac{a}{1} = \frac{b}{10} = \frac{c}{-8}$$

put in (1)

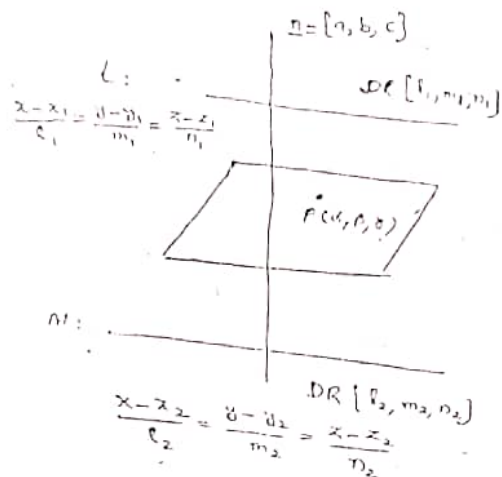
$$1(x-4)+10(y-8)-8(z-0)=0$$

$$x-4+10y-80-8z=0$$

$$x+10y-8z-84=0$$



(7)



Equation of the plane through the pt (α, β, γ) with normal $[a, b, c]$

$$a(x - \alpha) + b(y - \beta) + c(z - \gamma) = 0$$

normal of the plane $\perp$ to both the lines

$$a l_1 + b m_1 + c n_1 = 0 \quad \text{--- (2)}$$

$$a l_2 + b m_2 + c n_2 = 0 \quad \text{--- (3)}$$

Eliminate a, b, c from (2) (3)

$$\begin{vmatrix} x - \alpha & y - \beta & z - \gamma \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

(8)

Equation of the plane through the line which is line intersection of two planes is

$$(ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0$$

--- (A)

$$x + by + cz + d + ka'x + kb'y + kc'z + kd' = 0$$

$$(a + a'k)x + (b + b'k)y + (c + c'k)z + d + kd' = 0$$

(9)

Direction of the normal to the plane (A) is

$$[a + a'k, b + b'k, c + c'k]$$

plane (A) is parallel to the

$$\text{line } \frac{x}{l} = \frac{y}{m} = \frac{z}{n} \text{ whose}$$

$$\text{Direction } [l, m, n]$$

The normal of plane will be perpendicular to this line

So by condition of perpendicularity

$$l(a + a'k) + m(b + b'k) + n(c + c'k) = 0$$

$$la + la'k + mb + mb'k + nc + nc'k = 0$$

$$(la + mb + nc) + k(la' + mb' + nc') = 0$$

$$k(la' + mb' + nc') = -(la + mb + nc)$$

$$k = - \frac{la + mb + nc}{la' + mb' + nc'}$$

$$k = - \frac{k_1}{k_2} \text{ (say)}$$

Put in (A)

$$(ax + by + cz + d) - \frac{k_1}{k_2}(a'x + b'y + c'z + d')$$

$$k_2(ax + by + cz + d) - k_1(a'x + b'y + c'z + d') = 0$$

where

$$k_1 = al + bm + cn$$

$$k_2 = a'l + b'm + c'n$$

⑨ lines

$$L: \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$M: \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$$

are coplanar if

$$\begin{vmatrix} 2-1 & 3-2 & 4-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$\Rightarrow 1(15-16) - 1(10-12) + 1(8-9)$$

$$\Rightarrow -1 + 2 - 1$$

$$\Rightarrow 2 - 2$$

$$\Rightarrow 0$$

Hence L & M are coplanar.

⑩

$$\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} = p$$

A general pt on line ① is

$$\frac{x-1}{2} = p, \frac{y+1}{-3} = p, \frac{z+10}{8} = p$$

$$x-1=2p, y+1=-3p, z+10=8p$$

$$x=1+2p, y=-1-3p, z=-10+8p$$

$$(1+2p, -1-3p, -10+8p)$$

$$\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7} = q$$

A general pt on line ② is

$$\frac{x-4}{1} = q, \frac{y+3}{-4} = q, \frac{z+1}{7} = q$$

$$x-4=q, y+3=-4q, z+1=7q$$

$$x=4+q, y=-3-4q, z=-1+7q$$

$$(4+q, -3-4q, -1+7q)$$

⑪

The condition for the intersection of the lines are

$$2p+1 = q+4 \quad \text{--- (3)}$$

$$-3p-1 = -4q-3 \quad \text{--- (4)}$$

$$8p-10 = 7q-1 \quad \text{--- (5)}$$

Solving (3) & (4)

$$p=2$$

$$q=1$$

Put in (5)

$$8(2)-10 = 7(1)-1$$

$$6 = 6$$

$$LHS = RHS$$

Hence lines intersect.

To find pt of intersection

put $p=2$ in (1)

$$(5, -7, 6)$$

Equation of plane through line (1)

$$\begin{vmatrix} x-1 & y+1 & z+10 \\ 2 & -3 & 8 \\ 1 & -4 & 7 \end{vmatrix} = 0$$

$$\Rightarrow (x-1)(-21+32) + (y+1)(14-8) + (z+10)(-8+3) = 0$$

$$\Rightarrow 11x - 11 - 6y - 6 - 5z - 50 = 0$$

$$\Rightarrow 11x - 6y - 5z - 67 = 0$$

①

$$L: x+a = y = -12z$$

$$\frac{x+a}{1} = \frac{y}{1/2} = \frac{z}{-1/12}$$

$$\frac{x-(-a)}{12} = \frac{y-0}{6} = \frac{z-0}{-1}$$

$$pt P_1(-a, 0, 0)$$

$$OR \vec{b} = [12, 6, -1]$$

$$x = y + 2a = 6(x-a)$$

$$\frac{x}{1} = \frac{y+2a}{1} = \frac{x-a}{1/6}$$

$$\frac{x-a}{6} = \frac{y-(-2a)}{6} = \frac{x-a}{1}$$

$$pt P_2(0, -2a, a)$$

$$OR \vec{c} = [6, 6, 1]$$

$$\vec{u} = \vec{b} \times \vec{c}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 12 & 6 & -1 \\ 6 & 6 & 1 \end{vmatrix}$$

$$= 12\hat{i} - 18\hat{j} + 36\hat{k}$$

$$= 6(2\hat{i} - 3\hat{j} + 6\hat{k})$$

$$|\vec{u}| = \sqrt{(12)^2 + (-18)^2 + (36)^2}$$

$$= \sqrt{144 + 324 + 1296} = 42$$

As $\vec{u}$ is perpendicular to $\vec{b} \times \vec{c}$

So $\vec{u}$ lies along PQ

$$\vec{P_1P_2} = P_2 - P_1$$

$$= (0, -2a, a) - (-a, 0, 0)$$

$$= [a, -2a, a]$$

$$= a\hat{i} - 2a\hat{j} + a\hat{k}$$

Ex 8.6

(6.6)

$$\vec{P_1P_2} \cdot \vec{u} = (a\hat{i} - 2a\hat{j} + a\hat{k}) \cdot 6(2\hat{i} - 3\hat{j} + 6\hat{k})$$

$$= 6(2a + 6a + 6a)$$

$$= 6(14a)$$

$$= 84a$$

$$d = \frac{|\vec{P_1P_2} \cdot \vec{u}|}{|\vec{u}|}$$

$$d = \frac{84a}{42}$$

$$d = 2a$$

$\Rightarrow$ Length of the Shortest distance between the lines is the orthogonal projection of $\vec{P_1P_2}$ on $\vec{u}$

$\vec{u}$ = direction of vector of Common line

And Method

Equation of the plane through first line and parallel to second line is

$$\frac{x+a}{12} = \frac{y}{6} = \frac{z}{-1}$$

1st line

(0, -2a, a) and line

$$\frac{x}{6} = \frac{y+2a}{6} = \frac{z-a}{1}$$

$$\begin{vmatrix} x+a & y & z \\ 12 & 6 & -1 \\ 6 & 6 & 1 \end{vmatrix} = 0$$

$$2x - 3y + 6z + 2a = 0$$

Required distance = Perpendicular distance from a pt (0, -2a, a) of second line to this plane $2x - 3y + 6z + 2a = 0$

$$d = \frac{|0 - 3(-2a) + 6(a) + 2a|}{\sqrt{(2)^2 + (-3)^2 + (6)^2}} = \frac{14a}{\sqrt{49}} = \frac{14a}{7}$$

$$d = 2a$$

②

Equation of the plane through

The line which is line of intersection of two planes is

$$(ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0$$

$$ax + by + cz + d + ka'x + kb'y + kc'z + kd' = 0$$

$$(a + ka')x + (b + kb')y + (c + c'k)z + d + kd' = 0$$

The direction of the normals

to the plane ① is

$$[a + ka', b + kb', c + c'k]$$

NR of x-axis is $[1, 0, 0]$

plane ① and x-axis are parallel, so

then its normal will be

perpendicular to x-axis

so by condition of perpendicularity

$$(a + ka')(1) + 0(b + kb') + 0(c + c'k) = 0$$

$$a + ka' = 0$$

$$k = -\frac{a}{a'}$$

Put in ①

$$(ax + by + cz + d) - \frac{a}{a'}(a'x + b'y + c'z + d') = 0$$

$$a\cancel{a'}x + a'b'y + ac'z + ad' - a\cancel{a'}x - ab'y - ac'z - ad' = 0$$

$$(a'b - ab')y + (a'c - ac')z + a'd - ad' = 0$$

Shortest distance

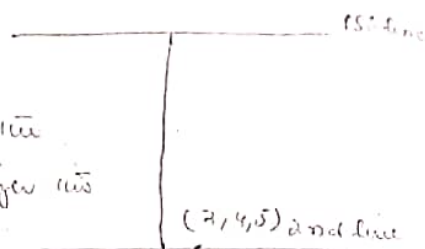
Per distance of any pt of x-axis from this plane is

Per distance from origin $(0, 0, 0)$

$$d = \frac{|a'd - ad'|}{\sqrt{(a'b - ab')^2 + (a'c - ac')^2}}$$

③

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad (1)$$



Equation of the plane through the

1st line

and

parallel

to the 2nd line

$$\left(\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5} \right)$$

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$(x-1)(5-16) - (y-2)(10-12) + (z-3)(8-9) = 0$$

$$-x + 1 + 2y - 4 - z + 3 = 0$$

$$-x + 1 + 2y - 4 - z + 3 = 0$$

$$x - 2y + z = 0$$

Required distance = Per distance from a pt $(2, 4, 5)$ of second line to the plane

$$x - 2y + z = 0$$

$$= \frac{|2 - 2(4) + 5|}{\sqrt{(1)^2 + (-2)^2 + (1)^2}}$$

$$d = \frac{1}{\sqrt{6}}$$

Equations of straight line perpendicular to both are found as

$$\begin{vmatrix} x-2 & y-4 & z-5 \\ 2 & 3 & 4 \\ -2 & 1 & 1 \end{vmatrix} = 0 = \begin{vmatrix} x-2 & y-4 & z-5 \\ 3 & 4 & 5 \\ 1 & -2 & 1 \end{vmatrix}$$

$$\Rightarrow 11x + 2y - 2z + 6 = 0 = 7x + y - 5z + 7$$

(4)

$$L: \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} = p$$

$$\frac{x-3}{1} = p, \quad \frac{y-5}{-2} = p, \quad \frac{z-7}{1} = p$$

$$x-3=p, \quad y-5=-2p, \quad z-7=p$$

$$x=3+p, \quad y=5-2p, \quad z=7+p$$

Any pt on line L is $(3+p, 5-2p, 7+p)$

$$M: \frac{x+1}{1} = \frac{y+1}{-6} = \frac{z+1}{1} = q$$

$$x+1=q, \quad \frac{y+1}{-6} = q, \quad \frac{z+1}{1} = q$$

$$x=-1+q, \quad y+1=-6q, \quad z=-1+q$$

$$y=-1-6q, \quad z=-1+q$$

pt on line M is $(-1+q, -1-6q, -1+q)$

$$DR_L [1, -2, 1]$$

$$DR_M [1, -6, 1]$$

$$PQ = (-1+q-3-p, -1-6q-5+2p, -1+q-7-p)$$

$$= (-q-p-4, -6q+2p-6, q-p-8)$$

PQ is $\perp$ to line L

$$1(-q-p-4) - 2(-6q+2p-6) + 1(q-p-8) = 0$$

$$-q-p-4 + 12q-4p+12 + q-p-8 = 0$$

$$-6p + 20q = 0 \quad \text{--- (1)}$$

PQ is $\perp$ to line M

$$1(-q-p-4) - 6(-6q+2p-6) + 1(q-p-8) = 0$$

$$-q-p-4 + 36q-12p+36 + q-p-8 = 0$$

$$20p - 86q = 0 \quad \text{--- (2)}$$

$$\text{Solving (1) \& (2)}$$

$$\Rightarrow p = 0$$

$$\text{or } q = 0$$

Put in eq (1)

$$P(3, 5, 7)$$

$$Q(-1, -1, -1)$$

$$PQ = (-4, -6, -8)$$

$$|PQ| = \sqrt{(-4)^2 + (-6)^2 + (-8)^2} = \sqrt{16 + 36 + 64} = \sqrt{116}$$

Eq of PQ is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-3}{-1-3} = \frac{y-5}{-1-5} = \frac{z-7}{-1-7}$$

$$\frac{x-3}{-4} = \frac{y-5}{-6} = \frac{z-7}{-8}$$

$$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

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(5)

Equation of the line joining
the pts $A(-3, 7, -13)$ & $B(-6, 1, -10)$
are (x_1, y_1, z_1) (x_2, y_2, z_2)

$$\frac{x+3}{-6+3} = \frac{y-7}{1-7} = \frac{z+13}{-10+13}$$

$$\frac{x+3}{-3} = \frac{y-7}{-6} = \frac{z+13}{3}$$

$$\frac{x+3}{-1} = \frac{y-7}{2} = \frac{z+13}{-1} = t$$

$$\frac{x+3}{-1} = t, \frac{y-7}{2} = t, \frac{z+13}{-1} = t$$

$$x+3 = -t, y-7 = 2t, z+13 = -t$$

$$x = -3-t, y = 7+2t, z = -13-t$$

Any pt on AB, $P(-3+t, 7+2t, -13-t)$

Reduce the Equation of other

Line L: $2x - y - 3z + 32 = 0$

$$3x + 2y - 15z - 8 = 0$$

To symmetric form

Put $z = 0$

$$2x - y + 32 = 0$$

$$3x + 2y - 8 = 0$$

$$4x - 2y + 64 = 0$$

$$3x + 2y - 8 = 0$$

$$7x + 56 = 0$$

$$7x = -56$$

$$x = -8$$

$$\Rightarrow y = 16$$

∴ pt $(-8, 16, 0)$

∴ Let a, b, c be the
DR of the normal of
the line

(6)

Both the normals of the
planes are to the line

$$2a - (b - 3c) = 0$$

$$3a + 2b - 15c = 0$$

$$\frac{a}{3} = \frac{b}{3} = \frac{c}{1}$$

Symmetric form is

$$\frac{x+8}{3} = \frac{y-16}{3} = \frac{z-0}{1}$$

$$\frac{x+8}{3} = \frac{y-16}{3} = \frac{z-0}{1} = s$$

$$\frac{x+8}{3} = s, \frac{y-16}{3} = s, \frac{z-0}{1} = s$$

$$x+8 = 3s, y-16 = 3s, z = s$$

$$x = -8+3s, y = 16+3s,$$

Any pt on line L is

$$Q(-8+3s, 16+3s, s)$$

DR of PQ

$$[3s-8+3-t, 16+3s-7-2t, s+13+t]$$

$$[3s-t-5, 3s-2t+9, s+t+13]$$

As PQ is ⊥ to AB & L.

So normal of PQ is ⊥ to AB & L

with DR $[1, 2, -1]$ & $[3, 3, 1]$

$$1(3s-t-5) + 2(3s-2t+9) - 1(s+t+13) = 0$$

$$\Rightarrow 8s - 6t = 0 \quad \text{--- (1)}$$

$$3(3s-t-5) + 3(3s-2t+9) + 1(s+t+13) = 0$$

$$\Rightarrow 19s - 8t = -25 \quad \text{--- (2)}$$

Solve (1) & (2)

$$s = -3$$

$$t = -4$$

Put $t = -4$ in P

$$P(-8-4, 7+2(-4), -13-(-4))$$

$$P(-12, -1, -9)$$

is required pt.



(6)

We first transform the Eqs of lines L & M to symmetric form

Transform L to symmetric form

$$\begin{aligned} 6x + 8y + 3z - 13 &= 0 \\ x + 2y + z - 3 &= 0 \end{aligned}$$

Put $z = 0$

$$\begin{aligned} 6x + 8y - 13 &= 0 \\ x + 2y - 3 &= 0 \end{aligned}$$

Solving $x = \frac{1}{2}, y = \frac{5}{4}$

So pt $P(\frac{1}{2}, \frac{5}{4}, 0)$

Let a, b, c be the direction ratios of the line L

Normals $[6, 8, 3]$ & $[1, 2, 1]$ of both the planes are $\perp$ to line L,

$$\begin{aligned} \text{So } 6a + 8b + 3c &= 0 \\ 1a + 2b + 1c &= 0 \end{aligned}$$

$$\frac{a}{2} = \frac{b}{-3} = \frac{c}{4}$$

Eq of L in symmetric form

$$\frac{x - \frac{1}{2}}{2} = \frac{y - \frac{5}{4}}{-3} = \frac{z - 0}{4}$$

pt $P(\frac{1}{2}, \frac{5}{4}, 0)$

Transform M to symmetric form

$$\begin{aligned} 3x - 9y + 3z &= 0 \\ x + y - z &= 0 \end{aligned}$$

put $z = 0$

$$\begin{aligned} 3x - 9y &= 0 \\ x + y &= 0 \end{aligned}$$

solving $x = 0$

(57)

Let a, b, c be the direction ratios of the line M

Normals of both the planes

$[3, -9, 3]$ & $[1, 1, -1]$ are $\perp$ to line M

$$\begin{aligned} \text{So } 3a - 9b + 3c &= 0 \\ 1a + 1b - 1c &= 0 \end{aligned}$$

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{3}$$

Eq of M in symmetric form

$$\frac{x - 0}{1} = \frac{y - 0}{2} = \frac{z - 0}{3}$$

pt $Q(0, 0, 0)$

$\vec{b} = [2, -3, 4]$

$\vec{c} = [1, 2, 3]$

$$\vec{u} = \vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 4 \\ 1 & 2 & 3 \end{vmatrix}$$

$$\vec{u} = -17\hat{i} - 2\hat{j} + 7\hat{k}$$

$$\begin{aligned} |\vec{u}| &= \sqrt{(-17)^2 + (-2)^2 + (7)^2} \\ &= \sqrt{289 + 4 + 49} = \sqrt{342} \end{aligned}$$

$$\vec{P_1P_2} = [\frac{1}{2}, \frac{5}{4}, 0] = \frac{1}{2}\hat{i} + \frac{5}{4}\hat{j} + 0\hat{k}$$

$$\begin{aligned} \vec{P_1P_2} \cdot \vec{u} &= (\frac{1}{2}\hat{i} + \frac{5}{4}\hat{j} + 0\hat{k}) \cdot (-17\hat{i} - 2\hat{j} + 7\hat{k}) \\ &= \frac{-17}{2} - \frac{5}{2} = -\frac{22}{2} = -11 \end{aligned}$$

$$d = \frac{|\vec{P_1P_2} \cdot \vec{u}|}{|\vec{u}|} = \frac{|-11|}{\sqrt{342}} = \frac{11}{\sqrt{342}}$$

ii) To find Equations of common perpendicular

is Eq of plane through L which is $\perp$ to common perpendicular

$$P_1 + kP_2 = 0$$



$$(6x+8y+3z-13)+k(x+2y+z-3)=0$$

$$(6+k)x+(8+2k)y+(3+k)z-13-3k=0$$

DR of normal of plane (A) is

$$[6+k, 8+2k, 3+k]$$

The plane contains the common perpendicular with normals $[-17, -2, 7]$

$$\text{So } -17(6+k)-2(8+2k)+7(3+k)=0$$

$$-102-17k-16-4k+21+7k=0$$

$$-14k-97=0$$

$$-14k=97$$

$$k = -\frac{97}{14}$$

Put in (A)

$$(6x+8y+3z-13)-\frac{97}{14}(x+2y+z-3)=0$$

$$13x+82y+55z-109=0$$

ii) Eq of plane through M

which is also Eq of common perpendicular

$$P_1 + k P_2 = 0$$

$$(3x-9y+5z)+k(x+y-z)=0$$

$$(3+k)x+(-9+k)y+(5-k)z=0$$

DR of normal of plane (B) is

$$[3+k, -9+k, 5-k]$$

The plane contains the common

perpendicular with normal $[-17, -2, 7]$

$$\text{So } -17(3+k)-2(-9+k)+7(5-k)=0$$

$$-51-17k+18-2k+35-7k=0$$

Put in (B)

$$(3x-9y+5z)+\frac{1}{13}(x+y-z)=0$$

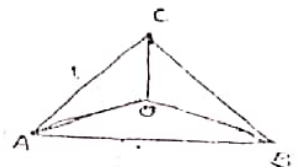
$$10x-29y+16z=0$$

(i) The planes

$$y+z=0, \quad z+x=0, \quad x+y=0$$

$$x+y+z=0$$

be Eqs of faces OCB, OAB and ABC respectively



Eq of line OA (being the line of intersection of OAB, OAC)

$$x+y=0$$

$$x+z=0$$

$$+x=y$$

$$z=-x$$

$$-y=1+x=-x \quad \text{Put } z=0$$

$$x=0$$

$$y=0$$

$$P_1(0,0,0)$$

Let a, b, c be the DR of line

The normals $[1,1,0]$ & $[1,0,1]$ of both the planes are $\perp$ to the line

$$\text{So } 1a+1b+0c=0$$

$$1a+0b+1c=0$$

$$\frac{a}{1} = \frac{b}{-1} = \frac{c}{-1}$$

$$\text{Eq of OA } \frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-0}{-1}$$

$$P_1(0,0,0)$$



Equation of line BC (line of intersection of planes OBC & ABC)

$$y + z = 0 \quad x + y + z = a$$

$$z = 0$$

$$y = 0, \quad x = a$$

$$P(a, 0, 0)$$

Let a, b, c be the DR of the line BC.

normals $[0, 1, 1]$ & $[1, 1, 1]$

If both the planes are $\perp$ to the line

$$0a + 1b + 1c = 0$$

$$1a + 1b + 1c = 0$$

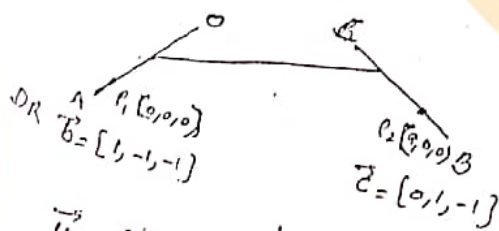
$$\frac{a}{0} = \frac{b}{1} = \frac{c}{-1}$$

Eg of BC

$$\frac{x-a}{0} = \frac{y-0}{1} = \frac{z-0}{-1}$$

$$P_1(a, 0, 0)$$

$$\vec{C} = [0, 1, -1]$$



$$\vec{U} = \vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 0 & 1 & -1 \end{vmatrix}$$

$$\vec{U} = 2\hat{i} + \hat{j} + \hat{k}$$

$$|\vec{U}| = \sqrt{(2)^2 + (1)^2 + (1)^2} = \sqrt{4+1+1} = \sqrt{6}$$

$$\vec{P_1P_2} = [a, 0, 0] = a\hat{i} + 0\hat{j} + 0\hat{k}$$

$$\vec{P_1P_2} \cdot \vec{U} = 2a$$

$$d = \frac{|\vec{P_1P_2} \cdot \vec{U}|}{|\vec{U}|} = \frac{2a}{\sqrt{6}}$$

Similarly, the distance between other opposite edges are $\frac{2a}{\sqrt{6}}$.

Eqs of shortest distance between BC and AD

$$\begin{vmatrix} x-a & y & z \\ 0 & 1 & -1 \\ +1 & -1 & -1 \end{vmatrix} = 0 = \begin{vmatrix} x & y & z \\ 0 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix}$$

$$x-a(-1-1) - y(0-1) + z(0-1) = 0$$

$$-2x + 2a - y - z = 0$$

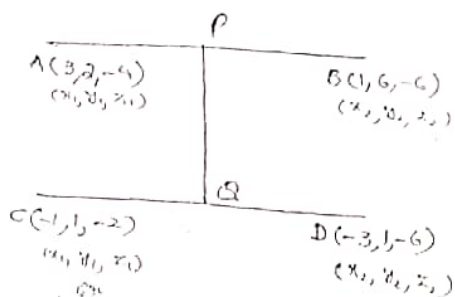
$$2x - y + z - 2a = 0$$

$$x(-1-1) - y(0-1) + z(0-1) = 0$$

$$-2x - y - z = 0$$

$$2x + y + z = 0$$

(3)



Any pt on AB is P

$$x = x_1 + (x_2 - x_1)t$$

$$y = y_1 + (y_2 - y_1)t$$

$$z = z_1 + (z_2 - z_1)t$$

$$\Rightarrow x = 3 + (1-3)t \Rightarrow x = 3-2t$$

$$y = 2 + (1-2)t \Rightarrow y = 2+t$$

$$z = -4 + (-2+4)t \Rightarrow z = -4+2t$$

$$\text{pt } P(3-2t, 2+t, -4+2t) \quad \text{--- (1)}$$

Any pt on CD is Q

$$x = x_1 + (x_2 - x_1)s$$

$$y = y_1 + (y_2 - y_1)s$$

$$z = z_1 + (z_2 - z_1)s$$

$$\Rightarrow x = -1 + (-3+1)s$$

$$x = -1-2s$$

$$y = 1 + (1-1)s$$

$$y = 1+0s$$

$$\Rightarrow y = 1$$

$$z = -2 + (-2+4)s$$

$$z = -2+2s$$

$$\text{pt } Q(-1-2s, 1, -2+2s) \quad \text{--- (2)}$$

$$\text{DR of PQ } [3-2t+1+2s, 2+t-1, -4-2t+2+2s]$$

$$[4-2t+2s, t+1, 4s-2t-2]$$

$$\text{DR of AB } [-2, 1, -2]$$

$$\text{DR of CD } [-2, 0, -2]$$

(5)

PQ is $\perp$ to AB & CD

$$-2(4-2t+2s) + 1(t+1) - 2(4s-2t-2) = 0$$

$$-2(4-2t+2s) + 0(t+1) - 4(4s-2t-2) = 0$$

$$\Rightarrow -8+4t-4s+1t+1-16s+8t+8 = 0$$

$$12t-20s = 0 \quad \text{--- (3)}$$

$$12t-20s = 0 \quad \text{--- (4)}$$

$$\text{from (3) & (4)}$$

$$t=0, s=0$$

$$\text{Put in (1)} \quad P(3, 2, -4) \\ Q(-1, 1, -2)$$

$$|PQ| = \sqrt{(3+1)^2 + (2-1)^2 + (-4+2)^2} \\ = \sqrt{16+1+4} \\ = \sqrt{21}$$

Eq of PQ

$$\frac{x-3}{-1-3} = \frac{y-2}{1-2} = \frac{z+4}{-2+4}$$

$$\frac{x-3}{-4} = \frac{y-2}{-1} = \frac{z+4}{2}$$

① $(2\sqrt{3}, 2, -2)$
 (x, y, z)

we are to find (r, θ, ϕ)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(2\sqrt{3})^2 + (2)^2 + (-2)^2}$$

$$= \sqrt{12 + 4 + 4} = \sqrt{20}$$

$$r = 2\sqrt{5}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{2}{2\sqrt{3}}\right)$$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\theta = \frac{\pi}{6}$$

$$z = -2$$

$$\text{So } (2\sqrt{5}, \frac{\pi}{6}, -2)$$

② $(\frac{16}{5}, \frac{12}{5}, 3)$
 (x, y, z)

we are to find (r, θ, ϕ)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{\left(\frac{16}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + 3^2}$$

$$= \sqrt{\frac{256}{25} + \frac{144}{25} + \frac{225}{25}}$$

$$r = \sqrt{\frac{625}{25}} = \sqrt{25} = 5$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{12/5}{16/5}\right)$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$z = 3$$

$$\text{So } (5, \tan^{-1}\left(\frac{3}{4}\right), 3)$$

③ $(5, \frac{\pi}{6}, 3)$
 (r, θ, ϕ)

we are to find (x, y, z)

$$x = r \cos \theta = 5 \cos \frac{\pi}{6} = 5 \left(\frac{\sqrt{3}}{2}\right) = \frac{5\sqrt{3}}{2}$$

$$y = r \sin \theta = 5 \sin \frac{\pi}{6} = 5 \left(\frac{1}{2}\right) = \frac{5}{2}$$

$$z = 3$$

$$\text{So } \left(\frac{5\sqrt{3}}{2}, \frac{5}{2}, 3\right)$$

④ $(6, \frac{\pi}{3}, -5)$

(r, θ, ϕ)

we are to find (x, y, z)

$$x = r \cos \theta = 6 \cos \frac{\pi}{3} = 6 \left(\frac{1}{2}\right) = 3$$

$$y = r \sin \theta = 6 \sin \frac{\pi}{3} = 6 \left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

$$z = -5$$

$$\text{So } (3, 3\sqrt{3}, -5)$$

⑤ $(1, 1, \sqrt{6})$

(x, y, z)

we are to find (r, θ, ϕ)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + 1 + 6} = \sqrt{8}$$

$$= 2\sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}(1)$$

$$= \tan^{-1}(1)$$

$$\theta = \frac{\pi}{4}$$

$$\phi = \cos^{-1}\left(\frac{z}{r}\right)$$

$$= \cos^{-1}\left(\frac{\sqrt{6}}{2\sqrt{2}}\right)$$

$$= \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{6}$$

$$\text{So } (2\sqrt{2}, \frac{\pi}{4}, \frac{\pi}{6})$$

⑥ $(1, 1, -\sqrt{6})$

(x, y, z)

we are to find (r, θ, ϕ)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + 1 + 6} = \sqrt{8} = 2\sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$\phi = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{-\sqrt{6}}{2\sqrt{2}}\right)$$

$$= \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$$

$$\text{So } (2\sqrt{2}, \frac{\pi}{4}, \frac{5\pi}{6})$$

c) $(-\sqrt{3}, 1, -2)$
 (x, y, z)

we are to find (ρ, θ, ϕ)

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{3 + 1 + 4} = \sqrt{8}$$

$$\rho = 2\sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \frac{5\pi}{6}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{-2}{2\sqrt{2}}\right)$$

$$= \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$\phi = \frac{3\pi}{4}$$

$$\text{so } (2\sqrt{2}, \frac{5\pi}{6}, \frac{3\pi}{4})$$

d) $(4, -4\sqrt{3}, 6)$
 (x, y, z)

we are to find (ρ, θ, ϕ)

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{16 + 48 + 36} = \sqrt{100} = 10$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-4\sqrt{3}}{4}\right)$$

$$= \tan^{-1}(-\sqrt{3})$$

$$= \frac{5\pi}{3}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{6}{10}\right)$$

$$= \cos^{-1}\left(\frac{3}{5}\right)$$

$$\text{so } (10, \frac{5\pi}{3}, \cos^{-1}(\frac{3}{5}))$$

4(a)

$$(5, \frac{\pi}{2}, \frac{\pi}{2})$$

$$(\rho, \theta, \phi)$$

we are to find (x, y, z)

$$x = \rho \sin \phi \cos \theta = 5 \sin \frac{\pi}{2} \cos \frac{\pi}{2} = 0$$

$$y = \rho \sin \phi \sin \theta = 5 \sin \frac{\pi}{2} \sin \frac{\pi}{2} = 5$$

$$z = \rho \cos \phi = 5 \cos \frac{\pi}{2} = 0$$

$$\text{so } (0, 5, 0)$$

b) $(4, \frac{\pi}{3}, \frac{2\pi}{3})$
 (ρ, θ, ϕ)

we are to find (x, y, z)

$$x = \rho \sin \phi \cos \theta = 4 \sin \frac{2\pi}{3} \cos \frac{\pi}{3}$$

$$= 4 \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2}$$

$$= \sqrt{3}$$

$$y = \rho \sin \phi \sin \theta = 4 \sin \frac{2\pi}{3} \sin \frac{\pi}{3}$$

$$= 4 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2}$$

$$= 3$$

$$z = \rho \cos \phi = 4 \cdot \cos \frac{2\pi}{3}$$

$$= 4 \cdot (-\frac{1}{2}) = -2$$

$$\text{so } (\sqrt{3}, 3, -2)$$

c) $(0, \frac{\pi}{15}, \frac{\pi}{5})$
 (ρ, θ, ϕ)

we are to find (x, y, z)

$$x = \rho \sin \phi \cos \theta = 0$$

$$y = \rho \sin \phi \sin \theta = 0$$

$$z = \rho \cos \phi = 0$$

$$\text{so } (0, 0, 0)$$

d) $(2, \frac{3\pi}{3}, \frac{3\pi}{4})$
 (ρ, θ, ϕ)

we are to find (x, y, z)

$$x = \rho \sin \phi \cos \theta = 2 \cdot \sin \frac{3\pi}{4} \cos \frac{3\pi}{4}$$

$$= 2 \cdot \frac{1}{\sqrt{2}} \cdot (-\frac{1}{\sqrt{2}}) = -\frac{1}{2}$$

$$y = \rho \sin \phi \sin \theta = 2 \cdot \sin \frac{3\pi}{4} \sin \frac{3\pi}{4}$$

$$= 2 \cdot (\frac{1}{\sqrt{2}}) \cdot (\frac{1}{\sqrt{2}}) = 1$$

$$= -\frac{\sqrt{3}}{2} = -\sqrt{\frac{3}{4}}$$

$$z = \rho \cos \phi = 2 \cos(\frac{3\pi}{4}) = 2 \cdot (-\frac{1}{\sqrt{2}}) = -\sqrt{2}$$

$$\text{so } (-\frac{1}{2}, 1, -\sqrt{2})$$

(5)

$$\rho \cos \phi = a$$

For transformation in rectangular coordinates

$$\text{put } \rho \cos \phi = x$$

$$\Rightarrow x = a.$$

(6)

$$\rho = a \cos \theta \sin \phi$$

$$\rho^2 = a^2 \cos^2 \theta \sin^2 \phi$$

$$\text{As } \rho^2 = x^2 + y^2 + z^2$$

$$x = \rho \sin \theta \cos \phi$$

$$x^2 + y^2 + z^2 = 2x$$

(7)

$$\rho = 7 \sin \theta \sin \phi$$

$$\rho^2 = 49 \sin^2 \theta \sin^2 \phi$$

$$\text{As } \rho^2 = x^2 + y^2 + z^2$$

$$y = \rho \sin \theta \sin \phi$$

$$x^2 + y^2 + z^2 = 7y$$

(8)

$$\rho^2 \cos 2\theta = a^2$$

$$\rho^2 (2 \cos^2 \theta - 1) = a^2$$

$$\rho^2 \left(\frac{2}{\sec^2 \theta} - 1 \right) = a^2$$

$$\rho^2 \left(\frac{2}{1 + \tan^2 \theta} - 1 \right) = a^2$$

$$\text{As } \rho^2 = x^2 + y^2 + z^2$$

$$\tan \theta = \frac{y}{x}$$

$$(x^2 + y^2 + z^2) \left(\frac{2}{1 + \frac{y^2}{x^2}} - 1 \right) = a^2$$

$$(x^2 + y^2 + z^2) \left(\frac{2}{\frac{x^2 + y^2}{x^2}} - 1 \right) = a^2$$

$$(x^2 + y^2 + z^2) \left(\frac{2x^2}{x^2 + y^2} - 1 \right) = a^2$$

$$(x^2 + y^2 + z^2) \left(\frac{2x^2 - x^2 - y^2}{x^2 + y^2} \right) = a^2$$

$$(x^2 + y^2 + z^2) \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = a^2$$

$$(x^2 + y^2 + z^2) (x^2 - y^2) = a^2 (x^2 + y^2)$$

(9)

$$x = r^2 \cos \theta$$

$$x = r^2 (2 \cos^2 \theta - 1)$$

$$= r^2 \left(\frac{2}{\sec^2 \theta} - 1 \right)$$

$$= r^2 \left(\frac{2}{1 + \tan^2 \theta} - 1 \right)$$

$$\text{As } r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$

$$= (x^2 + y^2) \left(\frac{2}{1 + \frac{y^2}{x^2}} - 1 \right)$$

$$= (x^2 + y^2) \left(\frac{2}{\frac{x^2 + y^2}{x^2}} - 1 \right)$$

$$= (x^2 + y^2) \left(\frac{2x^2 - x^2 - y^2}{x^2 + y^2} \right)$$

$$x^2 = x^2 - y^2$$

(10)

$$x = 1 + \sin \theta$$

$$x - 1 = \sin \theta$$

$$r(x - 1) = r \sin \theta$$

$$r^2 (x - 1)^2 = r^2 \sin^2 \theta$$

$$\text{As } r^2 = x^2 + y^2$$

$$y = r \sin \theta$$

$$(x^2 + y^2) (x - 1)^2 = y^2$$

(11)

$$(x + y)^2 - x^2 + 4 = 0$$

For cylindrical coordinates

$$\text{put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$(r \cos \theta + r \sin \theta)^2 - z^2 + 4 = 0$$

$$r^2 (\cos^2 \theta + \sin^2 \theta) - z^2 + 4 = 0$$

$$r^2 (\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta) - z^2 + 4 = 0$$

$$r^2 (1 + \sin 2\theta) - z^2 + 4 = 0$$

$$(x+y)^2 - z^2 + 4 = 0$$

i. For spherical coordinates

$$\text{Put } x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$x^2 + y^2 + z^2 - 2xy - z^2 + 4 = 0$$

$$x^2 + y^2 + z^2 - z^2 + 2xy - z^2 + 4 = 0$$

$$(x^2 + y^2 + z^2) - 2z^2 + 2xy + 4 = 0$$

$$\rho^2 + 2(\rho \cos \phi)^2 + 2(\rho \sin \phi \cos \theta)(\rho \sin \phi \sin \theta)$$

$$+ 4 = 0$$

$$\rho^2 - 2\rho^2 \cos^2 \phi + 2\rho^2 \sin^2 \phi \sin \theta \cos \theta + 4 = 0$$

(12)

$$x^2 + y^2 + 2z = 6$$

i. For cylindrical coordinates

$$\text{Put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta + 2z = 6$$

$$r^2 (\cos^2 \theta + \sin^2 \theta) + 2z = 6$$

$$r^2 (1) + 2z = 6$$

$$r^2 + 2z = 6$$

ii. For spherical coordinates

$$\text{Put } x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta + 2\rho \cos \phi = 6$$

$$\rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + 2\rho \cos \phi = 6$$

$$\rho^2 \sin^2 \phi (1) + 2\rho \cos \phi = 6$$

$$\rho^2 \sin^2 \phi + 2\rho \cos \phi = 6$$

(13)

$$x^2 - y^2 - z^2 = 1$$

(5)

i. For spherical coordinates

$$\text{Put } x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\rho^2 \sin^2 \phi \cos^2 \theta - \rho^2 \sin^2 \phi \sin^2 \theta - \rho^2 \cos^2 \phi = 1$$

$$\rho^2 \sin^2 \phi (\cos^2 \theta - \sin^2 \theta) - \rho^2 \cos^2 \phi = 1$$

$$\rho^2 \sin^2 \phi \cdot \cos 2\theta - \rho^2 \cos^2 \phi = 1$$

ii. For cylindrical coordinates

$$\text{Put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$r^2 \cos^2 \theta - r^2 \sin^2 \theta - z^2 = 1$$

$$r^2 (\cos^2 \theta - \sin^2 \theta) - z^2 = 1$$

$$r^2 \cos 2\theta - z^2 = 1$$

(14)

$$3x + y - 4z = 12$$

i. For cylindrical coordinates

$$\text{Put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$3r \cos \theta + r \sin \theta - 4z = 12$$

ii. For spherical coordinates

$$\text{Put } x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

so

$$3\rho \sin \phi \cos \theta + \rho \sin \phi \sin \theta - 4\rho \cos \phi = 12$$

Cell: 0321

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(5)

$$\frac{(x-1)^2}{4} - \frac{(y+2)^2}{1} = 4(x-y)$$

new origin $O'(4, -2, 1)$

let $P(x, y, z)$ be a pt on the surface

let (x', y', z') be the point relative to new origin

put $x = x' + 4, y = y' - 2, z = z' + 1$

Equation of surface relative to new origin

$$\frac{(z')^2}{4} - \frac{(y')^2}{1} = 4x'$$

$$x^2 - 9y^2 - 4z^2 - 6x + 18y + 16z + 20 = 0$$

new origin $O'(3, 1, 2)$

let $P(x, y, z)$ be a pt on the surface

let (x', y', z') be the point relative to new origin

put $x = x' + 3, y = y' + 1, z = z' + 2$

Equation of surface relative to new origin

$$x'^2 - 9(y'+1)^2 - 4(z'+2)^2 - 6(x'+3) + 18(y'+1) + 16(z'+2) + 20 = 0$$

$$x'^2 - 9y'^2 - 4z'^2 - 6x' + 18y' - 4z' - 16 = 0$$

$$x'^2 - 9y'^2 - 4z'^2 - 6x' + 18y' - 4z' - 16 = 0$$

$$x'^2 - 9y'^2 - 4z'^2 - 6x' + 18y' - 4z' - 16 = 0$$

$$x'^2 - 9y'^2 - 4z'^2 + 36 = 0$$

(6)

Ex. 8.8

(50)

$$x^2 + y^2 + z^2 - 2xz + 5z - 4 = 0$$

Trace in

Trace in xy -plane

put $z = 0$

$$x^2 + y^2 - 4 = 0$$

$$x^2 + y^2 = 4$$

$$x^2 + y^2 = 2^2$$

$$(x-0)^2 + (y-0)^2 = 2^2$$

which is a circle with

center at $(0, 0)$ and

$r = 2$.

(7)

$$xy + yz + zx = 1$$

Trace in xz -plane

put $y = 0$

$$zx = 1$$

which is rectangular hyperbola



$$(8) \quad x^2 + 4y^2 + z^2 + 4xy - 2xz - 2x - 4y + z + 1 = 0$$

Trace in xy -plane

put $z = 0$

$$x^2 + 4y^2 + 4xy - 2x - 4y + 1 = 0$$

$$(x + 2y - 1)^2 = 0$$

which is pair of coincident lines.

(9)

$$x^2 + xy - 3xz - 2 = 0$$

Trace in xz -plane

put $y = 0$

$$x^2 - 2 = 0 \quad (\text{meaningless})$$

No trace

(5)

$$x^2 + 4y^2 + 5xz - 2x + y - 3 = 0$$

InterceptsIntercepts on x -axis :- Putting $y = z = 0$

we have

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\Rightarrow x = 3, -1$$

Intercepts on y -axis :- Putting $x = z = 0$

we have

$$4y^2 + y - 3 = 0$$

$$(4y+3)(y-1) = 0$$

$$\Rightarrow y = -\frac{3}{4}, 1$$

Intercepts on z -axis :- Putting $x = y = 0$

$$-3 = 0$$

No intercept on z -axis

(6)

$$2x^2 - z^2 + xy - 8yz + y - x - 2 = 0$$

InterceptsIntercepts on x -axis :- Putting $y = z = 0$

$$2x^2 - 2 = 0$$

$$2x^2 = 2$$

$$x^2 = 1$$

$$x = \pm 1$$

Intercepts on y -axis :- Putting $x = z = 0$

$$y - 2 = 0$$

$$y = 2$$

Intercepts on z -axis :- Putting $x = y = 0$

$$-z^2 - z - 2 = 0$$

$$z^2 + z + 2 = 0$$

which gives imaginary roots

Hence no-intercepts on z -axis

EX. 8.9

(7)

$$x^2 + 2y^2 = 8$$

$$x = 0$$

As $x = 0$, so the curve is in xy -planeis To obtain surface of revolution about x -axiswe have to replace y^2 by $y^2 + z^2$

Then Eq of Surface of revolution

$$x^2 + 2(y^2 + z^2) = 8$$

is To obtain surface of revolution about y -axiswe have to replace x^2 by $x^2 + z^2$

So Eq of Surface of revolution

$$x^2 + z^2 + 2y^2 = 8$$

(8)

$$4x^2 - 9z^2 = 5$$

$$y = 0$$

As $y = 0$, so the curve is in xz -planeis To obtain surface of revolution about x -axiswe have to replace z^2 by $z^2 + y^2$

Then Eq of Surface of revolution is

$$4x^2 - 9(z^2 + y^2) = 5$$

is To obtain surface of revolution about z -axiswe have to replace x^2 by $x^2 + y^2$

Then Eq of Surface of revolution is

$$4(x^2 + y^2) - 9z^2 = 5$$

$$6y^2 + 6z^2 = 7 \quad , \quad x=0$$

As $x=0$, so the curve is in yz -plane

To obtain surface of revolution about y -axis

we have to replace z^2 by $z^2 + x^2$

then Eq of surface of revolution

$$6y^2 + 6(z^2 + x^2) = 7$$

To obtain surface of revolution about x -axis

we have to replace y^2 by $y^2 + x^2$

then Eq of surface of revolution

$$6(y^2 + x^2) + 6z^2 = 7$$

$$2x + 3y = 6 \quad , \quad z=0$$

As $z=0$, so the curve is in xy -plane

To obtain surface of revolution about x -axis

we have to replace y^2 by $y^2 + z^2$

or y by $\sqrt{y^2 + z^2}$

then Eq of surface of revolution

$$2x + 3\sqrt{y^2 + z^2} = 6$$

To obtain surface of revolution about y -axis

we have to replace x^2 by $x^2 + z^2$

or x by $\sqrt{x^2 + z^2}$

then Eq of surface of revolution

$$2\sqrt{x^2 + z^2} + 3y = 6$$

$$y=2$$

As $x=0$, so the curve is in yz -plane

To obtain surface of revolution about y -axis

we have to replace z^2 by $z^2 + x^2$

so Eq of surface of revolution

$$y=2$$

To obtain surface of revolution about x -axis

we have to replace y^2 by $y^2 + x^2$

or y by $\sqrt{y^2 + x^2}$

so Eq of surface of revolution

$$\sqrt{y^2 + x^2} = 2$$

(i)

$$x^2 + y^2 + z = 2$$

① xz -plane

In xz -plane, $y=0$

$$\text{so } x^2 + z = 2$$

②

when ② revolve about x -axis

① will generate

Eq of generatrix is

$$x^2 + z = 2$$

①

$$x^2 - 4y^2 - 4z^2 = 8$$

① xy -plane

In xy -plane, $z=0$

$$x^2 - 4y^2 = 8$$

②

when ② revolve about x -axis

① will generate

Eq of generatrix is

$$x^2 - 4y^2 = 8$$

②

$$x^2 y^2 + y^2 z^2 = 1$$

① yz -plane

In yz -plane, $x=0$

$$\text{so } y^2 z^2 = 1$$

$$yz = 1$$

②

when ② revolve about y -axis

① will generate

Eq of generatrix is

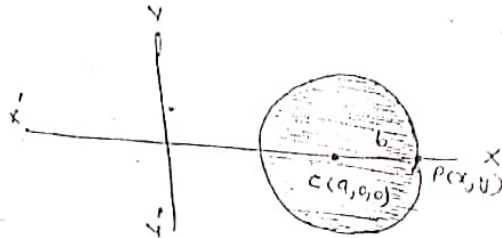
$$yz = 1$$



Torus

A surface generated by the rotation of a circle about an axis in its plane.

(10)



The shaded region revolve about y-axis to generate the torus.

Eq of circle

$$(x-a)^2 + (y-0)^2 = b^2$$

$$(x-a)^2 + y^2 = b^2$$

which revolve about y-axis

x^2 replace by $x^2 + z^2$

or x by $\sqrt{x^2 + z^2}$

$$(\sqrt{x^2 + z^2} - a)^2 + y^2 = b^2$$

$$x^2 + z^2 + a^2 - 2a\sqrt{x^2 + z^2} + y^2 = b^2$$

$$x^2 + y^2 + z^2 + a^2 - b^2 = 2a\sqrt{x^2 + z^2}$$

Eq of torus.

①

$$\rho = c$$

$$\rho^2 = c^2$$

$$x^2 + y^2 + z^2 = c^2$$

$$(x-0)^2 + (y-0)^2 + (z-0)^2 = c^2$$

which is Eq. of sphere with centre $(0, 0, 0)$ and radius as c .

②

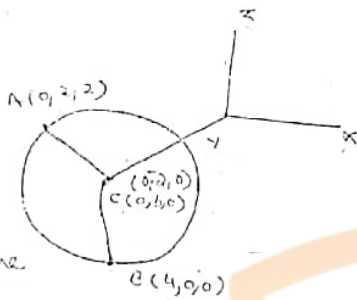
Let $C(0, b, 0)$ be the centre of sphere.

passes

through

 $(2, 2)$ $A(0, 2, 2)$

$C(0, b, 0)$ is the centre of sphere from Fig



$$|AC| = |BC|$$

$$(0)^2 + (2-b)^2 + (2-0)^2 = \sqrt{(4-0)^2 + (0-b)^2 + (0-0)^2}$$

$$\sqrt{(2-b)^2 + 4} = \sqrt{16 + b^2}$$

Squaring

$$4 + b^2 - 4b + 4 = 16 + b^2$$

$$-4b = 16 - 4 - 4$$

$$-4b = 8$$

$$b = -2$$

Therefore, the centre is $(0, -2, 0)$

and radius r is

$$r = |AC| = \sqrt{(0-0)^2 + (2+2)^2 + (0-2)^2}$$

$$= \sqrt{16 + 4}$$

$$= \sqrt{20}$$

So the Equation of sphere is

$$x^2 + y^2 + z^2 + (y+2)^2 + (z-0)^2 = (\sqrt{20})^2$$

$$x^2 + y^2 + z^2 + y^2 + 4y + 4 + z^2 = 20$$

$$x^2 + y^2 + z^2 + 4y - 16 = 0$$

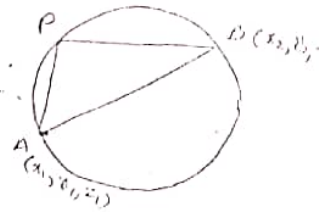
③

Let $P(x, y, z)$

be a point on the sphere

and $A(x_1, y_1, z_1)$

and $B(x_2, y_2, z_2)$



be points on

diameter. By elementary geometry

An angle described on diameter always an right angled.

i.e. $PA \perp PB$

The DR of PA are $x-x_1, y-y_1, z-z_1$

The DR of PB are $x-x_2, y-y_2, z-z_2$

By condition of perpendicularity

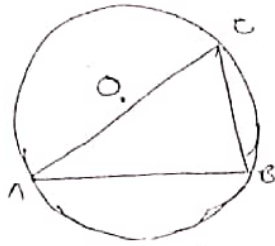
$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2)$$

$$+ (z-z_1)(z-z_2) = 0$$

which is required condition of sphere.

(4)

It is given that the sphere passes through the points A(-3, 6, 0), B(-2, -5, -1) and C(1, 4, 2)



which are the vertices of right angle triangle.

Therefore, the centre of sphere lies on the hypotenuse of the triangle ABC with right angle at B.

AC is diameter of sphere.

$$O = \text{mid pt of } AC = \left(\frac{-3+1}{2}, \frac{6+4}{2}, \frac{0+2}{2} \right) = (-1, 5, 1)$$

$$r = OA = \text{radius} = \sqrt{(-1+3)^2 + (5-6)^2 + (1-0)^2} = \sqrt{4+1+1} = \sqrt{6}$$

Therefore eq. of sphere is

$$(x+1)^2 + (y-5)^2 + (z-1)^2 = (\sqrt{6})^2$$

$$x^2 + 2x + 1 + y^2 + 25 - 10y + z^2 + 1 - 2z = 6$$

$$x^2 + y^2 + z^2 + 2x - 10y - 2z + 21 = 0$$

(5) (a)

$$x^2 + y^2 + z^2 - 6x + 4z = 0$$

Rearranging the terms

$$x^2 - 6x + y^2 + z^2 + 4z = 0$$

Completing the square of each variable, we get

$$x^2 - 6x + 9 + y^2 + z^2 + 4z + 4 = 9 + 4$$

$$(x-3)^2 + y^2 + (z+2)^2 = 13$$

which is the sphere with centre at (3, 0, -2)

$$\text{and } r = \sqrt{13}$$

(6)

$$x^2 + y^2 + z^2 + 2x - 4y - 6z + 5 = 0$$

$$x^2 + y^2 + z^2 + 2gx + 2fy + 2hz + d = 0$$

Comparing

$$d = 5$$

$$2g = 2, \quad 2f = -4, \quad 2h = -6,$$

$$g = 1, \quad f = -2, \quad h = -3$$

$$\text{Centre } (-g, -f, -h)$$

$$= (-1, 2, 3)$$

$$r = \sqrt{g^2 + f^2 + h^2 - d}$$

$$= \sqrt{1 + 4 + 9 - 5}$$

$$= \sqrt{9}$$

$$= 3$$

$$(7) 4x^2 + 4y^2 + 4z^2 - 4x + 8y + 24z + 1 = 0$$

Dividing by 4

$$x^2 + y^2 + z^2 - x + 2y + 6z + \frac{1}{4} = 0$$

$$x^2 + y^2 + z^2 + 2gx + 2fy + 2hz + d = 0$$

Comparing

$$2g = -1, \quad 2f = 2, \quad 2h = 6, \quad d = \frac{1}{4}$$

$$g = -\frac{1}{2}, \quad f = 1, \quad h = 3$$

$$\text{Centre } (-g, -f, -h) = \left(\frac{1}{2}, -1, -3 \right)$$

$$r = \sqrt{g^2 + f^2 + h^2 - d} = \sqrt{\frac{1}{4} + 1 + 9 - \frac{1}{4}}$$

$$r = \sqrt{10}$$

Let the Equation of the sphere is
 $x^2 + y^2 + z^2 + 2gx + 2fy + 2hz + d = 0$ — (1)
 Since pts $(0, 0, 0)$, $(0, 1, -1)$, $(-1, 2, 0)$ lie on the sphere
 so they must satisfy the equation

Put in (1)

$$\begin{aligned} d &= 0 \\ 2f - 2h &= -2 \quad \text{using } d=0 \quad \text{--- (2)} \\ -2g + 4f &= -5 \quad \text{--- (3)} \\ 2g + 4f + 6h &= -14 \quad \text{--- (4)} \end{aligned}$$

Solve (3) & (4)

$$\begin{aligned} -2g + 4f &= -5 \\ 2g + 4f + 6h &= -14 \end{aligned}$$

$$8f + 6h = -19$$

Solve (2) & (5)

$$\begin{aligned} 4 \times 2f - 2h &= -2 \\ 8f + 6h &= -19 \end{aligned}$$

$$8f - 8h = -8$$

$$8f + 6h = -19$$

$$-14h = 11$$

$$h = -\frac{11}{14}$$

Put in (2)

$$2f - 2h = -2$$

$$2f - 2\left(-\frac{11}{14}\right) = -2$$

$$2f + \frac{22}{14} = -2$$

$$2f = -2 - \frac{22}{14}$$

$$= \frac{-28 - 22}{14}$$

$$2f = -\frac{50}{14}$$

$$f = -\frac{25}{14}$$

Put

$$f = -\frac{25}{14} \text{ in (3)}$$

$$-2g + 4f = -5$$

$$-2g + 4\left(-\frac{25}{14}\right) = -5$$

$$-2g - \frac{100}{14} = -5$$

$$-2g = -5 + \frac{100}{14}$$

$$= \frac{-70 + 100}{14}$$

$$-2g = \frac{30}{14}$$

$$g = -\frac{15}{14}$$

Centre $(-g, -f, -h)$
 $= \left(\frac{15}{14}, \frac{25}{14}, \frac{11}{14}\right)$

$$\begin{aligned} r &= \sqrt{g^2 + f^2 + h^2} \\ &= \sqrt{\left(\frac{15}{14}\right)^2 + \left(\frac{25}{14}\right)^2 + \left(\frac{11}{14}\right)^2} \\ &= \frac{\sqrt{971}}{14} \end{aligned}$$

Equation of sphere is

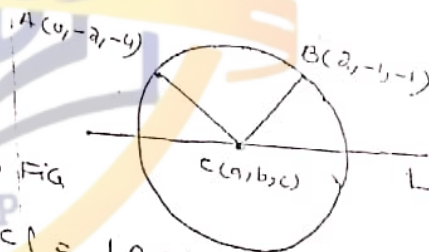
$$\begin{aligned} x^2 + y^2 + z^2 + 2\left(-\frac{15}{14}\right)x + 2\left(-\frac{25}{14}\right)y \\ + 2\left(-\frac{11}{14}\right)z + 0 = 0 \end{aligned}$$

$$x^2 + y^2 + z^2 - \frac{30}{14}x - \frac{50}{14}y - \frac{22}{14}z = 0$$

$$14x^2 + 14y^2 + 14z^2 - 30x - 50y - 22z = 0$$

(7)

Let (a, b, c) be the centre of sphere



From Fig

$$|AC| = |BC|$$

$$\begin{aligned} \sqrt{(a-0)^2 + (b+2)^2 + (c+4)^2} \\ = \sqrt{(a-2)^2 + (b+1)^2 + (c+1)^2} \end{aligned}$$

Squaring both sides

$$\begin{aligned} (a-0)^2 + (b+2)^2 + (c+4)^2 &= (a-2)^2 + (b+1)^2 + (c+1)^2 \\ a^2 + b^2 + 4 + 4b + c^2 + 8c &= a^2 + 4 - 4a + b^2 + 1 \\ &+ 2b + c^2 + 1 + 2c \end{aligned}$$

$$4a + 2b + 6c + 14 = 0$$

$$2a + b + 3c + 7 = 0$$

(8)

As Centre (a, b, c) lies on the line L

$$2x - 3y = 0$$

$$5y + 2z = 0$$

$$\text{So } 2a - 3b = 0, \quad 5b + 2c = 0$$

$$2a = 3b$$

$$a = \frac{3b}{2}$$

$$2c = -5b$$

$$c = -\frac{5b}{2}$$

Put in (1)

$$2\left(\frac{3b}{2}\right) + b + 3\left(-\frac{5b}{2}\right) + 7 = 0$$

$$6b + 2b - 15b + 14 = 0$$

$$-7b + 14 = 0$$

$$-7b = -14$$

$$b = 2$$

$$\text{So } a = 3$$

$$c = -5$$

$$\text{Centre } (3, 2, -5)$$

$$(x_1, y_1, z_1)$$

$$r = \text{radius} = \sqrt{(3-0)^2 + (2+2)^2 + (-5+4)^2}$$

$$= \sqrt{9 + 16 + 1}$$

$$r = \sqrt{26}$$

Eq of sphere is

$$(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2 = r^2$$

$$(x-3)^2 + (y-2)^2 + (z+5)^2 = (\sqrt{26})^2$$

$$x^2 + 9 - 6x + y^2 + 4 - 4y + z^2 + 25 + 10z = 26$$

$$x^2 + y^2 + z^2 - 6x - 4y + 10z + 12 = 0$$

$$c: x^2 + y^2 + z^2 - 9 = 0, \quad 2x + 3y + 4z - 5 = 0 \quad (2)$$

(9)

The given circle is intersection of the sphere and plane

Therefore the Equation of the sphere passes through this

given circle will be

$$S + kP = 0$$

$$(x^2 + y^2 + z^2 - 9) + k(2x + 3y + 4z - 5) = 0$$

It passes through $(1, 2, 3)$

then

$$(1+4+9-9) + k(2+6+12-5) = 0$$

$$5 + 15k = 0$$

$$15k = -5$$

$$k = -\frac{5}{15}$$

$$k = -\frac{1}{3}$$

Hence, Eq of sphere is

$$(x^2 + y^2 + z^2 - 9) - \frac{1}{3}(2x + 3y + 4z - 5) = 0$$

$$3(x^2 + y^2 + z^2 - 9) - (2x + 3y + 4z - 5) = 0$$

$$3x^2 + 3y^2 + 3z^2 - 27 - 2x - 3y - 4z + 5 = 0$$

$$3x^2 + 3y^2 + 3z^2 - 2x - 3y - 4z - 22 = 0$$

$$C: x^2 + y^2 + z^2 - 1 = 0 \Rightarrow 2x + 4y + 5z - 6 = 0$$

Equation of the sphere through the given circle is

$$S + kP = 0$$

$$(x^2 + y^2 + z^2 - 1) + k(2x + 4y + 5z - 6) = 0 \quad \text{--- (1)}$$

$$x^2 + y^2 + z^2 - 1 + 2kx + 4ky + 5kz - 6k = 0$$

$$x^2 + y^2 + z^2 + 2kx + 4ky + 5kz - 6k - 1 = 0$$

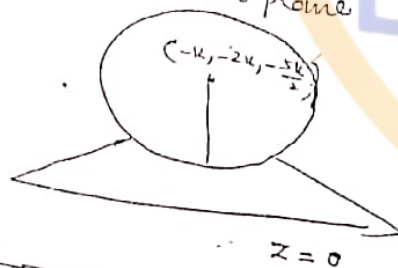
Center of sphere $(-k, -2k, -\frac{5k}{2})$

$$\text{Radius of sphere} = \sqrt{k^2 + 4k^2 + \frac{25k^2}{4} + 6k + 1}$$

This sphere touches the plane $z = 0$

The z -coordinate of the center of the sphere will be equal to the radius of sphere

i.e. Radius = Distance of the Center from the plane



$$\sqrt{k^2 + 4k^2 + \frac{25k^2}{4} + 6k + 1} = \left| -\frac{5k}{2} \right|$$

Squaring both sides

$$5k^2 + \frac{25k^2}{4} + 6k + 1 = \frac{25}{4}k^2$$

$$5k^2 + 6k + 1 = 0$$

$$(5k+1)(k+1) = 0$$

$$5k+1=0$$

$$5k=-1$$

$$k = -\frac{1}{5}$$

$$k+1=0$$

$$k=-1$$

Hence the Equation of sphere when $k=-1$

Put in (1)

$$x^2 + y^2 + z^2 - 1 - 2x - 4y - 5z + 6 = 0$$

$$x^2 + y^2 + z^2 - 2x - 4y - 5z + 5 = 0$$

if when $k = -\frac{1}{5}$

Put in (1)

$$(x^2 + y^2 + z^2 - 1) - \frac{1}{5}(2x + 4y + 5z - 6) = 0$$

$$5(x^2 + y^2 + z^2 - 1) - (2x + 4y + 5z - 6) = 0$$

$$5x^2 + 5y^2 + 5z^2 - 5 - 2x - 4y - 5z + 6 = 0$$

$$5x^2 + 5y^2 + 5z^2 - 2x - 4y - 5z + 1 = 0$$

Given circles are

$$C_1: x^2 + y^2 + z^2 - 9 = 0, x - 2y + 4z - 13 = 0$$

$$C_2: x^2 + y^2 + z^2 + 6y - 6z + 21 = 0, x + y + z + 2 = 0$$

Equation of the sphere through circle C_1

$$(x^2 + y^2 + z^2 - 9) + k(x - 2y + 4z - 13) = 0$$

$$x^2 + y^2 + z^2 - 9 + kx - 2ky + 4kz - 13k = 0$$

$$x^2 + y^2 + z^2 + kx - 2ky + 4kz - 13k - 9 = 0 \quad \text{--- (1)}$$

Equation of the sphere through circle C_2

$$S_1 + k'P_1 = 0$$

$$(x^2 + y^2 + z^2 + 6y - 6z + 21) + k'(x + y + z + 2) = 0$$

$$x^2 + y^2 + z^2 + 6y - 6z + 21 + k'x + k'y + k'z + 2k' = 0$$

$$x^2 + y^2 + z^2 + k'x + (6+k')y + (k'-6)z + 2k'+21 = 0 \quad \text{--- (2)}$$

The given circle will be on the same sphere if the equations ① & ② are identical.

Comparing ① & ②

$$K = K', \quad -2K = -6 + K', \quad 4K = -6 + K'$$

from (i) & (ii)

$$4K = -6 + K$$

$$4K - K = -6$$

$$3K = -6$$

$$K = -2$$

from (i) $K' = K = -2$

This represents that given circle lie on the same sphere.

Equation of sphere on which two circles lie is

Put $K = -2$ or $K' = -2$ in ①
or ②

$$x^2 + y^2 + z^2 - 2x + 4y - 8z + 21 - 4 = 0$$

$$x^2 + y^2 + z^2 - 2x + 4y - 8z + 17 = 0$$

$$\textcircled{11} \quad \begin{cases} x^2 + y^2 + z^2 + 7y - 2z + 2 = 0 \\ 2x + 3y + 4z - 8 = 0 \end{cases} \quad \textcircled{67}$$

Equation of the sphere passing through the given circle is

$$(x^2 + y^2 + z^2 + 7y - 2z + 2) + K(2x + 3y + 4z - 8) = 0 \quad \textcircled{1}$$

$$x^2 + y^2 + z^2 + 7y - 2z + 2 + 2Kx + 3Ky + 4Kz - 8K = 0$$

$$x^2 + y^2 + z^2 + 2Kx + 3Ky + 7y + 4Kz - 2z - 8K + 2 = 0$$

$$x^2 + y^2 + z^2 + 2Kx + (3K + 7)y + (4K - 2)z - 8K + 2 = 0$$

Centre of sphere is

$$\left(-K, -\frac{7+3K}{2}, -\frac{4K-2}{2} \right)$$

The given circle is on great circle of sphere & so the centre of sphere lies on the plane

$$2x + 3y + 4z - 8 = 0$$

$$2(-K) + 3\left(-\frac{7+3K}{2}\right) + 4\left(-\frac{4K-2}{2}\right) - 8 = 0$$

$$-4K - 21 - 9K - 16K + 8 = 0$$

$$-29K - 29 = 0$$

$$-29K = 29$$

$$K = -1$$

Put in ①

$$(x^2 + y^2 + z^2 + 7y - 2z + 2) - (2x + 3y + 4z - 8) = 0$$

$$x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0$$

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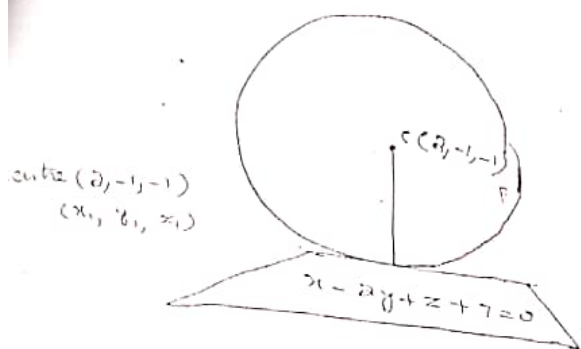
7430831

انور فوٹو سٹیب
نورنگر ہاسٹل 511-B سٹاٹس ٹاؤن کوجرانوالہ

Cell: 0321

7430831

انور فوٹو سٹیب
نورنگر ہاسٹل 511-B سٹاٹس ٹاؤن کوجرانوالہ



$r =$ Distance of the centre from the plane

$$= \frac{|2 - 2(-1) - 1 + 7|}{\sqrt{(1)^2 + (-2)^2 + (1)^2}}$$

$$r = \frac{10}{\sqrt{6}}$$

Equation of sphere

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

$$x - 2)^2 + (y + 1)^2 + (z + 1)^2 = \left(\frac{10}{\sqrt{6}}\right)^2$$

$$x^2 - 4x + y^2 + 2y + z^2 + 2z + 6 = \frac{100}{6}$$

$$x^2 + y^2 + z^2 - 4x + 2y + 2z + 6 = \frac{100}{6}$$

$$x^2 + y^2 + z^2 - 4x + 2y + 2z = \frac{100}{6} - 6$$

$$= \frac{50 - 18}{3}$$

$$x^2 + y^2 + z^2 - 4x + 2y + 2z = \frac{32}{3}$$

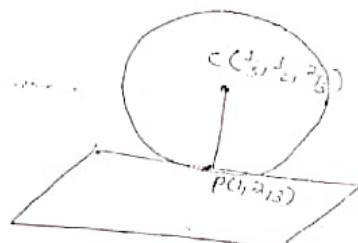
$$3x^2 + 3y^2 + 3z^2 - 12x + 6y + 6z = 32$$

(13)

$$3(x^2 + y^2 + z^2) - 2x - 3y - 4z - 22 = 0$$

$$3x^2 + y^2 + z^2 - \frac{2}{3}x - y - \frac{4}{3}z - \frac{22}{3} = 0$$

Centre of sphere $\left(\frac{1}{3}, \frac{1}{2}, \frac{2}{3}\right)$



$$DR \text{ of } CP = r = \left(\frac{2}{3}, \frac{3}{2}, \frac{1}{3}\right)$$

Equation of tangent plane to the sphere is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$\frac{2}{3}(x - \frac{1}{3}) + \frac{3}{2}(y - \frac{1}{2}) + \frac{1}{3}(z - \frac{2}{3}) = 0$$

$$4(x - 1) + 9(y - 2) + 14(z - 3) = 0$$

$$4x + 9y + 14z - 64 = 0$$

(14) (9)

Centre $(-2, 4, -6)$
 (x_1, y_1, z_1)

$r =$ Distance of the centre from the plane

$$r = 1 - 61$$

$$r = 6$$

Eq. of sphere

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

$$(x + 2)^2 + (y - 4)^2 + (z + 6)^2 = 6^2$$

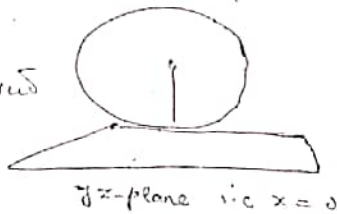
$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 4 + 16 + 36 = 36$$

$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 20 = 0$$

(6)

Centre $(-2, 4, -6)$
 (x_1, y_1, z_1)

$r =$ Distance of the
Centre from
the plane



xy -plane i.e. $z=0$

$$r = \frac{|-6|}{1}$$

$$r = 6$$

Eq of sphere is

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

$$(x + 2)^2 + (y - 4)^2 + (z + 6)^2 = 6^2$$

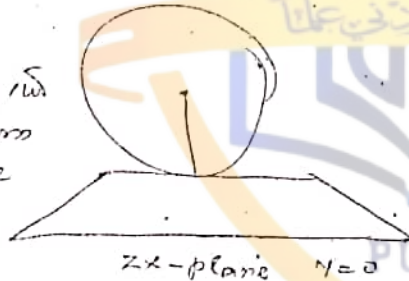
$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 4 + 36 + 36 = 36$$

$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 40 = 0$$

(7)

Centre $(-2, 4, -6)$
 (x_1, y_1, z_1)

$r =$ Distance of the
Centre from
the plane



xz -plane $y=0$

$$r = 4$$

Equation of sphere is

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

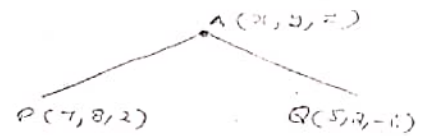
$$(x + 2)^2 + (y - 4)^2 + (z + 6)^2 = 4^2$$

$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 4 + 16 + 36 = 16$$

$$x^2 + y^2 + z^2 + 4x - 8y + 12z + 40 = 0$$

(15)

(6.8)



Let $A(x, y, z)$ be any pt on
the surface, then by given

condition

$$|AP| = |AQ|$$

$$\sqrt{(x-7)^2 + (y-8)^2 + (z-2)^2} = \sqrt{(x-5)^2 + (y-2)^2 + (z+1)^2}$$

Squaring both sides

$$(x-7)^2 + (y-8)^2 + (z-2)^2 = (x-5)^2 + (y-2)^2 + (z+1)^2$$

$\Rightarrow$

$$x^2 + 49 - 14x + y^2 + 64 - 16y + z^2 + 4 - 4z = x^2 + 25 - 10x + y^2 + 4 - 4y + z^2 + 36 + 12z$$

$$-14x + 10y - 16y + 4y - 4z - 12z = 0$$

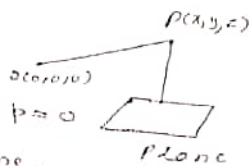
$$49 + 64 + 4 - 4 - 25 - 36 = 0$$

$$\Rightarrow -4x - 12y - 16z + 52 = 0$$

$$x + 3y + 4z - 13 = 0$$

(16)

Let $lx + my + nz - p = 0$
be a fixed plane.



Let $P(x, y, z)$ moves according
to the given condition

i.e
 $|OP|^2 \propto$ the distance of P to
 $lx + my + nz - p = 0$

$|OP|^2 = k$ (the distance of P to
 $lx + my + nz - p = 0$)

$$x^2 + y^2 + z^2 = k \frac{|lx + my + nz - p|}{\sqrt{l^2 + m^2 + n^2}}$$

$$\therefore (l^2 + m^2 + n^2 = 1)$$

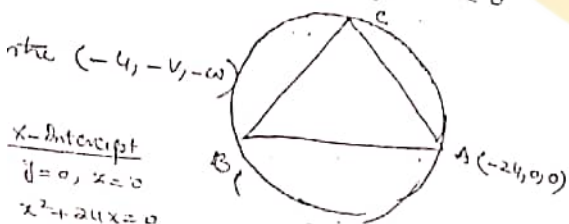
$$x^2 + y^2 + z^2 - klx - kmy - knz + kp = 0$$

This equation represents a
sphere. Hence the locus
of P is a sphere.

i)

Let the Equation of the
sphere passing through the
origin is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$$



x-Intercept

$$y = 0, z = 0$$

$$x^2 + 2ux = 0$$

$$x^2 = -2ux$$

$$x = -2u$$

$$\therefore A(-2u, 0, 0)$$

y-Intercept

$$x = 0, z = 0$$

$$y^2 + 2vy = 0$$

$$y^2 = -2vy$$

$$y = -2v$$

$$\therefore B(0, -2v, 0)$$

(17)

For z-Intercept

$$x = 0, y = 0$$

$$z^2 + 2wz = 0$$

$$z^2 = -2wz$$

$$z = -2w$$

$$\therefore C(0, 0, -2w)$$

radius of sphere is k .

$$\sqrt{4u^2 + 4v^2 + 4w^2} = k$$

$$4u^2 + 4v^2 + 4w^2 = k^2$$

①

Now the centroid of

~~triangle~~ triangle is

$$\left(\frac{-2u}{3}, \frac{-2v}{3}, \frac{-2w}{3} \right)$$

We show that Centroid of $\triangle ABC$

lies on the sphere

$$9(x^2 + y^2 + z^2) = 4k^2$$

LHS

$$9(x^2 + y^2 + z^2)$$

$$9\left(\frac{4u^2}{9} + \frac{4v^2}{9} + \frac{4w^2}{9}\right)$$

$$\Rightarrow 4u^2 + 4v^2 + 4w^2$$

$$\Rightarrow 4(u^2 + v^2 + w^2)$$

$$\Rightarrow 4k^2$$

$$\Rightarrow \text{RHS}$$

from ①

Hence the centroid of

the triangle lies on the

$$\text{Sphere } 9(x^2 + y^2 + z^2) = 4k^2$$

(18)

As the plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

meets the axes in A, B, C .

Therefore

$$A = (a, 0, 0)$$

$$B = (0, b, 0)$$

$$C = (0, 0, c)$$

Equation of the sphere through origin

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$$

It passes through A, B, C — (1)

$$\begin{cases} a^2 + 2ua = 0 \\ a^2 = -2ua \\ a = -2u \\ u = -\frac{a}{2} \end{cases} \quad \begin{cases} b^2 + 2vb = 0 \\ b^2 = -2vb \\ b = -2v \\ v = -\frac{b}{2} \end{cases} \quad \begin{cases} c^2 + 2wc = 0 \\ c^2 = -2wc \\ c = -2w \\ w = -\frac{c}{2} \end{cases}$$

① becomes

$$x^2 + y^2 + z^2 + 2\left(-\frac{a}{2}\right)x + 2\left(-\frac{b}{2}\right)y + 2\left(-\frac{c}{2}\right)z = 0$$

$$x^2 + y^2 + z^2 - ax - by - cz = 0$$

$$\text{Centre } M \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2} \right)$$

The Equation of Circumcircle is

$$x^2 + y^2 + z^2 - ax - by - cz = 0$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

N is foot of perpendicular from the centre of sphere M to the plane of the circumcircle. So NM is Normal to the plane of the circle.

Eq of NM is

$$\frac{x - \frac{a}{2}}{\frac{1}{a}} = \frac{y - \frac{b}{2}}{\frac{1}{b}} = \frac{z - \frac{c}{2}}{\frac{1}{c}} = t$$

$$x = \frac{a}{2} + \frac{t}{a}, \quad y = \frac{b}{2} + \frac{t}{b}, \quad z = \frac{c}{2} + \frac{t}{c}$$

are the coordinates of N

(19)

Let the Equation of plane be

$$lx + my + nz = p \quad \text{--- (1)}$$

It passes through (a, b, c)

$$la + mb + nc = p \quad \text{--- (2)}$$

It meets x -axis

$$\text{So } y = 0 \\ z = 0$$

$$lx = p$$

$$x = \frac{p}{l}$$

$$A \left(\frac{p}{l}, 0, 0 \right)$$

It meets y -axis

$$\text{So } x = 0$$

$$z = 0$$

$$my = p$$

$$y = \frac{p}{m}$$

$$B \left(0, \frac{p}{m}, 0 \right)$$

It meets z -axis

$$\text{So } x = 0$$

$$y = 0$$

$$nz = p$$

$$z = \frac{p}{n}$$

$$\text{So } C \left(0, 0, \frac{p}{n} \right)$$

Equation of sphere through origin i.e. $d = 0$

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$$

It passes through A, B, C

$$\frac{p^2}{l^2} + 2u \frac{p}{l} = 0$$

$$\Rightarrow u = -\frac{p}{2l}$$

$$\frac{p^2}{m^2} + 2v \frac{p}{m} = 0$$

$$\Rightarrow v = -\frac{p}{2m}$$

$$\frac{p^2}{n^2} + 2w \frac{p}{n} = 0$$

$$\Rightarrow w = -\frac{p}{2n}$$

Eq of sphere is

$$x^2 + y^2 + z^2 - \frac{p}{l}x - \frac{p}{m}y - \frac{p}{n}z = 0$$

centre $(\frac{p}{2l}, \frac{p}{2m}, \frac{p}{2n})$

to find the locus of centre

put

$$x = \frac{p}{2l}, \quad y = \frac{p}{2m}, \quad z = \frac{p}{2n}$$

$$l = \frac{p}{2x}, \quad m = \frac{p}{2y}, \quad n = \frac{p}{2z}$$

put in (1)

$$\frac{p}{2x} \cdot a + \frac{p}{2y} \cdot b + \frac{p}{2z} \cdot c = p$$

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$$

therefore, the equation of sphere becomes

$$x^2 + y^2 + z^2 + \frac{p}{l}x + \frac{p}{m}y + \frac{p}{n}z = 0$$

(2)

The planes which form the tetrahedron are

$x = 0$, the plane ABC

$y = 0$, the plane ACD

$z = 0$, the plane ABD

$lx + my + nz + p = 0$, the plane BCD



Solving these equations we

get $A(0,0,0)$

$B(p/l, 0, 0)$

$C(0, p/m, 0)$

$D(0, 0, p/n)$

Suppose the equation of the sphere circumscribing the tetrahedron

i.e. sphere through A, B, C, D

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$

it passes through A, B, C, D

$\Rightarrow d = 0$ using

$\Rightarrow \frac{p^2}{l^2} + 2 \frac{p}{l} u = 0 \Rightarrow 2u = -\frac{p}{l}$

$\Rightarrow \frac{p^2}{m^2} + 2 \frac{p}{m} v = 0 \Rightarrow 2v = -\frac{p}{m}$

$$w \quad x_1 = \frac{3x}{z}, \quad y_1 = \frac{3y}{z}$$

Put x_1 & y_1 in (1),

$$4 \left(\frac{3x}{z} \right)^2 + \left(\frac{3y}{z} - 2 \right)^2 = 4$$

$$4 \cdot \frac{9x^2}{z^2} + \frac{(3y-2z)^2}{z^2} = 4$$

$$w \quad 36x^2 + (3y-2z)^2 = 4z^2$$

(16) Directrix: $x^2 + 4y^2 - 2x + 8y - 4 = 0, z=3$

Vertex $(-1, 2, 1)$

Sol:- Let $P(x_1, y_1, z_1)$ be a point on the req. cone. Let line AP be the element of the cone intersects the directrix at $P(x_1, y_1, z_1)$

$$\therefore x_1^2 + 4y_1^2 - 2x_1 + 8y_1 - 4 = 0, z_1 = 3$$

The Eq of line AP is

$$\frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{z_1-1}$$

Since $z_1 = 3$

$$So \quad \frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{3-1}$$

$$\frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{2}$$

$$\frac{x+1}{x_1+1} = \frac{z-1}{2}$$

$$\frac{y-2}{y_1-2} = \frac{z-1}{2}$$

$$(x_1+1)(z-1) = 2(x+1), \quad (y_1-2)(z-1) = 2(y-2)$$

$$x_1+1 = \frac{2(x+1)}{z-1}$$

$$y_1-2 = \frac{2(y-2)}{z-1}$$

$$x_1 = \frac{2(x+1)}{z-1} - 1$$

$$y_1 = \frac{2(y-2)}{z-1} + 2$$

$$x_1 = \frac{2(x+1)-(z-1)}{z-1}$$

$$= \frac{2(y-2)+2(z-1)}{(z-1)}$$

$$= \frac{2x+2-z+1}{z-1}$$

$$= \frac{2y-4+2z-2}{(z-1)}$$

$$x_1 = \frac{2x-z+3}{z-1}$$

$$y_1 = \frac{2y+2z-6}{(z-1)}$$

Put x_1 & y_1 in (1)

$$\frac{(2x-z+3)^2}{(z-1)^2} + 4 \frac{(2y+2z-6)^2}{(z-1)^2} - 2 \frac{(2x-z+3)}{z-1} + 8 \frac{(2y+2z-6)}{z-1} - 4 = 0$$

$$-4 = 0$$

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(17) Directrix: $2x^2 + 3y^2 = 1, z=0$

Vertex $(3, 1, 2)$

Sol:- Let $P(x_1, y_1, z_1)$ be a point on the req. cone. Let line AP be the element of the cone intersects the directrix at $P(x_1, y_1, z_1)$.

$$\therefore 2x_1^2 + 3y_1^2 - 1 = 0, z_1 = 0$$

The Eq of line AP is

$$\frac{x-3}{x_1-3} = \frac{y-1}{y_1-1} = \frac{z-2}{z_1-2}$$

Since $z_1 = 0$

$$\frac{x-3}{x_1-3} = \frac{y-1}{y_1-1} = \frac{z-2}{-2}$$

$$\frac{x-3}{x_1-3} = \frac{z-2}{-2}$$

$$\frac{y-1}{y_1-1} = \frac{z-2}{-2}$$

$$(x_1-3)(z-2) = -2(x-3)$$

$$(y_1-1)(z-2) = -2(y-1)$$

$$x_1-3 = \frac{-2(x-3)}{z-2}$$

$$y_1-1 = \frac{-2(y-1)}{(z-2)}$$

$$x_1 = \frac{-2(x-3)}{(z-2)} + 3$$

$$y_1 = \frac{-2(y-1)}{(z-2)} + 1$$

$$= \frac{-2(x-3)+3(z-2)}{z-2}$$

$$y_1 = \frac{-2(y-1)+(z-2)}{z-2}$$

$$= \frac{-2x+6+3z-6}{z-2}$$

$$y_1 = \frac{-2y+2+z-2}{(z-2)}$$

$$x_1 = \frac{3z-2x}{z-2}$$

$$y_1 = \frac{z-2y}{(z-2)}$$

Put x_1 & y_1 in (1)

$$2 \left(\frac{3z-2x}{z-2} \right)^2 + 3 \left(\frac{z-2y}{z-2} \right)^2 - 1 = 0$$

Multiplying by $(z-2)^2$

$$2(3z-2x)^2 + 3(z-2y)^2 - (z-2)^2 = 0$$

$$2(9x^2 + 4x^2 - 12xz) + 3(4y^2 + z^2 - 4yz) - (z^2 + 4 - 4z) = 0$$

$$18x^2 + 8x^2 - 24xz + 12y^2 + 3z^2 - 12yz - z^2 - 4 + 4z = 0$$

$$w \quad 8x^2 + 12y^2 + 20z^2 - 12yz - 24xz + 4z - 4 = 0$$

$$Dividing by 4$$

$$2x^2 + 3y^2 + 5z^2 - 3yz - 6xz + z - 1 = 0$$

- (18) Let $P(x_1, y_1, z_1)$ be any point on the required cylinder. Then Eq of line through $P(x_1, y_1, z_1)$ (Generator) and parallel to $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = t \quad \text{--- (1)}$$

$$\Rightarrow \frac{x-x_1}{l} = t, \quad \frac{y-y_1}{m} = t, \quad \frac{z-z_1}{n} = t$$

$$x-x_1 = lt, \quad y-y_1 = mt, \quad z-z_1 = nt$$

$$x = x_1 + lt, \quad y = y_1 + mt, \quad z = z_1 + nt$$

Any pt on line (1) is $(x_1 + lt, y_1 + mt, z_1 + nt)$

If this pt lies on the curve (directrix) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is then

$$a(x_1 + lt)^2 + 2h(x_1 + lt)(y_1 + mt) + b(y_1 + mt)^2 + 2g(x_1 + lt) + 2f(y_1 + mt) + c = 0$$

As $z = 0$, this lies on the curve if $z_1 + nt = 0$

$$nt = -z_1$$

Put in above Eq

$$\Rightarrow a(x_1 - l\frac{z_1}{n})^2 + 2h(x_1 - l\frac{z_1}{n})(y_1 - m\frac{z_1}{n}) + b(y_1 - m\frac{z_1}{n})^2 + 2g(x_1 - l\frac{z_1}{n}) + 2f(y_1 - m\frac{z_1}{n}) + c = 0$$

Multiplying by n^2

$$a(nx_1 - lz_1)^2 + 2h(nx_1 - lz_1)(ny_1 - mz_1) + b(ny_1 - mz_1)^2 + 2gn(nx_1 - lz_1) + 2fn(ny_1 - mz_1) + cn^2 = 0$$

which is Eq of req. cylinder.

- (19) Let $P(x_1, y_1, z_1)$ be any pt on req. cylinder. Now Eq of line (Generator) through $P(x_1, y_1, z_1)$ and parallel to z -axis as direction cosines of z -axis $[0, 0, 1]$ is $\frac{x-x_1}{0} = \frac{y-y_1}{0} = \frac{z-z_1}{1} = t$ --- (1)

$$\Rightarrow x = x_1$$

$$y = y_1$$

$$z = z_1 + t$$

Any pt on line (1) is $(x_1, y_1, z_1 + t)$ If this pt lies on the curve (directrix)

$$x^2 + y^2 + z^2 = 1, \quad x + y + z = 1$$

$$\text{Then } x_1^2 + y_1^2 + (z_1 + t)^2 = 1 \quad \text{--- (2)}$$

$$x_1 + y_1 + z_1 + t = 1 \quad \text{--- (3)}$$

$$\text{From (3) } z_1 + t = 1 - x_1 - y_1$$

Put in (2)

$$x_1^2 + y_1^2 + (1 - x_1 - y_1)^2 = 1$$

$$x_1^2 + y_1^2 + 1 - 2x_1 - 2y_1 + 2x_1y_1 = 1$$

$$2x_1^2 + 2y_1^2 + 2x_1y_1 - 2x_1 - 2y_1 = 0$$

$$x_1^2 + y_1^2 + x_1y_1 - x_1 - y_1 = 0$$

Therefore locus of (x_1, y_1, z_1) is

$$x^2 + y^2 + xy - x - y = 0$$

- (20) Let $P(x_1, y, z)$ be a point on the cylinder.

$A(1, 3, 5)$ is a pt on its axis.

Let L be the foot of the $\perp$ from P on its axis

Then $AL =$ Projection of AP on its axis

$$= \frac{|2(x-1) + 2(y-3) - (z-5)|}{\sqrt{2^2 + 2^2 + (-1)^2}}$$

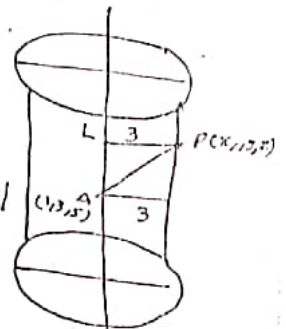
$$AL = \frac{2x + 2y - z - 3}{3}$$

From right angled triangle APL

$$(AP)^2 = (AL)^2 + (LP)^2$$

$$(\sqrt{(x-1)^2 + (y-3)^2 + (z-5)^2})^2 = \left(\frac{2x + 2y - z - 3}{3}\right)^2 + 9$$

$$\text{or } (x-1)^2 + (y-3)^2 + (z-5)^2 = \frac{(2x + 2y - z - 3)^2}{9} + 9$$



- (12) Given point $(3, 0, 5)$ lies in the xz -plane.

we have to find the circle in the xz -plane with center $(3, 5)$ and radius 2.

So this circle has Eq $(x-3)^2 + (z-5)^2 = 2^2$

this Eq represents a right circular cylinder.

- (13) Major axis: $2a=8$

Semi major axis: $a=4$

Foci $(0, \pm 3, 0)$, Directrix yz -plane $c=3$

For an Ellipse

$$a^2 = b^2 + c^2$$

$$4^2 = b^2 + 3^2$$

$$16 = b^2 + 9$$

$$16-9 = b^2$$

$$b^2 = 7$$

Thus required Eq of right Elliptic cylinder is

$$\frac{y^2}{7} + \frac{z^2}{16} = 1$$

$$\frac{y^2}{4^2} + \frac{z^2}{7} = 1$$

$$\frac{y^2}{16} + \frac{z^2}{7} = 1$$

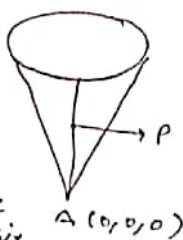
$$7y^2 + 16z^2 = 112$$

- (14) Directrix: $y^2 + z^2 = 1$, $x=2$

Vertex $A(0, 0, 0)$

Sol:-

Let $P(x_1, y_1, z_1)$ be a point on the req. cone and let line AP be the element of the cone intersects the directrix at $P(x_1, y_1, z_1)$



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$$y_1^2 + z_1^2 = 1, x_1 = 2$$

The Eq of line AP is

$$\frac{x-0}{x_1-0} = \frac{y-0}{y_1-0} = \frac{z-0}{z_1-0}$$

$$\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1}$$

Since $x_1 = 2$

$$\frac{x}{2} = \frac{y}{y_1} = \frac{z}{z_1}$$

$$\frac{x}{2} = \frac{y}{y_1}, \quad \frac{x}{2} = \frac{z}{z_1}$$

$$\frac{y_1}{y} = \frac{2}{x}, \quad \frac{z_1}{z} = \frac{2}{x}$$

$$y_1 = \frac{2y}{x}, \quad z_1 = \frac{2z}{x}$$

Put y_1 & z_1 in (1),

$$\left(\frac{2y}{x}\right)^2 + \left(\frac{2z}{x}\right)^2 = 1$$

$$\frac{4y^2}{x^2} + \frac{4z^2}{x^2} = 1$$

$$4y^2 + 4z^2 = x^2$$

- (15) Directrix: $4x^2 + (y-2)^2 = 4$, $z=3$

Sol

Vertex $A(0, 0, 0)$
Let $P(x_1, y_1, z_1)$ be a point on the req. cone and let line AP be the element of the cone intersects the directrix at $P(x_1, y_1, z_1)$.

$$4x_1^2 + (y_1-2)^2 = 4, z_1 = 3$$

The Eq of line AP is

$$\frac{x-0}{x_1-0} = \frac{y-0}{y_1-0} = \frac{z-0}{z_1-0}$$

$$\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1}$$

Since $z_1 = 3$

$$\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{3}$$

$$\frac{x}{x_1} = \frac{z}{3}, \quad \frac{y}{y_1} = \frac{z}{3}$$

$$\frac{x_1}{x} = \frac{3}{z}, \quad \frac{y_1}{y} = \frac{3}{z}$$

Exercise 8.10

- (1) Directrix $C: x^2 + y^2 = 9$, element of cylinder
 $\vec{n} = [1, -2, 1]$
Solution:- The parametric Eqs of a line L through a point (x_1, y_1, z_1) on C & parallel to $\vec{n}$

$$L: \begin{cases} x = x_1 + at = x_1 + t \\ y = y_1 + bt = y_1 - 2t \\ z = z_1 + ct = z_1 + t \end{cases}$$

So $x = x_1 + t$ — (i)
 $y = y_1 - 2t$ — (ii)
 $z = z_1 + t$ — (iii)

Put $t = z - z_1$ in (i) & (ii)

$$x = x_1 + z - z_1, \quad y = y_1 - 2z + 2z_1$$

or $x_1 = x - z + z_1, \quad y_1 = y + 2z - 2z_1$

The Equation of the directrix is $x^2 + y^2 - 9 = 0$ and

(x_1, y_1, z_1) lies on it

Hence $x_1^2 + y_1^2 - 9 = 0$

The Eq of cylinder is

$$(x - z + z_1)^2 + (y + 2z - 2z_1)^2 - 9 = 0$$

$$x^2 + z^2 - 2xz + 2xz_1 + 4z^2 + 4yz - 9 = 0$$

$$x^2 + y^2 + 5z^2 - 2xz + 4yz - 9 = 0$$

- (2) Directrix $C: x + z = 4$, element of cylinder
 $\vec{n} = [0, 2, -1]$
Sol:- The parametric Eqs of a line L through a point (x_1, y_1, z_1) on C & parallel to $\vec{n}$

$$L: \begin{cases} x = x_1 + at = x_1 \\ y = y_1 + bt = y_1 + 2t \\ z = z_1 + ct = z_1 - t \end{cases}$$

So $x = x_1$ — (i)
 $y = y_1 + 2t$ — (ii)
 $z = z_1 - t$ — (iii)

From (ii) $y = 2t$
 $t = \frac{y}{2}$

Put in (i) & (iii)

$$x = x_1, \quad z = z_1 - \frac{y}{2}$$

or $x_1 = x, \quad z_1 = z + \frac{y}{2}$

The Eq of the directrix is

$$x + z = 4 \text{ and } (x_1, y_1, z_1) \text{ lies on it.}$$

Hence $x_1 + z_1 = 4$

The Eq of cylinder is

$$x + (z + \frac{y}{2}) = 4$$

$$2x - y + 2z = 8$$

- (3) Directrix $C: \frac{x^2}{4} + \frac{y^2}{9} = 1$, element of cylinder
 $\vec{n} = [1, 1, 1]$
Sol:- The parametric Eqs of a line L through a pt (x_1, y_1, z_1) on C & parallel to $\vec{n}$

$$L: \begin{cases} x = x_1 + at = x_1 + t \\ y = y_1 + bt = y_1 + t \\ z = z_1 + ct = z_1 + t \end{cases}$$

So $x = x_1 + t$ — (i)
 $y = y_1 + t$ — (ii)
 $z = z_1 + t$ — (iii)

from (i) $t = x - x_1$ put in (ii) & (iii)

$$y = y_1 + x - x_1, \quad z = z_1 + x - x_1$$

$$y_1 = y - x + x_1, \quad z_1 = z - x + x_1$$

The Eq of the directrix is

$$\frac{x^2}{4} + \frac{y^2}{9} = 1 \text{ and } (x_1, y_1, z_1) \text{ lies on it.}$$

Hence $\frac{x_1^2}{4} + \frac{y_1^2}{9} = 1$

The Eq of cylinder is

$$\frac{(x - x_1)^2}{4} + \frac{(y - x_1)^2}{9} = 1$$

$$9(x - x_1)^2 + 4(y - x_1)^2 = 36$$

$$9(x^2 + x_1^2 - 2xx_1) + 4(y^2 + x_1^2 - 2yx_1) = 36$$

$$13x^2 + 4y^2 + 9x_1^2 - 18xx_1 - 8yx_1 = 36$$

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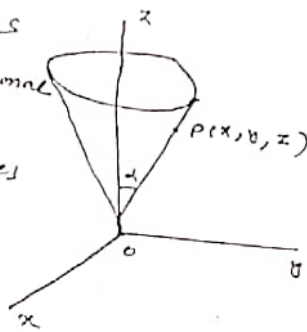
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21) Let $P(x, y, z)$ be any pt on the cone.

The direction cosines of OP are proportional to x, y, z

The direction cosines of z (z -axis) are $0, 0, 1$

If α is the semi-vertical angle of the cone, then



$$\cos \alpha = \frac{x(0) + y(0) + z(1)}{\sqrt{x^2 + y^2 + z^2} \sqrt{0+0+1}}$$

$$\cos \alpha = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\sqrt{x^2 + y^2 + z^2} = \frac{z}{\cos \alpha}$$

$$x^2 + y^2 + z^2 = \frac{z^2}{\cos^2 \alpha}$$

$$x^2 + y^2 + z^2 = z^2 \sec^2 \alpha$$

$$x^2 + y^2 = z^2 \sec^2 \alpha - z^2$$

$$x^2 + y^2 = z^2 (\sec^2 \alpha - 1)$$

$$x^2 + y^2 = z^2 \tan^2 \alpha$$

2) The general Eq of the cone of 2nd degree is

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

Since this cone passes through the co-ordinate axes.

Therefore the pts at unit distance from origin line on cone

Put $(1, 0, 0)$ in (1)

$$a = 0$$

Put $(0, 1, 0)$ in (1)

$$b = 0$$

Put $(0, 0, 1)$ in (1)

$$c = 0$$

Put a, b, c in (1), we get

$2fyz + 2gzx + 2hxy = 0$ is req. cone.

23) The Eq of line through origin is

$$\frac{x}{A} = \frac{y}{B} = \frac{z}{C} = t \text{ --- (1)}$$

$$\Rightarrow x = At, y = Bt, z = Ct$$

Any pt on line (1) is (At, Bt, Ct)

This pt lies on the directrix of req. cone which is

$$lx + my + nz = p$$

$$ax^2 + by^2 + cz^2 = 1$$

$$\text{So } l(At) + m(Bt) + n(Ct) = p \text{ --- (2)}$$

$$a(A^2t^2) + b(B^2t^2) + c(C^2t^2) = 1 \text{ --- (3)}$$

$$\text{From (2), } t(lA + mB + nC) = p$$

$$t = \frac{p}{lA + mB + nC}$$

Put in (3)

$$(aA^2 + bB^2 + cC^2)t^2 = 1$$

$$(aA^2 + bB^2 + cC^2) \cdot \frac{p^2}{(lA + mB + nC)^2} = 1$$

$$\frac{p^2}{(lA + mB + nC)^2} = \frac{1}{(aA^2 + bB^2 + cC^2)}$$

$$\left(\frac{lA + mB + nC}{p} \right)^2 = aA^2 + bB^2 + cC^2$$

$$\text{from (1), } A = \frac{x}{t}$$

$$B = \frac{y}{t}$$

$$C = \frac{z}{t}$$

$$\left(\frac{\frac{x}{t} \cdot l + \frac{y}{t} \cdot m + \frac{z}{t} \cdot n}{p} \right)^2 = a \frac{x^2}{t^2} + b \frac{y^2}{t^2} + c \frac{z^2}{t^2}$$

$$\frac{1}{t^2} \left(\frac{xl + my + nz}{p} \right)^2 = \frac{1}{t^2} (ax^2 + by^2 + cz^2)$$

$$\left(\frac{xl + my + nz}{p} \right)^2 = ax^2 + by^2 + cz^2$$

is req. Eq of cone.

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Q24) By First of all we will find the eq. of line through origin is the eq. of directrix of req. cone.

For this, Given plane is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

meets the co-ordinate axes in A, B, C.

The coordinates of A, B, C are

$$A(a, 0, 0) \quad B(0, b, 0) \quad C(0, 0, c)$$

Let eq. of sphere through origin is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$$

it passes through A, B, C is

$$4u = -\frac{a}{2}, \quad v = -\frac{b}{2}, \quad w = -\frac{c}{2}$$

$$\therefore x^2 + y^2 + z^2 - ax - by - cz = 0$$

So eq. of directrix of cone is

$$x^2 + y^2 + z^2 - ax - by - cz = 0, \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The eq. of line through origin is

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = t \quad \left\{ \begin{array}{l} l = \frac{x}{t}, m = \frac{y}{t}, n = \frac{z}{t} \end{array} \right\} \quad (2)$$

Any pt on this line is (lt, mt, nt)

This pt lies on (1), (2)

$$(lt)^2 + (mt)^2 + (nt)^2 - a(lt) - b(mt) - c(nt) = 0$$

$$(l^2 + m^2 + n^2)t^2 - t(al + bm + cn) = 0 \quad (3)$$

$$\text{and } \frac{lt}{a} + \frac{mt}{b} + \frac{nt}{c} = 1$$

$$t\left(\frac{l}{a} + \frac{m}{b} + \frac{n}{c}\right) = 1$$

$$t = \frac{1}{\frac{l}{a} + \frac{m}{b} + \frac{n}{c}} \quad \text{put in (3)}$$

$$(l^2 + m^2 + n^2) \frac{1}{\left(\frac{l}{a} + \frac{m}{b} + \frac{n}{c}\right)^2} - \frac{al + bm + cn}{\frac{l}{a} + \frac{m}{b} + \frac{n}{c}} = 0$$

$$(l^2 + m^2 + n^2) - (al + bm + cn)\left(\frac{l}{a} + \frac{m}{b} + \frac{n}{c}\right) = 0$$

Eliminate l, m, n from (3), (4)

$$\left(\frac{x^2}{t^2} + \frac{y^2}{t^2} + \frac{z^2}{t^2}\right) - \left(\frac{ax}{t} + \frac{by}{t} + \frac{cz}{t}\right)\left(\frac{x}{at} + \frac{y}{bt} + \frac{z}{ct}\right) = 0$$

$$(x^2 + y^2 + z^2) - (ax + by + cz)\left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c}\right) = 0$$

$$(x^2 + y^2 + z^2) - \left(x^2 + y^2 + z^2 + \frac{ax^2}{b} + \frac{ay^2}{c} + \frac{bx^2}{c} + \frac{by^2}{a} + \frac{cx^2}{a} + \frac{cy^2}{b}\right) = 0$$

$$yz\left(\frac{b}{c} + \frac{c}{b}\right) + zx\left(\frac{c}{a} + \frac{a}{c}\right) + xy\left(\frac{a}{b} + \frac{b}{a}\right) = 0$$

is req. eq. of

Q25) The eq. of line through origin is

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = t \quad (1)$$

$$x = lt \quad y = mt \quad z = nt$$

Any pt on this line is (lt, mt, nt)

Given eq. of directrix is

$$x = a, \quad y^2 + z^2 = b^2$$

$$\therefore lt = a, \quad m^2 t^2 + n^2 t^2 = b^2 \quad (2)$$

$$\text{for (2), } t = \frac{a}{l} \quad \text{put in (2)}$$

$$(m^2 + n^2) \frac{a^2}{l^2} = b^2$$

$$(m^2 + n^2) a^2 = b^2 l^2$$

$$l = \frac{x}{t}, \quad m = \frac{y}{t}, \quad n = \frac{z}{t}$$

$$a^2 \left(\frac{y^2}{t^2} + \frac{z^2}{t^2} \right) = b^2 \frac{x^2}{t^2}$$

$$a^2 (y^2 + z^2) = b^2 x^2$$

is eq. of req. cone.

part II

For trace of this cone in xy-plane

$$\text{put } z = 0$$

$$a^2 y^2 = b^2 x^2$$

$$a^2 y^2 - b^2 x^2 = 0$$

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 0$$

This represents hyperbola.

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